TOUCH AND CROSSING: DUAL PRIMITIVE MODALITIES OF THRESHOLD MONITORS

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Abstract. Threshold monitors of real-valued signals fire on one of two operational semantics: the *crossing* modality, with events at upward transits of a threshold, or the *touch* modality, with events at local extrema. The runtime-verification and signal-processing literatures treat these as competing encoder families, usually presenting one as primitive and the other as a degenerate case. We argue that this framing is wrong: touch and crossing are *dual primitives* with incomparable minimal resource requirements — crossing is causal and carries one bit of memory, whereas touch is memoryless but needs a three-sample window and one step of lookahead — so neither monitor can simulate the other without genuinely new structure, and bridging them costs exactly one added bit of structure in each direction: persistent state one way, window or lookahead the other.

We develop three consequences. First, the two modalities sit on a ladder indexed by the number of distinct verdicts a monitor may emit: a verdict-free baseline, a degenerate binary rung, a ternary rung that hosts the touch/crossing duality, and a four-valued rung that organises both along orthogonal touch/crossing and weak/strong axes. The two-, three-, and four-valued rungs themselves are the known runtime-verification verdict progression (the two-/three-/four-valued LTL semantics); our contribution at this layer is not the ladder but the strict tuning hierarchy L1 $<$ L2 $<$ L3 of tuning powers that realise the rungs. Second, the touch–crossing join is a *literal parameter coincidence* at the four-valued rung: a single augmented monitor realises both modalities as exact parameter choices, and the three classical engineering encoders — edge detector, Schmitt trigger, and extremum detector — appear as named special cases of its ternary subfamily. Third, the duality is operational: a pointwise count inequality (upward crossings never outnumber touches, up to a trace-boundary tick) separates the two on every signal; a linear-time algorithm synthesises tuning parameters realising any *feasible* target finite event pattern, with an asymmetric feasibility criterion; and a safety / cosafety / liveness / coliveness classification places both modalities inside the existing temporal-property landscape without new logical machinery.

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1. Introduction

A threshold monitor reads a real-valued signal $Q : \mathbb{N} \to \mathbb{R}$ and emits, at selected positions, an event of type validate or invalidate. Two operational semantics for these events appear ubiquitously in the literature, often unstated and conflated:

Crossing modality.:
A validate event fires whenever Q transits an upward threshold θ — i.e., whenever Q was strictly below θ at the previous position and is at least θ at the current position. A subsequent validate requires a prior invalidate (a downward transit), enforcing alternation. This is the semantics of edge detectors, Schmitt triggers, and the atomic propositions of Signal Temporal Logic.

Touch modality.:
A validate event fires whenever Q attains a strict local maximum (with value at or above an optional barrier θ). Successive validate events require no invalidate between them — every local maximum is a fresh event. This is the semantics of extremum and peak detectors.

In practice the two modalities produce *different* event-traces on the same input signal. Section 6 makes this precise: the touch count N_T pointwise dominates the crossing count N_X, with equality only for signals whose above-θ excursions each contain a single local maximum. A signal that oscillates above θ — staying in the positive region while exhibiting multiple peaks — produces *several* touch events but only *one* crossing event. The two modalities are therefore not stylistic alternatives: they encode different properties of the signal.

This raises the question this paper answers. Fixing a single real-valued signal and the threshold-event *firing* semantics as the object of study — not the verdict alphabet, the property classification, or any visualisation of the monitor — we ask:

> *Is threshold-event firing governed by a single primitive modality, or by irreducibly distinct ones; and if distinct, how do they combine into a single monitor?*

A common framing — sometimes implicit in the way extremum detectors are presented in event-based sensing — suggests that touch is more fundamental, with crossing reducing to "touches at the threshold level". Equally, the dominance of the Schmitt-trigger and edge-detector traditions in monitoring suggests that crossing is the primitive and touch is the "no-state" degeneration of the crossing rule.

We argue that *neither view is correct*. Touch and crossing are dual primitives at the same structural level. A touch-native monitor and a crossing-native monitor have *incomparable* minimal capability sets: the former needs 3-window pattern recognition but no persistent state, while the latter needs only one bit of sticky state and reads a single sample per step (window 1). Neither set strictly contains the other. To obtain the behaviour of one modality starting from the minimal monitor of the other, one must *augment* the monitor with genuinely new structure — and the augmentation cost is the same in both directions, measured by the size of the smallest extension. We make this precise in Sections 9.1, 9.2 and 10.

The paper contributes:

(1) Formal definitions of the touch and crossing modalities as instant-level predicates over a value-trace (Sections 2, 4 and 5).

(2) A continuous-to-discrete *sampling correspondence* (Proposition 2.10) that positions the work relative to Signal Temporal Logic: continuous level-monitoring (STL-native crossing) and the discrete event stream of runtime verification are two samplings of one signal, with the touch/crossing duality as the shared invariant and the sub-tick events (those occurring strictly between two sampling instants) as the explicit case the correspondence does not capture.
(3) A pointwise count inequality $N_X(Q, \theta) \leq N_T(Q, \theta)$ and a characterisation of the equality case (Section 6).
(4) Two explicit transformations realising the dual-primitive duality (Sections 9.1 and 9.2), each adding capabilities that are demonstrably absent from the source monitor.
(5) The *augmented monitor* $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ of Section 10, which generalises the two modalities by enabling and disabling each threshold across segments (touch-only with state-machine alternation, n=3 verdict alphabet), and its *quaternary extension* $M^{(4)}(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \pi)$ of Section 10.2, which adds a central crossing threshold and a n=4 verdict alphabet (touch events at the outer barriers, crossing events at the central threshold). The quaternary monitor realises both M_T and M_C as exact parameter points (Theorem 10.14), settling the touch–crossing join as a literal parameter coincidence. The three classical threshold-encoder families (edge detector, Schmitt trigger, extremum detector) — historically presented as competing encoders, yet structurally sharing the same threshold-comparison core — coincide as the three named parameter points Coincident, Schmitt, Extremum of the n=3 subfamily (Proposition 10.5). The contribution is the unifying parameter space, not the existence of the individual encoders.
(6) A classification of the monitor's validation predicate within the CLASP partition (Section 13.5).

This paper provides the formal language needed to talk about encoding choices in the calibration analysis of temporal implication monitoring: the eight observational window classes of a downstream calibration analysis are calibrated under a fixed parameter point of the augmented monitor, and their migration under tuning operations is exactly motion in the parameter space $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$.

Related work. The classical foundations underlying this paper are the safety/liveness dichotomy of Lamport [15] and Alpern–Schneider ([1], recognising version [17]), the linear-time temporal logic of Pnueli [16], the temporal-property hierarchy of Manna–Pnueli [12], and the LTL complexity analysis of Sistla–Clarke [18]. We position the present work along four axes that are not subsumed by these foundations.

Runtime verification and verdict alphabets. Bauer, Leucker, and Schallhart introduced three-valued LTL_3 [4] and four-valued RV-LTL [19]; Leucker and Schallhart [8] survey the landscape; Diekert and Leucker [20] characterise monitorability topologically (the boundary has empty interior) and show that every $G_\delta \cap F_\sigma$ set is monitorable. Aceto, Achilleos, Francalanza, Ingólfsdóttir, and Lehtinen [21] give an operational reformulation of monitorability for regular properties; Camerlo and Dagnino [30] determine the descriptive-set-theoretic complexity of the family of monitorable subsets of a countable space. Earlier foundational work by Havelund and Roşu [28] gave an algorithm that synthesises efficient monitors for past-time LTL safety properties over finite traces. The introduction of Bartocci, Falcone, Francalanza, and Reger [25], the survey of specification-based monitoring of cyber-physical

systems by Bartocci et al. [26], and the Sánchez et al. survey [27] document the modern state of the field. These works fix the *monitorability boundary* at the verdict-stream layer; the present paper crosses that layer in the opposite direction by asking which *kinematic* event modalities a threshold monitor can express on a real-valued signal.

Signal monitoring and quantitative semantics. Signal Temporal Logic was introduced by Maler–Nickovic [10]; robust satisfaction over real-valued signals is the subject of Donzé–Maler [5]; quantitative edit-distance monitoring is in Jakšić et al. [23]; Tabuada and Neider [22] introduce robust LTL (rLTL), whose five truth values form a linearly ordered lattice. Henzinger and Saraç [29] formalise quantitative and approximate monitoring. These works ground the *quantitative* reading of the touch/crossing distinction that surfaces in Section 14.

Read structurally, this body of work occupies the *continuous* side of monitoring: STL evaluates a real-valued signal directly through level-comparison predicates $f(x) \geq c$ — the crossing modality in continuous time. The present paper works on the *discrete* side: the integer-tick event stream of runtime verification over the ternary alphabet $\{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$. The two sides are linked by the continuous-time refinement of Definition 2.9, under which the discrete monitor is a ϕ_{tick}-phased sampling of the continuous signal: *crossing* is the sampled level-transit (STL-native), while *touch* is the sampled local extremum, which STL's level predicates do not express natively. The touch/crossing duality can thus be read as the invariant of a correspondence between continuous-signal monitoring (STL) and discrete-event monitoring (the runtime-verification event stream), with the events that fall strictly between samples (Section 2) as the case that sampling does not capture rather than a defect of either modality. This is the precise sense in which our contribution stands relative to STL: we do not refine STL's quantitative semantics but identify the extremum-based primitive that its level-transit semantics leaves implicit, and locate both primitives on a single continuous-to-discrete bridge, formalised as the sampling correspondence of Proposition 2.10.

Enforcement and parametric adaptation. Falcone, Fernández, and Mounier [6] characterise what can be *enforced* at runtime by enforcement monitors that edit the monitored execution (suppressing or inserting events). This frames enforcement as a transducer applied to the execution. The augmented monitor of Section 10 is structurally distinct: it implements enforcement-via-adaptation on the monitor's parameters $(\theta_{\text{u}}, \theta_{\text{d}}, \pi)$ rather than on the monitored execution. A systematic development of this adaptation-based enforcement is left to future work.

Physical and circuit roots. The touch and crossing modalities have direct circuit analogues: edge detectors and timing-recovery loops [13], Schmitt triggers [14], dynamic vision sensors [9], and extremum-seeking control [7]. We recover the three classical encoder families (edge, Schmitt-trigger, extremum) as specific parameter points of the augmented monitor (Proposition 10.5), exposing the underlying algebra that ties them together.

Closest prior framings. The framing closest to ours is the runtime-verification treatment of LTL and STL. Bauer–Leucker–Schallhart's LTL_3 [4], which takes the monitored letters as given, and Maler–Nickovic's STL [10], whose atomic predicates are level comparisons, both develop their machinery around a single kind of atomic observation. Extremum-versus-crossing detection is of course classical signal processing; what is, to our knowledge, novel within the RV literature is the specific *framing* of the two as *incomparable* primitives in a

monitor-resource partial order (Definition 2.4), neither reducible to the other without added structure. The classical electronics constructions of Schmitt [14] and event-camera designs [9] each use one of the modalities natively, but treat the choice as a hardware-implementation matter rather than a duality statement; the timing-recovery literature separates threshold-crossing and derivative/peak methods as distinct classes [13], again without a duality statement. Bartocci et al.'s CPS-RV survey [26] enumerates both predicate families separately without flagging the duality.

Paper structure and detailed contributions. The remainder of the paper develops the contributions above in six layers; we preview them here as a roadmap.

Modality ladder (Section 3). The four rungs $|\mathbb{V}| \in \{1, 2, 3, 4\}$ partition the modality landscape: trivial (modality-free baseline, constant signal with binary detection state – the joint signal–threshold degenerate baseline), binary degenerate, ternary touch/crossing duality, quaternary 2×2 lattice along touch/crossing and weak/strong axes. Each rung is characterised by an explicit signal/threshold model and, where non-degenerate, a count proposition. The section opens at the ternary focus rung; the two *degenerate* rungs ($|\mathbb{V}|$=1, 2) are developed separately in Section 11, once the modalities and the tuning-layer hierarchy they presuppose are in place.

The augmented monitor (Section 10). The monitor $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ is the join of the two minimal n=3 monitors over a shared ternary event alphabet $\Sigma = \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$. A coverage matrix (Table 2) records the structural asymmetries between Model A, Model B, and the augmented join across native primitive, parameter space, direction–event coupling, count inequality, robustness margin, plateau conventions, and convention algebra cardinality.

Tuning taxonomy (Appendices A to D). A three-layer taxonomy of tuning operations: threshold and wall motion (moving-threshold rule, signal-direction rule, realisability inequality under motion); semantic-aggregation tunings (tick anatomy and tick-width tuning); tick-level signal copy-paste (general, continuity-preserving, and direction-coherent variants); and input-signal modification. A strict expressive hierarchy of the layers, $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$, is proved (Proposition 8.1).

Plateau conventions (Appendix E). Five plateau-touch conventions C_1–C_5 (Strict, Weak, Sample, Edge, DirRes) resolve the ambiguity of event firing on a touch plateau (DirRes later splits into its entry and exit refinements, giving the six-element timing set $\mathcal{T}$ of Definition E.4). They assemble into a convention algebra $\mathcal{C} = \mathcal{T}^{|W|} \times \mathcal{S}$ of cardinality $|\mathcal{C}_A| = 12$ (Model A) and $|\mathcal{C}_B| = 72$ (Model B); a touch-as-crossing duality theorem (Theorem E.1) connects them.

Event-stream algebra and LTL$_3$ semantics. The monitor output is an event stream $E_M(\sigma) \in V^\omega$ equipped with concatenation, windowing, type-projection, type-swap, and parallel composition. Each plateau convention induces an LTL$_3$ verdict assignment; the (V, A) parameter-pair reading makes explicit that V is fixed at the ternary event alphabet while A varies along the touch/crossing duality and the convention algebra — which parallels the verdict-alphabet ladder studied in [3].

Synthesis and CLASP classification (Sections 13.3 and 13.5). Given a signal Q and a target finite event pattern E^*, a synthesis algorithm produces a tuning-parameter assignment realising E^* on Q in time $O(n + T)$: Model A admits a linear scan with a signal-direction check (and an infeasibility verdict when E^* is not realisable); Model B realises every *separable-contact* target under moving walls, returning an infeasibility verdict only for a constant segment or two same-type events at adjacent ticks (targets that require signal

modification). CLASP classification: single events in Cosafety; under ALT alternation, unbounded balanced sequences in Mixed. The generalisation from finite event patterns to LTL trace properties via online parametric adaptation — an enforcement-via-adaptation mechanism distinct from edit-based runtime enforcement — is beyond the scope of this paper and is left to future work.

Robustness and noise. The two modalities admit canonical robustness margins on their native primitives: ρ_A linear in $|Q - \theta|$, ρ_B quadratic at extrema. The Schmitt-trigger hysteresis construction recovers as a robustness-restoring tuning of the augmented monitor.

A note on length. This paper exceeds the customary LMCS length guideline because it develops the touch/crossing duality end to end: the capability calculus, the augmented-monitor parameter space and its join, the strict tuning hierarchy, the synthesis algorithm, and the convention algebra are each compact but jointly necessary for the self-contained argument.

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2. Preliminaries

Definition 2.1 (Value trace)**.** A *value trace* is a sequence $\tau \in V^*$ (or V^ω for the infinite case), where V is a totally-ordered set. We write τ_n for the value at position $n \in \mathbb{N}$, and assume $\tau_n \in V$ is well-defined for all n.

For concreteness we take $V = \mathbb{R}$ throughout, with the standard order. All results extend to any totally-ordered V and to the element-value alphabet $\Sigma_{E,V} = E \times V$, which pairs each event symbol with its underlying value.

Definition 2.2 (Threshold and barrier)**.** A *threshold* is a value $\theta \in V$ at which monitor events are conditioned. A *barrier* is a threshold used to admit or reject events without imposing alternation.

The same parameter θ serves both roles in different modalities: *threshold* when used as a level-transit reference, *barrier* when used to constrain which local extrema fire events.

Definition 2.3 (Monitor)**.** A *monitor* on V is a function $M : V^* \to \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}^*$ mapping value traces to event traces, where ε denotes "no event". We assume M is causal: $M(\tau_{0:n})$ depends only on $\tau_{0:n}$ and on a bounded amount of state maintained by M.

We will be precise about the capability set of each monitor: the window size of its memory, the state it maintains, and whether it admits arbitrary lookahead.

Canonical event-alphabet notation. Throughout this paper we use a single fixed notation for event symbols. The ternary event alphabet is $\Sigma := \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ with $|\Sigma| = 3$; we write Σ_n when the rung must be made explicit (so $\Sigma_3 \equiv \Sigma$ in this paper's focus chapter, and Σ_4 is the quaternary alphabet of Section 10.2). We use the macros $\mathsf{validate}$ and $\mathsf{invalidate}$ inside formal statements and the abbreviations V and I inside event-trace literals or compact tables; the abbreviations are aliases of the macros, not separate symbols. The Greek ε denotes the silent (no-event) symbol and is also written "$-$" in event-stream literals when symbolic compactness is needed (e.g. Table 2). We keep ε in its *standard* role — the empty, no-event emission (the empty output word of a transducer, and the silent ε-move used by the subset construction in the determinisation below) — and do *not* overload it with a verdict

meaning. The three-valuedness of the monitor is carried instead by the *verdict* reading: under Definition 12.3 the emission stream maps to the LTL$_3$ truth-domain $\mathbb{B}_3 = \{\top, \bot, ?\}$ ($\mathsf{validate} \mapsto \top$, $\mathsf{invalidate} \mapsto \bot$, $\varepsilon \mapsto ?$), and it is the third verdict ? — the *inconclusive* value of three-valued runtime verification — that is the genuine indeterminate output ("no definite decision at this tick"), a first-class value and not mere silence. Every tick thus yields one of the three verdicts $\{\top, \bot, ?\}$: ? is the indeterminate value, whereas ε keeps its conventional reading as the silent emission that maps to it. (We avoid writing ε for the inconclusive verdict — in the literature ε is the empty string and "?" the inconclusive truth value; the two are distinct.)

Definition 2.4 (Capability triple)**.** The *capability triple* of a monitor M is the triple $\kappa(M) = (w, b, k) \in \mathbb{N}^3$ where

- $w \in \mathbb{N}_{>0}$ is the *window size*: the smallest integer such that the firing rule at position n depends only on the samples $\tau_{n-w+1}, \ldots, \tau_n$ (causal window) or $\tau_{n-w+1}, \ldots, \tau_{n+k}$ (windowed, anti-causal lag k);
- $b \in \mathbb{N}$ is the *persistent state bit-width*: $b = \lceil \log_2 |\mathcal{S}| \rceil$ for the automaton state set $\mathcal{S}$ of Definition 2.6, excluding the bounded sliding window;
- $k \in \mathbb{N}$ is the *anti-causal lag*: the number of future samples required to commit the firing decision at n. A causal monitor has $k = 0$; the touch-native monitor of Definition 5.3 has $k = 1$.

Two capability triples are *incomparable* when neither componentwise dominates the other in $\{(w_1, b_1, k_1) \leq (w_2, b_2, k_2)\} \iff \{w_1 \leq w_2 \wedge b_1 \leq b_2 \wedge k_1 \leq k_2\}$. We evaluate κ on the *canonical native automaton* of Definition 2.6 (window and persistent state are interchangeable carriers of finite history—a Myhill–Nerode trade—so κ is fixed on that canonical form, not as an infimum over realisations). The capability triples of Definitions 4.1 and 5.3 are then $\kappa(M_C) = (1, 1, 0)$ and $\kappa(M_T) = (3, 0, 1)$: M_C reads only the current sample (window 1) but carries one bit of sticky state and is causal; M_T is stateless but needs a width-3 window and one step of anti-causal lag. Each dominates the other on one coordinate (M_T on window and lag, M_C on state), so the triples are incomparable; this is the formal content of the "incomparable capability sets" claim of Section 1.

Remark 2.5 (Provenance of the three axes)**.** The three coordinates of κ are individually classical. The state-bit measure b is the Myhill–Nerode / automaton-memory index: the number of bits needed to record the residual history a monitor must distinguish is the $\log_2$ of its Nerode-class count [41]. The anti-causal lag k is a lookahead-latency measure, related to the anticipation of verdicts in predictive runtime verification [39] (which uses a model of the system) and to the delay before an LTL$_3$ verdict can be committed [4]. The window size w is the bounded-history span of a sliding-window detector. Our contribution is not any of these measures in isolation but their *packaging* into a single partial order: collecting the three axes into one tuple, ordered componentwise, is what makes *incomparability* — rather than a scalar cost — the expressible relation between two monitors.

Definition 2.6 (Monitor as parallel state automaton)**.** At every rung n a monitor is operationally realised as a finite-state automaton $\mathcal{A}_n = (\mathcal{S}_n, s_0, \delta_n, \mathcal{F}_n)$ in parallel with the value trace, where:

- $\mathcal{S}_n$ is a finite *state set* encoding the local relation of the signal to the threshold(s) and any auxiliary control channels;
- $s_0 \in \mathcal{S}_n$ is the initial state;

- $\delta_n : \mathcal{S}_n \times \mathcal{I}_n \to \mathcal{S}_n$ is the transition rule, where the input $\mathcal{I}_n$ is the tick-local configuration (τ_m, θ_m, d_m) – signal value, threshold value, and a binary detection-control channel $d_m \in \{\text{ON}, \text{OFF}\}$ that is implicitly ON at rungs $n \geq 2$;
- $\mathcal{F}_n \subseteq \mathcal{S}_n \times \mathcal{S}_n \to \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ assigns an event type to each *event-firing transition* (and ε to the rest).

The monitor fires an event of type $\mathcal{F}_n(s_{m-1}, s_m)$ at tick m when the automaton takes the transition $s_{m-1} \to s_m$.

Remark 2.7 (Ladder specialization of the automaton)**.** The state set, transition rule, and event-firing transitions specialize differently at each rung:

- $n = 1$ *(trivial)*: $\mathcal{S}_1 = \{\text{OFF}, \text{ON}\}$ (detection state); δ_1 driven by d_m; event-firing transition OFF $\to$ ON fires validate (V-mode), invalidate (I-mode), or stays silent (OFF-mode). The signal-threshold component of the configuration is degenerate ($\tau_m \equiv \theta_m \equiv$ const), so only d_m carries variability.
- $n = 2$ *(binary)*: $\mathcal{S}_2 = \{\theta_0, \theta_1\}$; δ_2 driven by $\tau_m \in \{\theta_0, \theta_1\}$; transitions $\theta_0 \to \theta_1$ fire V, $\theta_1 \to \theta_0$ fire I. Intermediate states are forbidden by the jumping-signal construction.
- $n = 3$, crossing modality: $\mathcal{S}_3^C = \{S_\downarrow, S_\uparrow\}$, the value-class of τ_m latched against the single threshold θ ($S_\downarrow$ below $\tau_m < \theta$, $S_\uparrow$ above $\tau_m \geq \theta$), with $s_0 = S_\downarrow$. The value-class is recomputed from the current sample (window 1) and the only stored state is the 1-bit latch, so $\kappa = (1, 1, 0)$. Event-firing transitions: validate on the upward crossing $S_\downarrow \to S_\uparrow$ ($\tau_m \geq \theta$) and invalidate on the downward crossing $S_\uparrow \to S_\downarrow$ ($\tau_m < \theta$); the two self-loops emit ε. The transition function is total, and the latch is causal; drawn as the two-state loop of Figure 3a. This is Figure 8 with the band collapsed ($\theta_\text{u} = \theta_\text{d} = \theta$); its ternary direction-resolved realisation, with the added engaged on-level state S_0, is Figure 11.
- $n = 3$, touch modality: $\mathcal{S}_3^T = \{\bullet\}$, a single stateless control state, with $\kappa = (3, 0, 1)$. The detector stores no state; it reads the three-sample window $(\tau_{m-1}, \tau_m, \tau_{m+1})$ and fires by *pattern*: validate on a strict local maximum at level ($\tau_{m-1} < \tau_m > \tau_{m+1}$ with $\tau_m \geq \theta_\text{u}$) and invalidate on a strict local minimum at level ($\tau_{m-1} > \tau_m < \tau_{m+1}$ with $\tau_m \leq \theta_\text{d}$); every other window emits ε. Detecting the extremum at τ_m needs the future sample τ_{m+1}, so the detector is anti-causal with lag 1 — the window, not a stored state, carries the history; drawn as the stateless single-state detector of Figure 3b.
- $n = 3$, Schmitt *(hysteretic crossing)*: $\mathcal{S}_3^S = \{L, M^\uparrow, U, M^\downarrow\}$, the product of the value-class of τ_m against the two trigger levels $\theta_\text{d} < \theta_\text{u}$ — L *(below)* $\tau_m \leq \theta_\text{d}$, M *(in-band)* $\theta_\text{d} < \tau_m < \theta_\text{u}$, U *(above)* $\tau_m \geq \theta_\text{u}$ — with a one-bit output latch. In the open band the latch is free, so M splits into the two reachable states $M^\uparrow$ (latch low, ascending branch) and $M^\downarrow$ (latch high, descending branch), giving the four states above with $s_0 = L$. The value-class $L/M/U$ is *recomputed* from the current sample (window 1), not stored; by the window-exclusion clause of Definition 2.4 only the 1-bit latch counts toward b, so $b = 1$ (not $\lceil \log_2 4 \rceil$). The four-state set is thus the un-minimised realisation: $M^\uparrow$ and $M^\downarrow$ emit identically to the external event stream until the latch flips and Myhill–Nerode-merge there. Hence $\kappa = (1, 1, 0)$. Event-firing transitions: validate on entering U from a low-latch state ($L \to U$, $M^\uparrow \to U$) and invalidate on entering L from a high-latch state ($U \to L$, $M^\downarrow \to L$); all other edges — band-internal stays and the silent fall-backs $M^\uparrow \to L$, $M^\downarrow \to U$ — emit ε. The states form a hysteresis loop (left rail $L, M^\uparrow$ output low; right rail $U, M^\downarrow$ output high; to flip one must traverse the whole band, so chatter inside M does not fire), drawn in Figure 8; the monitor-faithful (extremum) realisation

and its mapping are Figure 9. When $\theta_u = \theta_d$ the band M is empty, $M^\uparrow/M^\downarrow$ vanish, and $\mathcal{S}_3^S$ degenerates to the crossing latch $\{S_\downarrow, S_\uparrow\}$ of Definition 4.1[1] — Schmitt is "crossing with a band" (Remark 10.4).

- $n = 4$ *(quaternary)*: product of the touch and crossing automata of $n = 3$, with the five-valued alphabet $\Sigma^{(4)}$ tagging each event-firing transition by its source predicate, drawn in Figure 12.

The detection-state channel d_m is implicitly ON for $n \geq 2$ because the signal automaton already carries the variability needed to drive transitions. At $n = 1$ the signal automaton is degenerate, so the detection channel is the sole driver of event-firing transitions – this is the operational content of the discontinuous-threshold model of Remark 11.2.

Remark 2.8 (Automata for each interpretation of crossing and touch)**.** Remark 2.7 specialises the automaton by *rung*; we now specialise it by *interpretation*, so that every distinct reading of crossing and of touch used in this paper appears as an explicit automaton $\mathcal{A} = (\mathcal{S}, s_0, \delta, \mathcal{F})$. We write $S_\downarrow, S_\uparrow$ for the two crossing sticky-states, $\bullet$ for a single (stateless) control state, and "wall" for the predicate $\tau_m = \theta_u$ (upper) or $\tau_m = \theta_d$ (lower). All of these are realised in the proof-of-concept of Section 13.3.

Crossing interpretations.

- *Level-transit* (CROSSING, Definition 4.1): $\mathcal{S} = \{S_\downarrow, S_\uparrow\}$, $s_0 = S_\downarrow$; firing transitions $S_\downarrow \xrightarrow{\tau_m \geq \theta} S_\uparrow$ (validate) and $S_\uparrow \xrightarrow{\tau_m < \theta} S_\downarrow$ (invalidate). One bit of sticky state, reads the current sample only (window 1), causal, $\kappa = (1, 1, 0)$.
- *Hysteretic* (SCHMITT, Proposition 10.5): same two states, but the guards use two thresholds — $S_\downarrow \xrightarrow{\tau_m \geq \theta_u} S_\uparrow$ (validate) and $S_\uparrow \xrightarrow{\tau_m \leq \theta_d} S_\downarrow$ (invalidate), with the open band (θ_d, θ_u) firing nothing (consistent with the closed trigger levels of Definition 10.2). Same $\kappa = (1, 1, 0)$ for this level-transit realisation; the gap suppresses chatter. (Externally this is a crossing event stream; *internally*, when realised via the augmented monitor of Definition 10.2, the SCHMITT firing rule is a touch predicate at the two trigger levels, so its internal κ is touch-shaped — see Remark 10.4.)

Touch interpretations (by plateau convention of Definition E.4; they agree on a strict, isolated extremum and differ only on a wall plateau):

- STRICT (Definition 5.1): stateless, $\mathcal{S} = \{\bullet\}$; fires on the 3-window strict extremum $\tau_{m-1} < \tau_m > \tau_{m+1}$, $\tau_m \geq \theta_u$ (validate , symmetric for invalidate). Anti-causal, $\kappa = (3, 0, 1)$; never fires on a plateau.
- WEAK: stateless, $\mathcal{S} = \{\bullet\}$; fires at *every* wall-contact sample ($\tau_m = \theta_u$ gives validate). A 1-window value check on the plateau body.
- EDGE: $\mathcal{S} = \{\text{off}, \text{on}\}$ (off/on the wall); fires once, on the off $\to$ on entry edge (or on $\to$ off exit edge). One bit of plateau state.
- DIRRES: $\mathcal{S} = \{\text{off}, \text{on}\} \times \{+, -\}$ (plateau bit $\times$ approach-direction bit); fires at the entry or the exit edge, with the type resolved by the sign of the pre-plateau slope. Two bits of state — this product is the un-minimised form; its engaged states are the ternary $\{S_-, S_0, S_+\}$ of Figure 11, the canonical automaton on which $b = \lceil \log_2 3 \rceil = 2$ is evaluated.

[1]The collapse holds up to the plateau-at-θ convention of Definition 4.2: at $\theta_u = \theta_d = \theta$ the Schmitt invalidate guard $\tau_m \leq \theta$ is closed, whereas Definition 4.1 uses the strict $\tau_m < \theta$; the two differ only on a plateau sample exactly at θ, where Definition 4.2 fires no event.

- SAMPLE(f_s): $\mathcal{S} = \{\text{off}, \text{on}\} \times \mathbb{N}$ (plateau bit × a length counter c); fires at the $\lfloor f_s\, c \rfloor$ sampled positions of a length-c plateau. The counter makes this the only interpretation whose state is *unbounded* in the plateau length — a finite automaton only per fixed plateau length, which is why SAMPLE is a one-parameter family rather than a single point (Remark E.5).

Named encoders as parameter points. The five named encoders of Proposition 10.5 and Remark 10.7 are not new automata but parameter points of the above: COINCIDENT $M(\theta, \theta, \text{ALT})$ collapses the two touch walls onto one level and reduces to the level-transit crossing automaton; EXTREMUM $M(-\infty, -\infty, \text{FREE})$ is STRICT touch with the value filter removed (every strict extremum fires); SCHMITT is the hysteretic crossing automaton above; CROSSING $M^{(4)}(+\infty, \theta, -\infty, \text{ALT})$ is the central-threshold level-transit automaton; and TRIO $M^{(4)}(\theta_{\text{u}}, \theta_{\text{cross}}, \theta_{\text{d}}, \text{FREE})$ is the product automaton whose event-firing transitions are the disjoint union of the touch and crossing transitions (Theorem 10.14), drawn in Figure 12.

Reading off the value count. Across the catalogue the monitor's number of possible values is read directly from each automaton: the *verdict* alphabet $\mathbb{V}_n$ is the set of value-classes its guards distinguish ($|\mathbb{V}| = 3$ at the focus rung, 4 at the quaternary rung), while the *emission* alphabet is the set of distinct firing labels together with ε ($|\Sigma| = 3$ at $n = 3$, $|\Sigma^{(4)}| = 5$ at the Trio rung). The two grow at different rates: a touch barrier contributes a single firing label, whereas a crossing threshold contributes two (one up-, one down-transit). The quaternary rung adds a central *crossing* threshold θ_{cross} to the two outer touch barriers, so it gains one verdict band but two firing labels $\mathsf{validate}_{\text{cross}}, \mathsf{invalidate}_{\text{cross}}$ — which is why it carries $|\mathbb{V}| = 4$ but $|\Sigma^{(4)}| = 5$.

Finite- vs infinite-trace scope. Most results in this paper are *finite-trace* results: they are decidable on a bounded prefix of τ. A smaller set of claims concern the *infinite trace* V^ω and are obtained by an explicit finite-to-infinite lift, whose CLASP cell classification we take from [3]. Specifically:

- **Finite-trace claims** (decidable on a bounded prefix, no finite-to-infinite lift needed): the count inequality (Proposition 6.1), the touch-as-crossing duality (Theorem E.1), the join statement at the quaternary rung (Theorem 10.14), the synthesis algorithm and its termination (Section 13), and all instant-level CLASP classifications (Proposition 13.10).
- **Infinite-trace claims** (require lifting a finite witness to an ω-property, with an explicit account of the unverified suffix): ω-balanced alternation patterns (Proposition 13.11), the Liveness/Cosafety/Mixed lattice classification of emitted event languages, and the universal-property conjecture (Conjecture 10.12). These claims are anchored at the CLASP partition of [3]; each cell assignment is checked directly against the class definitions there.

Where a claim is infinite-trace we state explicitly, in the proof or remark that follows it, which part of the infinite suffix no finite prefix determines. Unmarked claims are finite-trace.

Continuous-time refinement and tick-frame phase. Definition 2.3 fixes the monitor as a function on integer-indexed traces. Several figures and synthesis constructions in this paper (notably Figure 22, Figure 35 (b), Figure 29, and Figure 19) present events firing at *non-integer* times such as $t^* \approx 2.694$ or target instants $t \in \{1.5, 2.5, 4.5\}$. These belong to the

following continuous-time refinement, used purely as an illustrative extension of the discrete monitor.

Definition 2.9 (Continuous-time refinement)**.** Given an integer-indexed value trace $\tau : \mathbb{N} \to V$, its *continuous-time refinement* is a piecewise-smooth extension $\tilde{\tau} : \mathbb{R}_{\geq 0} \to V$ such that $\tilde{\tau}(n) = \tau_n$ for every $n \in \mathbb{N}$. In this paper $\tilde{\tau}$ is realised by the explicit piecewise formula of the figure caption (e.g. a half-cosine on $[n-1/2, n+1/2]$ for the landscape Q_v of Figure 27). The *tick-frame phase* $\phi_{\text{tick}} \in [0, 1)$ shifts the integer-tick sampling grid so that the discrete monitor of Definition 2.3 reads $\tilde{\tau}$ at positions $t_n^{\phi} := n + \phi_{\text{tick}}$. The canonical phase $\phi_{\text{tick}} = 0$ recovers $\tau_n = \tilde{\tau}(n)$; $\phi_{\text{tick}} = 1/2$ aligns tick centres with half-integers, the setting used in Figure 19.

Sub-tick events — those firing at $t \in \mathbb{R} \setminus (\mathbb{N} + \phi_{\text{tick}})$ — are not events of the discrete monitor in the formal sense of Definition 2.3. They are points at which a continuous-time extension of the firing predicate is satisfied between two consecutive tick samples; figures invoking such events do so to illustrate the *moving-threshold rule* of Appendix A at sub-tick resolution. Where a result of this paper claims a sub-tick event we either (i) note that the event is sub-tick and quote it as a continuous-time consequence, or (ii) realise it discretely by setting ϕ_{tick} so that the event time becomes a tick centre.

Continuous firing instants. For a continuous-time refinement $\tilde{\tau}$ and a threshold θ, a *continuous crossing instant* is a time t^* at which $\tilde{\tau}$ transits θ (i.e. $\operatorname{sgn}(\tilde{\tau} - \theta)$ changes sign at t^*), typed validate on an upward transit and invalidate on a downward one; a *continuous touch instant* is a time t^* at which $\tilde{\tau}$ has a strict local extremum meeting the wall constraint of Definition 5.1, typed by which wall it meets. These are the continuous-time analogues of the discrete predicates of Definitions 4.1 and 5.1.

The monitor-automaton of Definition 2.6 reading the sampled trace $\left(\tilde{\tau}(t_k^{\phi})\right)_k$ is precisely the *discrete sampler* of the continuous signal: its tick-local input (τ_m, θ_m, d_m) is the sample of $\tilde{\tau}$ at t_m^{ϕ}. The following statement makes the continuous-to-discrete map exact under an adequate-sampling hypothesis, and states explicitly the cases it does not capture.

Proposition 2.10 (Continuous-to-discrete sampling correspondence)**.** *Let $\tilde{\tau}$ be a continuous-time refinement (Definition 2.9) sampled at $t_k^{\phi} = k + \phi_{\text{tick}}$, and let $\mathcal{A}_n$ be the rung-n monitor automaton of Definition 2.6 run on the sample sequence $\left(\tilde{\tau}(t_k^{\phi})\right)_k$. Call $\tilde{\tau}$* adequately sampled *(for the rung-n modality at phase ϕ_{tick}) when*

(A1) (separation) *every open inter-sample interval $(t_k^{\phi}, t_{k+1}^{\phi})$ contains at most one continuous firing instant of the modality (Section 2); and*

(A2) (extrema on grid) *for the touch modality, every continuous touch instant of $\tilde{\tau}$ occurs at a sample position t_k^{ϕ}; and*

(A3) (strict on-grid separation) *for the touch modality, at every sampled touch extremum t_k^{ϕ} the two neighbouring samples are strictly separated from it ($\tilde{\tau}(t_{k-1}^{\phi}) \neq \tilde{\tau}(t_k^{\phi})$ and $\tilde{\tau}(t_{k+1}^{\phi}) \neq \tilde{\tau}(t_k^{\phi})$, with no on-grid peak plateau $\tilde{\tau}(t_{k-1}^{\phi}) = \tilde{\tau}(t_k^{\phi})$), so the strict 3-window touch predicate of Definition 5.1 fires. This is the same* STRICT*-induced regularity the paper adopts globally until Appendix E (Remark 5.2).*

Under (A1)–(A3)*:*

(1) *the event stream emitted by $\mathcal{A}_n$ coincides, in type and order, with the sequence of continuous firing instants of $\tilde{\tau}$ — the crossing instant at t^* is reported at the first sample $t_k^\phi \geq t^*$, the touch instant at its (sampled) extremum; and*
(2) *the touch/crossing duality is preserved by the sampling: the crossing automaton $\mathcal{A}_3^C$ and the touch automaton $\mathcal{A}_3^T$ sample the same $\tilde{\tau}$ into event streams whose capability triples (Definition 2.4) remain incomparable, exactly as in continuous time.*

When (A1)*,* (A2)*, or* (A3) *fails, the omitted firing instants — two or more inside one inter-sample interval, an off-grid touch extremum, or a sampled extremum tied to a neighbour (an on-grid peak plateau, silent under* STRICT*) — are exactly the cases the correspondence does not capture: the bijection of clause (1) holds modulo this excluded set, which is exactly the sub-tick stratum discussed above together with the on-grid tie of Remark 5.2.*

Proof sketch. (1) Crossing: by (A1) each continuous transit t^* is the unique modality instant in its inter-sample interval, so $\operatorname{sgn}(\tilde{\tau} - \theta)$ takes opposite values at the two enclosing samples and is constant on the sampled history between consecutive transits; the two-window crossing predicate of $\mathcal{A}_3^C$ (the stateless Myhill–Nerode form of the native crossing monitor of Definition 4.1, sticky-state alternation) therefore fires exactly once, at the first sample with the flipped sign, with the type fixed by the transit direction. Touch: by (A2) the continuous extremum sits at a sample t_k^ϕ, by (A1) its two neighbours $t_{k\pm1}^\phi$ enclose no further extremum, and by (A3) those neighbours are strictly separated from $\tilde{\tau}(t_k^\phi)$ (no on-grid peak plateau), so $\big(\tilde{\tau}(t_{k-1}^\phi), \tilde{\tau}(t_k^\phi), \tilde{\tau}(t_{k+1}^\phi)\big)$ is a strict 3-window extremum and the stateless touch predicate of $\mathcal{A}_3^T$ (Definition 5.1) fires there. The maps $t^* \mapsto$ (firing sample) are injective by (A1) and exhaustive by construction, giving the type-and-order bijection. (2) The two automata read the *same* sampled trace; their capability triples $\kappa(\mathcal{A}_3^C) = (1, 1, 0)$ and $\kappa(\mathcal{A}_3^T) = (3, 0, 1)$ (Definition 2.4) are properties of the automata, not of the sampling, hence incomparable after sampling as before (here κ is the canonical native-form value of Definition 2.4: the crossing modality's canonical automaton is window-1 with one-bit sticky state, giving $(1, 1, 0)$; its stateless 2-window realisation is the Myhill–Nerode-equivalent $(2, 0, 0)$, and we report the canonical form throughout). The cases the correspondence does not capture are exactly the contrapositive of (A1)–(A3): any firing instant interior to an inter-sample interval, any off-grid touch extremum, or any on-grid peak plateau (a sampled extremum tied to a neighbour, strict-touch-silent) is unsampled or strict-silent and so excluded; quoting it requires either a finer phase ϕ_{tick} (Definition 2.9), a non-strict plateau-touch convention (Appendix E), or the explicit sub-tick annotation above. □

This proposition is the formal core of the continuous-to-discrete reading announced in the related-work discussion of Section 1: STL-style continuous level-monitoring and the discrete event stream of runtime verification are two samplings of one signal, and the touch/crossing duality is the invariant they share. The sampled reading on which this bridge relies matches the input model of efficient robust STL monitoring, which computes robustness over piecewise-linear signals given by their sampling points (an offline algorithm) [40]. With the primitives, the automaton model, and this sampling bridge in place, we now situate touch and crossing on the verdict-alphabet ladder, opening at the ternary focus rung where the two modalities first become distinguishable.

3. The modality ladder by verdict-alphabet cardinality

Before we develop the touch and crossing modalities in detail, we situate them on a ladder indexed by the cardinality of the monitor's verdict alphabet. The ladder mirrors the verdict-alphabet ladder of [3], but the question it answers is different: there the cardinality $|\mathbb{V}|$ indexes the verdict monitors whose relation to the CLASP cells is studied at Layer 1 (monitorability itself is not a function of the cell [4, Lemma 3.8]); here the same parameter controls *how many threshold-event modalities* the monitor can express. The two enumerations are independent but structurally parallel. The rung cardinalities $|\mathbb{V}| \in \{2, 3, 4\}$ are themselves the standard runtime-verification verdict progression — two-valued LTL, three-valued LTL_3 [4], and four-valued RV-LTL [19]; what is new here is not the ladder of verdict counts but the strict tuning hierarchy $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$ (Proposition 8.1) of the tuning powers that realise the modalities populating each rung.

A *modality* in the present sense is an instant-level predicate $\mu : V^* \to \{0, 1\}$ that selects, from a value trace τ, the positions at which a validate event should fire. Two modalities are *distinguishable* if there exists a signal on which they fire at different positions. We enumerate the modalities expressible at verdict-alphabet cardinality $|\mathbb{V}| \in \{1, 2, 3, 4\}$, recording at each rung the modality count, the structural mechanism that makes the additional modalities expressible, and the canonical exemplars. The two *degenerate* rungs at the bottom of the ladder — the trivial rung $|\mathbb{V}|$=1 and the binary rung $|\mathbb{V}|$=2 — are specialisations that presuppose the touch and crossing modalities they collapse, together with the tuning powers that realise them; we therefore defer their detailed treatment to Section 11, after the modalities and the tuning-layer hierarchy are established. We open directly at the *focus rung*, $|\mathbb{V}|$=3, where the two modalities first become distinguishable, and then ascend to $|\mathbb{V}|$=4.

3.1. The ternary rung ($|\mathbb{V}| = 3$): the touch/crossing duality.

With verdict alphabet $\mathbb{V}_3 = \{-1, 0, +1\}$ (or equivalently $\{\bot, ?, \top\}$ as in LTL_3 [4]) the monitor can express three mutually exclusive value-classes at a single instant: *above* threshold $(+1)$, *at or near* threshold (0), and *below* threshold (-1). The neutral 0-symbol is the structural enabler: it lets the monitor emit a verdict that distinguishes the transit-event (a $-1 \to +1$ transition crossing through 0) from the attainment-event (the signal arriving at the boundary without committing to either side).

Proposition 3.1 (The ternary rung hosts two modalities)**.** *At $|\mathbb{V}| = 3$ the value-class predicates available at one instant are $\{v = +1, v = 0, v = -1\}$ (and their Boolean combinations). Combined with a one-step lookback this yields two distinguishable instant-level threshold modalities, both expressible without unbounded state:*

1. Crossing*: the transit event $v_{n-1} = -1 \wedge v_n = +1$ (and dually $v_{n-1} = +1 \wedge v_n = -1$ for* invalidate*).*
2. Touch*: the attainment event $v_n = 0$ as a discrete proxy for "the signal is at the threshold", or in finer formulations the local-extremum event read from a three-window (v_{n-1}, v_n, v_{n+1}) pattern over the ternary alphabet.*

The two are distinguishable: a signal that touches the threshold from above and returns above $(+1, 0, +1)$ fires a touch event but no crossing event. These two modalities also exhaust the rung up to threshold tuning: every other 3-valued threshold modality reduces to one of them via a choice of $(\theta_u, \theta_d, \pi)$ in the augmented monitor family (Theorem 10.10).

Proof sketch. Crossing is encoded by a length-2 pattern over $\mathbb{V}_3$ with sticky-state alternation (Definition 4.1). Touch in the discrete proxy form is encoded by the length-1 pattern $v_n = 0$; in the local-extremum form it is encoded by the length-3 pattern $v_{n-1} \neq +1 \wedge v_n = 0 \wedge v_{n+1} \neq +1$ or the dual minima pattern. The signal $(+1, 0, +1)$ satisfies the touch pattern — it matches the discrete proxy $v_n = 0$ and the dual *from-above* extremum pattern $v_{n-1} \neq -1 \wedge v_n = 0 \wedge v_{n+1} \neq -1$ (a tangency that reaches θ from above and returns), though not the *from-below* pattern stated first — whereas the crossing pattern requires the prior verdict to be -1, which is absent here. Hence the modalities are distinguishable. □

The dive into the formal definitions, the count-inequality $N_X \leq N_T$, and the structural duality of the two modalities occupies the remainder of this paper (Sections 4 to 7, 9.1, 9.2 and 10). The ternary rung is the *focus rung* of this paper: Proposition 3.1 states the count; the body of the paper develops the structure. Figure 1 previews the side-by-side comparison of the two modalities on a continuous sinusoidal signal, anticipating the formal development of the touch and crossing event semantics in Sections 4 and 5.

3.2. **The quaternary rung ($|\mathbb{V}| = 4$): three thresholds, four bands, 2×2 modality lattice.** *Signal model.* We keep the continuous signal model of the focus rung — a sinusoid, or more generally a continuous real-valued trajectory $\tau : \mathbb{N} \to \mathbb{R}$ sampled at ticks — and *enrich the threshold structure.* Instead of a single threshold we now fix *three* thresholds

$$\theta_{\mathrm{LL/L}} \ < \ \theta_{\mathrm{L/H}} \ < \ \theta_{\mathrm{H/HH}},$$

each playing a distinct semantic role:

- $\theta_{\mathrm{L/H}}$ *(central threshold)*: divides the value domain into a *low* half-plane and a *high* half-plane. It is the polarity boundary and plays the role of the single threshold of the focus rung.
- $\theta_{\mathrm{LL/L}}$ *(lower outer threshold)*: further refines the low side by separating *low-low* (well below) from *low* (just below central).
- $\theta_{\mathrm{H/HH}}$ *(upper outer threshold)*: symmetrically refines the high side by separating *high* from *high-high.*

The three thresholds partition the value domain into four horizontal bands LL, L, H, HH (low-low, low, high, high-high). A sinusoidal signal of varying amplitude can cross any subset of the three thresholds during a single excursion, so the geometric model is realisable and non-trivial: the modality count is not an artefact of alphabet labelling.

Verdict alphabet. $\mathbb{V}_4 = \{\mathrm{LL}, \mathrm{L}, \mathrm{H}, \mathrm{HH}\}$, equivalently the RV-LTL four-valued domain [19] (there $\{\bot, \bot_p, \top_p, \top\}$, linearly ordered; written here $\{\bot_s, \bot_w, \top_w, \top_s\}$) via

$$\mathrm{LL} \leftrightarrow \bot_s, \quad \mathrm{L} \leftrightarrow \bot_w, \quad \mathrm{H} \leftrightarrow \top_w, \quad \mathrm{HH} \leftrightarrow \top_s,$$

assigned by band:

$$\mathbb{V}_4 = \big\{\, v < \theta_{\mathrm{LL/L}} \colon \mathrm{LL};\ \ \theta_{\mathrm{LL/L}} \leq v < \theta_{\mathrm{L/H}} \colon \mathrm{L};\ \ \theta_{\mathrm{L/H}} \leq v < \theta_{\mathrm{H/HH}} \colon \mathrm{H};\ \ v \geq \theta_{\mathrm{H/HH}} \colon \mathrm{HH} \,\big\}.$$

The assignment is symmetric across the central threshold. Values inside the central strip $\mathrm{L} \cup \mathrm{H}$ (between the outer thresholds) carry the revocable weak verdicts $\bot_w, \top_w$; values in the outer bands LL, HH carry the absorbing strong verdicts $\bot_s, \top_s$ in the RV-LTL reading. Figure 2 illustrates the signal model.

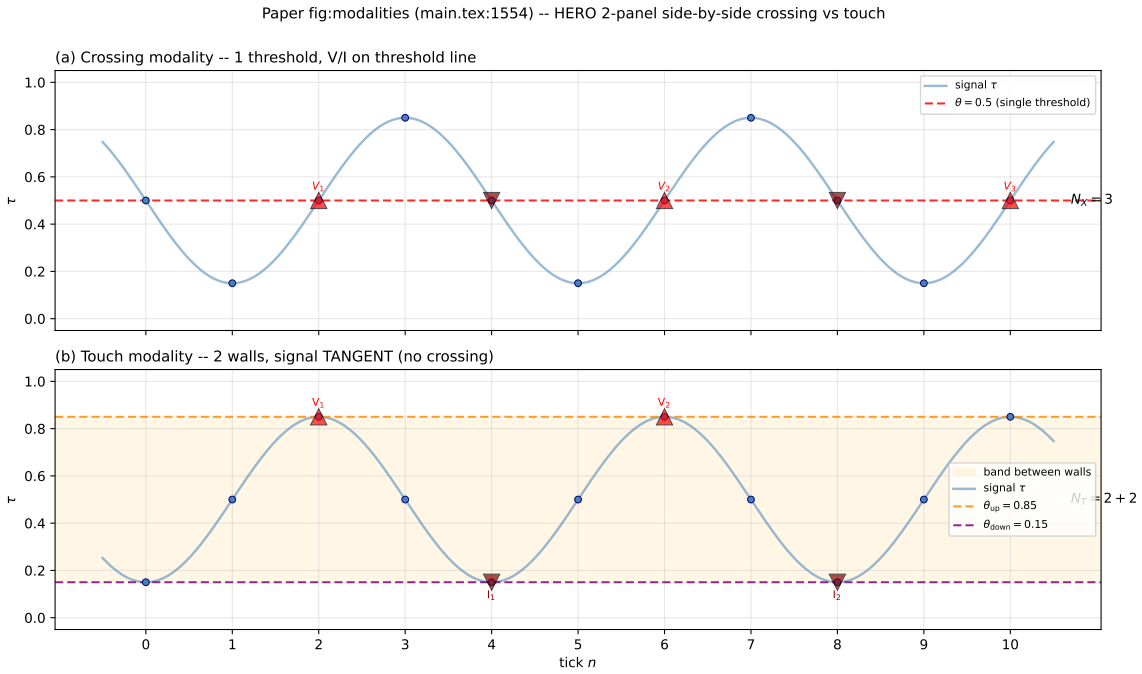


FIGURE 1. Side-by-side comparison of the two modalities on the same underlying sinusoid (shifted by a quarter period between panels so that events align in x). **(a)** The *crossing modality* fires validate at upward transits of $\theta = 0.5$ and invalidate at downward transits. **(b)** The *touch modality* fires validate at maxima tangent to $\theta_{\mathrm{u}} = 0.85$ and invalidate at minima tangent to $\theta_{\mathrm{d}} = 0.15$. The interior events align across the two panels — validate at $x = 2, 6$ and invalidate at $x = 4, 8$ in both — so the modalities fire at the same plotted x-coordinate but at *structurally different* signal locations: panel (a) at mid-slope (level transits), panel (b) at slope-zero (extrema). The causal crossing detector *additionally* fires at the boundary transits $x = 0, 10$, where the anti-causal touch detector cannot: the endpoint extrema have an undefined outer neighbour and do not fire under STRICT (Remark 5.2). (The pointwise count inequality $N_X \leq N_T$ of Section 6 compares the two on a *single* signal, not across these phase-shifted panels.) The two minimal automata — the crossing latch and the touch detector — are the two corners of Figure 3.

Proposition 3.2 (The quaternary rung hosts a 2×2 modality lattice)**.** *On the continuous signal model with three thresholds, four distinguishable threshold-event modalities arise. They arrange as a* 2×2 *lattice along two orthogonal axes — a kinematic axis (touch vs crossing) and an absorption axis (weak vs strong):*

	weak	strong
touch	Touch_w	Touch_s
crossing	Cross_w	Cross_s

with the semantics

- Touch_w**::** *the signal attains a local extremum inside the central strip* $\mathrm{L} \cup \mathrm{H}$ *(between the outer thresholds). The verdict visits* $\top_w$ *or* $\bot_w$ *at the extremum but does not enter the outer bands* LL *or* HH.
- Touch_s**::** *the signal attains a local extremum in one of the outer bands* LL *or* HH. *The verdict reaches* $\bot_s$ *or* $\top_s$.
- Cross_w**::** *the signal transits the central threshold* $\theta_{\mathrm{L/H}}$ *without entering an outer band. The verdict transitions* $\mathrm{L} \to \mathrm{H}$ *(upward) or* $\mathrm{H} \to \mathrm{L}$ *(downward), staying inside the revocable strip.*
- Cross_s**::** *the signal transits one of the outer thresholds* $\theta_{\mathrm{LL/L}}$ *or* $\theta_{\mathrm{H/HH}}$. *The verdict transitions* $\mathrm{L} \to \mathrm{LL}$ *or* $\mathrm{H} \to \mathrm{HH}$ *(or the symmetric upward exits) — a band-step into the outer region.*

The four modalities are pairwise distinguishable.

Proof sketch. We exhibit signals that realise each of the four cells of the 2×2 lattice while leaving the other three cells inactive, witnessing pairwise distinguishability.
Inner-strip witness. A sinusoid of small amplitude oscillating across $\theta_{\mathrm{L/H}}$ but staying inside the central strip $\mathrm{L} \cup \mathrm{H}$ fires Cross_w at each central-threshold transit and Touch_w at each local extremum, while never reaching LL or HH and therefore never firing Cross_s or Touch_s. This distinguishes the weak modalities from the strong.
Outer-band witness. Raising the amplitude of the same sinusoid past $\theta_{\mathrm{H/HH}}$ (or below $\theta_{\mathrm{LL/L}}$) introduces Cross_s at each outer-threshold transit and Touch_s at each extremum in the outer band, without removing the weak variants which still fire at the inner threshold and inner-strip extrema. This distinguishes touch from crossing on the strong side.
Axis orthogonality. The two axes (kinematic touch/crossing; band depth weak/strong) do not interact: by independent excursions of the signal (slow build-up to the outer band; brief inner-strip oscillation without reaching the outer band), any combination of the four modality firings is realisable on the same monitor. Hence the cells of the 2×2 lattice are pairwise distinguishable. □

Remark 3.3 (Why the count doubles from $|\mathbb{V}| = 3$ to $|\mathbb{V}| = 4$)**.** The third symbol of $\mathbb{V}_3$ unlocks the touch/crossing *kinematic axis*: it introduces a neutral value at which attainment and transit can be told apart. The fourth symbol of $\mathbb{V}_4$ unlocks an *orthogonal* axis — band depth — by adding the two outer thresholds $\theta_{\mathrm{LL/L}}$ and $\theta_{\mathrm{H/HH}}$ that split each side of the central threshold into a near band (L or H) and a far band (LL or HH). The two axes do not interact geometrically: the same sinusoidal trajectory can touch and cross in either the central strip or the outer bands, and the four modality cells are realised by independent excursions of the signal. Heuristically, the count therefore *multiplies* by 2 rather than incrementing by 1.

Remark 3.4 (Connection to the augmented monitor and the (V, A) pair)**.** The augmented monitor $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ of Section 10 is the $|\mathbb{V}| = 3$ join of touch and crossing relative to a single central threshold $\theta = \theta_{\mathrm{L/H}}$ (the symmetric case collapses $\theta_{\mathrm{u}} = \theta_{\mathrm{d}}$). The $|\mathbb{V}| = 4$ extension keeps the same operating algebra A (pattern matching plus sticky-state alternation per threshold) and enriches the alphabet V by introducing the two outer thresholds $\theta_{\mathrm{LL/L}}$ and $\theta_{\mathrm{H/HH}}$. Read through the (V, A) pair (Remark 12.4), the move from $|\mathbb{V}| = 3$ to $|\mathbb{V}| = 4$ is a pure V-side enrichment. In the RV-LTL reading of Section 13.5 the near/far band split coincides exactly with the current/permanent distinction of $\top_w$ vs $\top_s$ and $\bot_w$ vs $\bot_s$.

A concrete realisation of this V-side enrichment is the *quaternary augmented monitor* $M^{(4)}(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \pi)$ of Definition 10.13, under the threshold correspondence $\theta_{\mathrm{u}} \leftrightarrow \theta_{\mathrm{H/HH}}$

(upper touch barrier), $\theta_{\text{cross}} \leftrightarrow \theta_{\text{L/H}}$ (central crossing threshold), and $\theta_{\text{d}} \leftrightarrow \theta_{\text{LL/L}}$ (lower touch barrier). The outer touch barriers and the central crossing threshold realise the $\text{Touch}_s/\text{Cross}_w$ pair of the 2×2 lattice and embed both native modalities M_T, M_C as exact parameter points (Theorem 10.14).

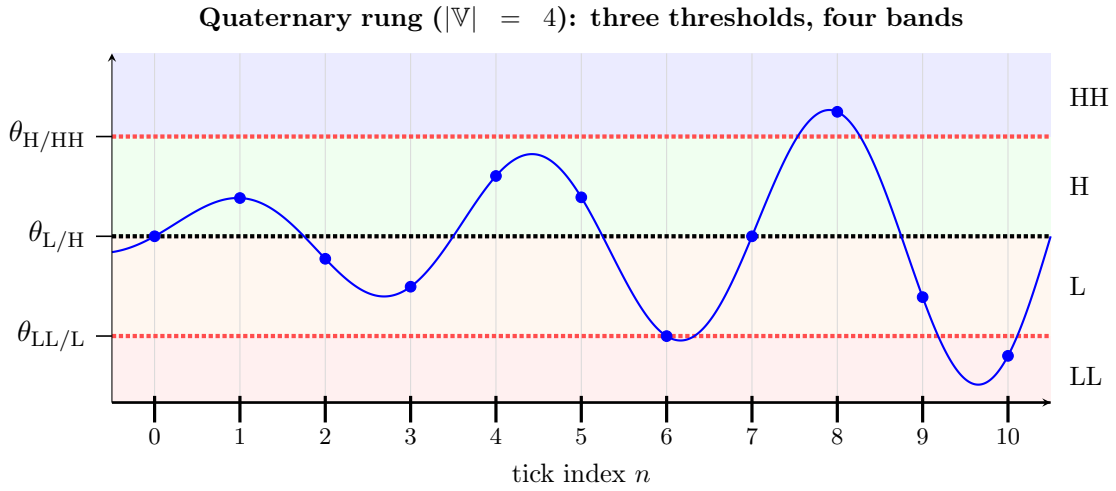


FIGURE 2. Quaternary rung ($|\mathbb{V}| = 4$) signal model. Three thresholds $\theta_{\text{LL/L}} < \theta_{\text{L/H}} < \theta_{\text{H/HH}}$ partition the value domain into four bands $\text{LL}, \text{L}, \text{H}, \text{HH}$. The central threshold $\theta_{\text{L/H}}$ is the polarity boundary (low/high); the two outer thresholds further separate near-band from far-band on each side. A sinusoidal signal of growing amplitude visits all four bands and triggers four distinguishable modalities arranged in a 2×2 lattice along orthogonal kinematic (touch/crossing) and band-depth (near/far) axes (Proposition 3.2). The verdict alphabet $\mathbb{V}_4 = \{\text{LL}, \text{L}, \text{H}, \text{HH}\}$ matches the RV-LTL domain $\{\bot_s, \bot_w, \top_w, \top_s\}$ via $\text{LL} \leftrightarrow \bot_s$, $\text{L} \leftrightarrow \bot_w$, $\text{H} \leftrightarrow \top_w$, $\text{HH} \leftrightarrow \top_s$.

3.3. **Summary and scope of this paper.** The remainder of this paper develops the $|\mathbb{V}| = 3$ row of Table 1: formal definitions of the touch and crossing modalities (Sections 4 and 5), the pointwise count inequality $N_X \leq N_T$ (Section 6), structural asymmetries between the two (Section 7), and the augmented monitor as the join of the two minimal monitors (Section 10). The two degenerate rungs $|\mathbb{V}| = 1$ and $|\mathbb{V}| = 2$ are then treated as κ-minimal specialisations of that machinery in Section 11. The $|\mathbb{V}| = 4$ row is recorded in this section for structural completeness; a full development of the 2×2 modality lattice under hysteresis and saturation thresholds is left to future work.

4. The crossing modality

Definition 4.1 (Crossing-native monitor)**.** The *crossing-native monitor* with threshold θ is the monitor M_C^θ that maintains a one-bit sticky state $s \in \{0, 1\}$, initialised to $s := 0$, and

TABLE 1. Modality count per verdict-alphabet rung. The third symbol of $\mathbb{V}_3$ unlocks the touch/crossing kinematic axis; the fourth symbol of $\mathbb{V}_4$ unlocks the orthogonal weak/strong absorption axis. The remainder of this paper is the deep dive on the $|\mathbb{V}| = 3$ rung; the $|\mathbb{V}| = 4$ row is stated here for structural completeness, with full development deferred to future work. The two degenerate rungs $|\mathbb{V}| = 1, 2$ are listed for ladder completeness and detailed in Section 11.

Rung	**Signal/threshold model**	#	**Modalities expressible**	**Structural enabler**
$\lvert\mathbb{V}\rvert=1$ trivial	Constant signal on the threshold; binary detection state (ON/OFF), no value-class motion.	0	None (modality-free baseline). Touch and crossing do not arise.	Joint signal–threshold degeneracy; the only freedom is the detection state (Section 11).
$\lvert\mathbb{V}\rvert=2$ binary	Piecewise-constant signal at two thresholds $\theta_0<\theta_1$, jumping between them. No continuous trajectory.	1	Threshold-membership $b_n=1$. Touch and crossing *collapse*.	Signal never inhabits the gap (θ_0,θ_1), so kinematic transit is unsampled (Proposition 11.5).
$\lvert\mathbb{V}\rvert=3$ Kleene	Continuous signal. One central threshold, three regions (above/at/below).	2	Touch (Section 5); Crossing (Section 4).	Neutral 0-symbol distinguishes attainment from transit (Proposition 3.1). **Focus rung.**
$\lvert\mathbb{V}\rvert=4$ RV-LTL	Continuous signal. Central threshold separates low/high; two outer thresholds further separate LL/L and H/HH. Four bands LL, L, H, HH.	4	Touch_w, Touch_s, Cross_w, Cross_s (2×2 lattice).	Outer thresholds split each polarity into a near (L, H) and a far (LL, HH) band (Proposition 3.2 and Remark 3.3).

emits at each position n:

$$M_C^\theta(\tau_n) = \begin{cases} \mathsf{validate} & \text{if } s=0 \text{ and } \tau_n \geq \theta; \quad s:=1 \\ \mathsf{invalidate} & \text{if } s=1 \text{ and } \tau_n < \theta; \quad s:=0 \\ \varepsilon & \text{otherwise.} \end{cases}$$

Equivalently, M_C^θ emits validate at every upward transit of θ and invalidate at every downward transit. The capability set is minimal: 1 threshold, 1 bit of sticky state, 1 step of memory. The monitor is causal: it emits its decision at position n using only the history up to n. The two sticky states and their two firing transitions form the minimal level-crossing latch, drawn (beside the touch detector) in Figure 3a.

Definition 4.2 (Signal-direction rule)**.** The *signal-direction rule* is the constraint, on the crossing-native monitor M_C^θ of Definition 4.1 with possibly piecewise-constant θ, that the type of every emitted event is fixed by the sign of the discrete derivative of Q at the firing time: an event at position n is a

$$\mathsf{validate} \iff \operatorname{sgn}\big(Q(n)-Q(n-1)\big)=+1, \qquad \mathsf{invalidate} \iff \operatorname{sgn}\big(Q(n)-Q(n-1)\big)=-1.$$

A flip of $\operatorname{sgn}(Q-\theta)$ at n for which $\operatorname{sgn}(Q(n)-Q(n-1))=0$ (i.e. $Q(n)=Q(n-1)$, a plateau sample) fires no event.

Remark 4.3 (Strengthening relative to Definition 4.1)**.** For *static* θ the firing predicates of Definition 4.1 use the weak inequality $\tau_n \geq \theta$ on the upward transit; a configuration $\tau_{n-1} < \theta$, $\tau_n = \theta$, $\tau_{n+1} = \theta$ satisfies the validate predicate even though $Q(n) = Q(n-1)$ is not yet defined relative to the rule. Definition 4.2 *strengthens* Definition 4.1 on this plateau-at-θ configuration by adding the explicit non-zero discrete-derivative requirement. For signals that never touch θ tangentially, the two definitions agree; for the threshold-motion regime of Appendix A (where the wall is permitted to land *on* the signal at a tick), the signal-direction rule is the operative semantics. All results downstream of Appendix A adopt Definition 4.2; Section 4 (this section) presents the static definition for didactic clarity and continuity with the classical Schmitt-trigger literature.

The crossing modality has been the standard semantics of threshold-based event detection in signal processing since at least the introduction of the Schmitt trigger in [14]. It underlies edge-triggered sampling, the atomic propositions of Signal Temporal Logic [10], and the boolean signals used in runtime verification of cyber-physical systems [26, Sec. 2.1].

Panel (a) of Figure 1 (below) illustrates the modality on a sinusoidal signal with three above-θ excursions, each contributing one validate/invalidate pair. The signal is chosen to be a quarter-period shift of the touch-modality sinusoid in panel (b), so that the upward transits of the crossing threshold sit at the same x-coordinates as the upper-wall touches of the touch modality, and the downward transits sit at the same x-coordinates as the lower-wall touches. This alignment lets the reader compare the two modalities event by event on a common time axis.

5. The touch modality

Definition 5.1 (Validation and invalidation predicates)**.** Let $\theta \in V \cup \{-\infty\}$. The *validation predicate* $\mathsf{validate}(\theta)$ holds at position n iff

$$\tau_{n-1} < \tau_n \ \wedge\ \tau_{n+1} < \tau_n \ \wedge\ \tau_n \geq \theta,$$

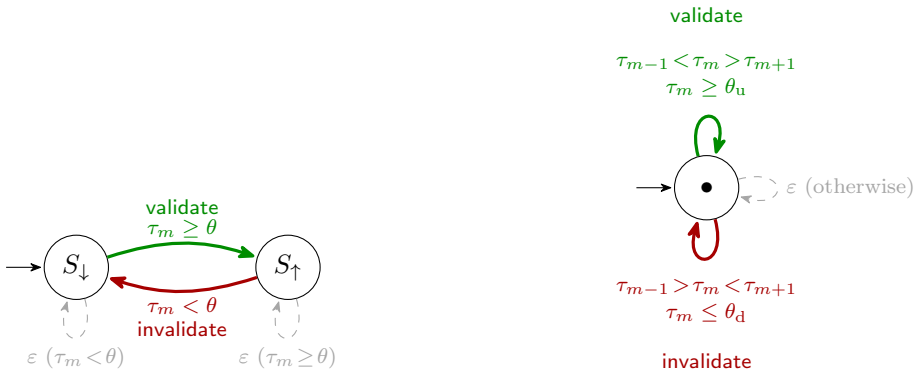
i.e. τ_n is a strict local maximum at value $\geq \theta$ (a tangency to the upper wall, in the band view of panel (b) of Figure 1). Symmetrically, the *invalidation predicate* $\mathsf{invalidate}(\theta)$ holds at position n iff

$$\tau_{n-1} > \tau_n \ \wedge\ \tau_{n+1} > \tau_n \ \wedge\ \tau_n \leq \theta,$$

i.e. τ_n is a strict local minimum at value $\leq \theta$ (a tangency to the lower wall).

A $\mathsf{validate}(\theta)$ event marks a strict local maximum with value at least θ (a "wall touched from below"); a $\mathsf{invalidate}(\theta)$ event marks a strict local minimum with value at most θ. This rule reads a three-sample window and stores no state; it is drawn as a stateless pattern detector in Figure 3b.

Remark 5.2 (Boundary and ties under Strict)**.** The two strict inequalities $\tau_{n-1} < \tau_n$ and $\tau_n > \tau_{n+1}$ have two structural consequences that must be acknowledged: *(boundary non-firing)* the trace endpoints $n = 0$ and $n = T-1$ never fire under Strict because one of the two neighbours is undefined; a maximum at the trace boundary is structurally silent. *(tied peaks)* two consecutive equal samples at the peak of an excursion (e.g. $\tau = (\ldots, a, b, b, a, \ldots)$ with $b > a$) fail *both* inequalities at the two top positions and fire no



(A) Crossing: minimal 2-state latch, $\kappa = (1, 1, 0)$.

(B) Touch: minimal stateless 3-window detector, $\kappa = (3, 0, 1)$.

FIGURE 3. The two *incomparable minimal corners* of the capability duality. **(a)** The CROSSING latch is *causal* and *state-based*: a single threshold θ splits the sample τ_m into two value-classes latched in the sticky states $S_\downarrow/S_\uparrow$ ($\kappa = (1, 1, 0)$), firing green validate on the upward transit and red invalidate on the downward transit, the self-loops emitting ε. **(b)** The TOUCH detector is *anti-causal* (lag 1) and *stateless, window-based*: one control state • ($\kappa = (3, 0, 1)$) decides purely by the shape of the three-sample window $(\tau_{m-1}, \tau_m, \tau_{m+1})$, the future sample τ_{m+1} supplying the history. Both operate at the focus rung with verdict alphabet $\mathbb{V}_3 = \{-1, 0, +1\}$ ($|\mathbb{V}| = 3$) and emission alphabet $\{\varepsilon, \mathsf{validate}, \mathsf{invalidate}\}$ ($|\Sigma| = 3$). These two corners — and their distance — are the basis of the capability duality and its join. Their fuller, more expressive *value-on-state* (Moore) automata — which render the verdict value-classes as explicit states — are Figure 31 (crossing) and Figure 30a (touch).

event, even though a peak is unambiguously present. Both consequences are dissolved by the non-strict conventions of Definition E.4: the WEAK/EDGE/DIRRES conventions fire at least one event per plateau-on-wall, and the boundary omission is recovered by treating the trace endpoints as plateau edges. The tied-peaks case is distinct from the plateau-on-wall case treated in Appendix E: plateau-on-wall requires equality with θ_u (a value-coincidence event), whereas tied peaks are a pure signal-shape phenomenon that does not require any wall contact.

Definition 5.3 (Touch-native monitor)**.** The *touch-native monitor* with barrier θ is the stateless monitor M_T^θ that emits at position n:

$$M_T^\theta(\tau_n) = \begin{cases} \mathsf{validate} & \text{if } \mathsf{validate}(\theta) \text{ holds at } n, \\ \mathsf{invalidate} & \text{if } \mathsf{invalidate}(\theta) \text{ holds at } n, \\ \varepsilon & \text{otherwise.} \end{cases}$$

The capability set is again minimal but *different from crossing*: 1 barrier, no persistent state, 3 steps of memory (positions $n-1$, n, $n+1$). The monitor is *anti-causal*: deciding whether τ_n is a local maximum requires knowing τ_{n+1}, introducing a one-step online lag.

The touch modality is the natural semantics of extremum-based event detection: derivative- and peak-based timing recovery is one of the classical method classes reviewed by Mueller and Müller [13], and extremum-seeking control [7] steers a plant towards the maximum of an unknown equilibrium map.

Panel (b) of Figure 1 illustrates the touch modality on a sinusoidal signal whose envelope is bounded by the two walls. The walls play the role of *repelling barriers* for the trajectory: the signal approaches them at its extrema, touches them tangentially, and reverses direction without ever crossing. Each tangency to θ_{u} fires validate; each tangency to θ_{d} fires invalidate. In contrast to crossing, the events are decided locally from a 3-window pattern with no persistent state, and no alternation is enforced: a sequence of touches at θ_{u} can occur with arbitrarily many touches at θ_{d} interleaved — or none — without affecting the firing rule.

Hysteretic touch (Schmitt). Adding one bit of *alternation* state to this stateless rule — so that a validate touch cannot recur until an intervening invalidate — gives the SCHMITT encoder, the *hysteretic touch*: the same wall extrema, now latched, with $\kappa = (3, 1, 1) =$ touch $(3, 0, 1)$ plus a one-bit latch. At the alternation schedule it *is* the augmented monitor of Section 10, where we develop it as the join of touch and crossing (its monitor-faithful automaton is Figure 9); here we note only that SCHMITT belongs to the touch family — one alternation bit above the stateless detector.

Convention foreshadowing. The 3-window firing rule above is the STRICT convention: touch fires only on a *strict* local extremum. This is the simplest of five legitimate conventions for the touch predicate, distinguished by their behaviour on plateaus (intervals on which $Q \equiv \theta_{\mathrm{u}}$ or θ_{d}):

Convention	**Fires on a plateau at the wall**
STRICT	never (no strict extremum on a flat segment)
WEAK	at every plateau sample
SAMPLE(f_s)	at $\lfloor f_s L \rfloor$ samples for plateau length L
EDGE	once, at the plateau's first or last sample
DIRRES	at entry or exit, resolved by direction of approach

Until Appendix E all results, figures, and count statements in this paper adopt STRICT; the no-plateau hypotheses that appear in Proposition 6.1 and downstream statements are STRICT-induced and disappear under the alternative conventions tabulated above. Appendix E catalogues the conventions formally (the *convention algebra*, with $|\mathcal{C}_B| = 72$ distinct combinations across the two walls), characterises their relative event counts, and discusses when each is operationally appropriate.

6. The count inequality

Having fixed both modalities and their conventions, we first compare them *quantitatively*: on any trace the number of crossing events is bounded by the number of touch events. The underlying fact is classical real analysis — between two consecutive level crossings of a continuous function lies at least one local extremum, the discrete level-crossing counterpart of Rolle's theorem; our contribution is its use as a *monitor-separation* statement, counting the events that two resource-incomparable monitors emit on the same signal.

Proposition 6.1 (Pointwise count inequality)**.** *Let τ be a value trace and θ a threshold; assume (i) τ has no* plateau *on any above-θ excursion (i.e. no two consecutive equal samples within an interval where $\tau \geq \theta$), and (ii) τ returns below θ after its final above-θ excursion (the trace does not end inside an above-θ excursion). Then*

$$N_X(\tau, \theta) \;\leq\; N_T(\tau, \theta),$$

where N_X and N_T are the validate *counts of M_C^θ and M_T^θ respectively. Equality holds iff between every pair of consecutive strict local maxima of τ at value $\geq \theta$, the trace descends below θ. The plateau-on-excursion assumption (i) is discharged in Corollary 6.2 below by passing to any non-strict plateau-touch convention of Appendix E; the trailing-boundary assumption (ii) is likewise discharged there by the explicit $\mathbf{1}[\tau$ ends inside an excursion$]$ term, which gives the unconditional form. Hypothesis (ii) is needed because a final-sample maximum is structurally strict-touch-silent (Remark 5.2).*

Proof. Under the no-plateau hypothesis, each upward transit $\tau_{n-1} < \theta \leq \tau_n$ is the left endpoint of a maximal interval on which $\tau \geq \theta$. By hypothesis (ii) every such excursion is *closed* — it has a right endpoint at which τ descends below θ — so the excursion's supremum is attained at an interior position, both of whose neighbours lie strictly below it. Since the trace is integer-indexed, finite-supported on each excursion, and has no two consecutive equal samples, this supremum is achieved at a unique position m that is a *strict* local maximum with value $\geq \theta$ in the sense of Definition 5.1 ($\tau_{m-1} < \tau_m$ and $\tau_{m+1} < \tau_m$, both inequalities strict). (The excluded case — a trace ending inside an excursion — can place the supremum at the final sample $m = T - 1$, whose right neighbour is undefined; such a boundary maximum is strict-touch-silent by Remark 5.2, which is exactly the $\mathbf{1}[\cdot]$ term of Corollary 6.2.) The map sending each upward transit to this strict local maximum is therefore well-defined and injective, hence $N_X \leq N_T$.

For equality, observe that distinct above-θ excursions contribute disjoint sets of local maxima. Equality $N_X = N_T$ therefore requires each excursion to contain exactly one local maximum — equivalently, the trace must descend below θ between any two consecutive local maxima at value $\geq \theta$. □

Corollary 6.2 (Plateau-robust count inequality)**.** *For* any *value trace τ and threshold θ, and for any non-strict plateau-touch convention $C \in \{\textsc{Weak}, \textsc{Sample}, \textsc{Edge}, \textsc{DirRes}\}$ of Definition E.4,*

$$N_X(\tau, \theta) \;\leq\; N_T^C(\tau, \theta) \;+\; \mathbf{1}[\tau \text{ ends inside an excursion}],$$

where N_T^C is the touch-event count under convention C. The strict convention $C = \textsc{Strict}$ requires the no-plateau hypothesis of Proposition 6.1; under any of the four non-strict conventions a plateau on an above-θ excursion contributes at least one touch event, restoring the injection from upward transits to touch events on that excursion (with the boundary $+1$ absorbing a final excursion that has not yet returned below θ).

Proof sketch. Each above-θ excursion is either plateau-free (handled by Proposition 6.1), or contains at least one plateau. A non-strict convention fires at least one touch event per plateau-on-excursion: Weak fires at every plateau position, Sample fires at the sampled position, Edge fires at both plateau edges, DirRes fires at the entry edge. Either way, the excursion contributes ≥ 1 touch event, so the injection from upward transits is restored. The boundary $+1$ accounts for a trace whose final excursion has not yet descended back below θ

at the end of the sampling window: the upward transit is counted, the corresponding touch may not yet have fired. □

A signal that stays above θ for an extended interval while exhibiting multiple peaks satisfies $N_X = 1$ (one crossing for the whole excursion) but may have $N_T \gg 1$ (one touch per peak). This is the signature numerical divergence that prevents the two modalities from being relabellings of each other.

7. Asymmetries between A and B

The count inequality is a coarse summary of a deeper asymmetry: each modality observes aspects of the signal that the other *cannot express at all*, even with arbitrary tuning of its threshold. Before showing in Sections 9.1 and 9.2 how each modality can simulate the other by augmentation, we make explicit what each gains and what each loses with respect to the other.

7.1. **What crossing captures and touch does not.**

Dwell-time semantics. Each validate event of M_C^θ marks the *onset* of an above-θ excursion, regardless of how the signal behaves inside. The count N_X is therefore the number of distinct excursions: a measure of how many times the signal has *transitioned* into the above-θ region. Touch has no native notion of excursion — it counts peaks, not transitions. Recovering a dwell-time interpretation from a touch stream requires an ad-hoc grouping rule that the modality itself does not supply.

Causal decisions at zero lag. The crossing monitor decides at the transit instant itself: its canonical native form reads only the present sample plus one bit of sticky state (window 1, the equivalent stateless 2-window realisation $\mathcal{A}_3^C$ trades that sticky bit for a $+1$ window by the Myhill–Nerode correspondence of Definition 2.4). The touch monitor cannot certify a local maximum until the *next* sample arrives — an unavoidable one-step anti-causal lag. For decision systems where the event must be acted upon immediately (trip-circuit relays, real-time alarms, transient-driven control), crossing has an irreducible timing advantage.

Persistent state as a fact about the present. The sticky state $s \in \{0, 1\}$ of M_C^θ records a queryable fact about the trace at any time: *the signal is currently above θ*. Touch carries no such state; after a peak event, no information remains about whether the signal is currently above or below any reference level. Properties that condition on the present level ("while the signal is high…") are crossing-expressible and not touch-expressible.

7.2. **What touch captures and crossing does not.**

Multi-peak resolution within an excursion. A signal that oscillates above θ produces $N_X = 1$ but $N_T = k \geq 1$, where k counts the local maxima inside the excursion. Crossing collapses this oscillation structure; touch resolves it. No choice of θ for M_C recovers the multi-peak count of M_T on such a signal — the information is irreversibly lost at the moment of θ-thresholding.

Envelope and amplitude information. Under the two-wall view of panel (b) of Figure 1, the parameters $\theta_{\mathrm{u}}, \theta_{\mathrm{d}}$ encode the signal's envelope: the peak amplitude and the trough depth. Crossing has a single threshold and is blind to amplitude beyond "above or below". The two-wall touch monitor carries strictly more information about the signal's range than any single-threshold crossing monitor.

Topology-awareness (shape). Touch events are determined by the sign of the second discrete difference: a maximum is a curvature-defined feature, invariant under uniform vertical translation of the signal. Crossing events are tied to an extrinsic reference level θ and shift if the signal's baseline drifts. For tasks where shape matters more than absolute level (vibration analysis, frequency tracking, asynchronous vision), touch is the natural modality.

No alternation cost. Touch events fire independently: arbitrarily many validate in succession require no intervening invalidate, and vice versa. Crossing's sticky state forces strict alternation. Properties that demand "three peaks in a row without a valley between" are touch-expressible and not crossing-expressible.

7.3. **Domains of natural application.** The asymmetric powers explain why the two modalities have populated different engineering and verification communities. Crossing dominates wherever *level-threshold semantics* is the primary concern: Signal Temporal Logic atomic propositions, Schmitt-trigger circuits, supervisory alarm systems, and the runtime-verification framework of cyber-physical systems. Touch dominates wherever *shape and oscillation* is primary: extremum-seeking control, frequency- and vibration-monitoring instruments, and the underlying machinery of Mueller–Müller and Gardner timing-error detectors.

Neither modality is universally preferable. Choosing between them is a substantive epistemic decision about *which property of the signal is to be observed* — a decision that the augmented monitor of Section 10 makes explicit as a choice of parameter point.

8. Anatomy and tuning taxonomy

The asymmetries of the previous section show that the two modalities are not freely interchangeable; how far a monitor can be reshaped — within a modality and toward the other — is governed by its tuning operations, which we now catalogue. The remainder of the paper is a guided tour of the tuning operations that the augmented monitor admits. Before walking the catalogue, it is useful to fix two pieces of shared infrastructure: the *anatomy* of the input signal (which gives the parameter space its concrete coordinates) and the *taxonomy* of tuning operations themselves (which determines the order in which the sections that follow are organised). The anatomy is presented at the start of Appendix B together with the band-boundary tuning that exercises one of its knobs; the taxonomy is given here.

The tuning operations available to the augmented monitor fall into three layers, each one strictly more expressive than the previous in the sense that any event-stream reachable by a layer is also reachable by the layers above it, but not conversely.

Layer 1 — Monitor-internal tunings.:

These operations leave the input signal Q *exactly unchanged* and act only on the monitor's parameter family: the thresholds and walls $\theta, \theta_{\mathrm{u}}, \theta_{\mathrm{d}}$, the gates G, the band boundaries $(b_{\mathrm{PX}}, b_{\mathrm{XB}}, b_{\mathrm{BP}})$, the tick widths w_i, and the schedule π. Layer 1 splits naturally

TABLE 2. Coverage matrix of the dual-modality landscape. Model A (crossing) and Model B (touch) coincide in the CLASP cell of their event stream (bottom two rows) but diverge structurally in every other aspect (rows above). The augmented monitor is the join of the two algebras over the shared event alphabet $V = \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ of Definition 10.2. The table makes explicit that the touch/crossing distinction is a non-CLASP one — it is visible in the algebra A, not in the verdict alphabet V.

Aspect	**Model A (crossing)**	**Model B (touch)**	**Augmented (join)**
Native primitive	level crossing	local extremum touch	both
Wall parameter	$\theta \in \mathbb{R}$	$(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}) \in \mathbb{R}^2$	$(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$
Direction–event coupling	tight (sign of $Q - \theta$)	decoupled (extremum kind)	decoupled
Count inequality	$N_X \leq N_T$ (Proposition 6.1)	—	$N_{\mathrm{aug}} \geq \max(N_X, N_T)$
Robustness margin	ρ_A linear in $\lvert Q - \theta \rvert$	ρ_B quadratic near extrema	both available
Plateau conventions	trivial (no plateau in crossing)	five conventions C_1–C_5 (Appendix E)	inherits from B
Convention algebra cardinality	$\lvert\mathcal{C}_A\rvert = 12$	$\lvert\mathcal{C}_B\rvert = 72$ (Definition E.4)	inherits
Layer-1 CLASP cell of event stream	Cosafety	Cosafety	Cosafety (Proposition 13.10)
CLASP cell under ALT	Mixed	Mixed	Mixed (Proposition 13.11)

into two halves: *threshold/wall motion* (Appendix A — interruption, movement-away, movement-toward) which reshapes *when* events fire on the signal as-is; and *semantic-aggregation tunings* (Appendix B — band-boundary tuning and tick-width tuning) which reshape *how* the events that do fire are organised over the time axis.

Layer 2 — Tick-level signal copy-paste.:
These operations rearrange *observed segments* of Q by copying the content of one tick window into another, overwriting the destination. No new sample value is fabricated — every value present in the output is a value already present in the canonical input — but the order in which the monitor sees the segments is no longer canonical. Layer 2 is treated in Appendix C and admits three sub-regimes (general, continuity-preserving, direction-coherent) that progressively relax the cosmetic constraints on the splice.

Layer 3 — Signal modification.:
These operations *synthesise* Q, replacing its content with values never observed in the canonical record. Layer 3 is treated in Appendix D and combines naturally with the threshold-segment movement of Layer 1 to give an arbitrary-precision event-placement instrument.

A separate concern, orthogonal to the three layers, is the *convention* under which touch events on plateaus and walls are detected and labelled. This concern is treated in Appendix E as a standalone section, because the available conventions interact with all three layers (they

refine the detection rule that Layers 1–3 manipulate) and the choice between them is itself a tuning parameter.

Proposition 8.1 (Strict expressive hierarchy of tuning layers)**.** *Fix a canonical signal Q and write $\mathcal{L}_k(Q)$ for the set of event streams (over $\{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}^{\mathbb{N}}$) realisable on Q by the augmented monitor of Definition 10.2 under the operations of Layer k, $k \in \{1, 2, 3\}$. Then*

$$\mathcal{L}_1(Q) \subsetneq \mathcal{L}_2(Q) \subsetneq \mathcal{L}_3(Q),$$

with the containment chain strict for every non-constant Q.

Proof sketch. $\mathcal{L}_1 \subseteq \mathcal{L}_2$ is immediate because every Layer 1 operation can be re-expressed as a trivial (identity) copy-paste of Q followed by the same Layer 1 manipulation. $\mathcal{L}_2 \subseteq \mathcal{L}_3$ is immediate because every copy-paste of Q is a special case of free signal modification (one that uses only values already present in Q).

Strictness $\mathcal{L}_1 \subsetneq \mathcal{L}_2$. On the canonical sinusoid of Equation (B.1), the copy-paste operation of Figure 25 substitutes the maximum tick 3 for the minimum tick 5 and produces the event stream containing a $\mathsf{validate}$ at $t = 5$ in Model B; no Layer 1 operation (threshold/wall motion alone) can produce a $\mathsf{validate}$ at $t = 5$ on the canonical Q under the strict-extremum convention, because Q has no strict local maximum near $t = 5$ and Layer 1 cannot create one without modifying Q (Appendix E).

Strictness $\mathcal{L}_2 \subsetneq \mathcal{L}_3$. On any signal whose value range is strictly bounded — say $Q(\mathbb{N}) \subseteq [\theta_{\mathrm{d}}, \theta_{\mathrm{u}}]$ — Layer 2 copy-paste cannot produce any output sample outside that range, since every output value is also an input value. An event stream that requires a $\mathsf{validate}$ touch at the wall $\theta'_{\mathrm{u}} > \max Q$ is therefore unrealisable under Layer 2 but realisable under Layer 3 by setting $Q'(t) := \theta'_{\mathrm{u}}$ at the target time. Layer 3 thus strictly contains Layer 2 on any signal that fails to attain the wall. □

The concrete tuning operations that instantiate this taxonomy — threshold and wall motion, semantic-aggregation tunings, tick-level signal copy-paste, input-signal modification, and the touch convention algebra — are developed in Appendices A to E.

9. Dual augmentations between touch and crossing

The count inequality of Section 6 and the asymmetries of Section 7 show that the touch and crossing monitors are not freely interchangeable: each lacks capabilities the other has natively. Where the tuning taxonomy of Section 8 measured how far a single monitor can be reshaped *within* its own machinery ($\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$), this section instead *adds* machinery: it makes the asymmetry constructive in both directions, and we exhibit the minimal additional machinery that promotes M_T to crossing behaviour and the symmetric machinery that promotes M_C to touch behaviour. The two constructions are formally dual (Section 10 will identify them as the two ascents of a capability join).

9.1. **From touch to crossing.** The procedure adds genuinely new structure to M_T: a one-bit hysteretic state and a finite-state alternation rule. Without this addition the touch monitor cannot enforce the alternation requirement of crossing, which is the defining feature of the modality.

Definition 9.1 (Touch-with-alternation monitor)**.** Let $\theta \in V$. The *touch-with-alternation monitor* $M^{\theta}_{T\to C}$ is the monitor that wraps two copies of the touch detector, $\mathsf{validate}(\theta)$ and $\mathsf{invalidate}(\theta)$, with a one-bit state $s \in \{S_{\text{up}}, S_{\text{down}}\}$ initialised to S_{up} — an alternation *phase* (S_{up} awaits the next validate, S_{down} the next invalidate), not to be confused with the below/above sticky states of the crossing latch (Definition 4.1, drawn $S_{\downarrow}/S_{\uparrow}$) — and rule:

- if $s = S_{\text{up}}$ and $\mathsf{validate}(\theta)$ holds at n: emit validate, set $s := S_{\text{down}}$;
- if $s = S_{\text{down}}$ and $\mathsf{invalidate}(\theta)$ holds at n: emit invalidate, set $s := S_{\text{up}}$;
- otherwise: emit ε.

Proposition 9.2 (Touch-with-alternation simulates crossing)**.** *For every value trace τ and every threshold θ, the event sequence emitted by $M^{\theta}_{T\to C}$ coincides with that emitted by M^{θ}_{C} in* type and order*: exactly one* validate *per above-θ excursion and one* invalidate *per below-θ excursion, in the same alternating sequence and with the same counts. The two monitors differ in event* timing*. M^{θ}_{C} timestamps each event at the threshold transit (when τ first reaches θ, per Definition 4.1), whereas $M^{\theta}_{T\to C}$ timestamps it at the excursion extremum — the first strict maximum $\geq \theta$ for* validate*, resp. the first strict minimum $\leq \theta$ for* invalidate *— which lags the transit by the excursion's rise (resp. fall) time and is certified with the touch detector's one-step anti-causal decision latency. The simulation is therefore exact at the level of the event sequence (types, order, and counts), not tick-by-tick. The count match assumes each above-θ excursion contains an* interior *strict maximum (and dually each below-θ excursion a strict minimum) — the no-plateau-at-peak hypothesis (i) of Proposition 6.1, the same assumption its dual Proposition 9.4 carries explicitly: a flat-topped excursion fires no strict touch under Definition 5.1 while M^{θ}_{C} still fires once at the up-transit, so it is excluded on this ground (or restored by a non-strict convention, Corollary 6.2). Separately, a final excursion truncated by the trace end — its extremum on the last sample, where the strict touch test is undefined — may cost $M^{\theta}_{T\to C}$ one event that M^{θ}_{C} fires causally at the transit, the same trailing-boundary correction as Corollary 6.2.*

Proof sketch. Each above-θ excursion of τ contains at least one strict local maximum $\geq \theta$; the first such maximum, detected by $\mathsf{validate}(\theta)$, fires validate and flips s to S_{down}. Subsequent local maxima within the same excursion are gated out by the state. When the excursion ends, the trace contains at least one strict local minimum $\leq \theta$ (the first descent into the below-θ region); this fires invalidate and flips s back. The resulting validate/invalidate events match those of M^{θ}_{C} in type and order; each is timestamped at the excursion extremum (certified with the $+1$ anti-causal decision latency), displaced from M^{θ}_{C}'s threshold transit by the excursion's rise (resp. fall) time. □

Figure 4 illustrates the construction.

The transformation strictly increases the capability set: M^{θ}_{T} has *no* persistent state and *no* finite-state rule, both required in $M^{\theta}_{T\to C}$. The added complexity is not free.

9.2. **From crossing to touch.** The symmetric construction starts from the crossing-native monitor M^{θ}_{C} and adds a *gating* mechanism that disables the alternation requirement and re-enables the threshold selectively. The added structure is again non-trivial: the gate is itself a small monitor on the signal's rises — the consecutive-sample comparison $\tau_{n-1} < \tau_n$ that the touch extremum test of Definition 5.1 is itself built from — maintaining its own bounded history.

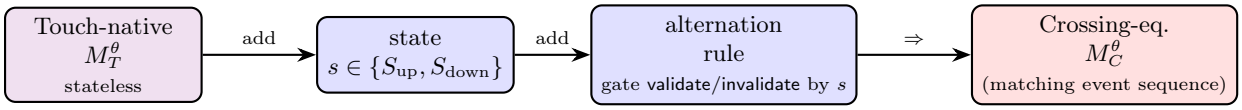


FIGURE 4. From touch to crossing: starting from the stateless touch-native monitor M_T^θ, two augmentations (a one-bit hysteretic state plus an alternation rule that gates the touch detectors by the state) produce a monitor whose event sequence coincides with that of the crossing-native monitor M_C^θ in type and order (each event timestamped at the excursion extremum, not the threshold transit). Neither augmentation is present in M_T^θ.

Definition 9.3 (Gated crossing monitor)**.** A *gate function* is a stream $G : \mathbb{N} \to \{0, 1\}$ synchronous with τ. The *gated crossing monitor* $M_{C\to T}^{\theta,G}$ behaves like M_C^θ except that at each rising edge of G (i.e. each n with $G(n-1) = 0$ and $G(n) = 1$), the state s is reset to 0 and a validate event is emitted if and only if $\tau_n \geq \theta$ at that position.

Proposition 9.4 (Gated crossing simulates touch; symmetric companion to Proposition 9.2)**.** *Let $G^* : \mathbb{N} \to \{0, 1\}$ be the indicator stream $G^*(n) := \mathbb{1}[\tau_{n-1} < \tau_n]$ (the rising edge of the value trace itself). Then for every value trace τ and every threshold θ, the* validate *events emitted by $M_{C\to T}^{\theta,G^*}$ coincide with those of M_T^θ in* type, order, and count*: exactly one* validate *per above-θ ascent, which equals N_T, the number of local maxima above θ, in the same order. (As in Proposition 6.1 we assume no plateau* within *an ascent (the rising run of an above-θ excursion): a flat step $\tau_{n-1} = \tau_n$ on a rising run splits it into two rising edges of G^* and would over-count by one; with strict ascents the rising-edge count equals N_T.) The two monitors differ in event* timing*. $M_{C\to T}^{\theta,G^*}$ timestamps each* validate *at the threshold* up-crossing *(when τ first reaches θ within the ascent), whereas M_T^θ timestamps it at the excursion* peak *(the strict local maximum); the up-crossing* leads *the peak by the ascent's rise time. The simulation is therefore exact at the level of the event sequence (type, order, and count), not tick-by-tick. This is the mirror image of Proposition 9.2: there the touch-built monitor fires at the extremum and* lags *the transit; here the crossing-built monitor fires at the transit and* leads *the extremum.*

Proof sketch. G^* is 0 on every descent and 1 on every ascent of τ. A rising edge of G^* — a transition $0 \to 1$ — occurs precisely at each local minimum: the position immediately after which the trace begins to rise again. At each such transition the gate resets s to 0. The next validate fires at the *up-crossing* of θ — the first sample $\tau_n \geq \theta$ after the rising-edge reset — and not at the local maximum of the ascent. Hence each above-θ ascent contributes exactly one validate, so the total validate count is N_T, the number of local maxima above θ, in the same order. Timing displacement: the up-crossing leads the corresponding peak (where M_T^θ fires) by the ascent's rise time. □

The added structure here is the gate G^*, which is itself a small monitor (it computes $\mathbb{1}[\tau_{n-1} < \tau_n]$ from a 2-window of the value trace) plus a reset hook into the state s of M_C^θ.

Figure 5 illustrates the construction.

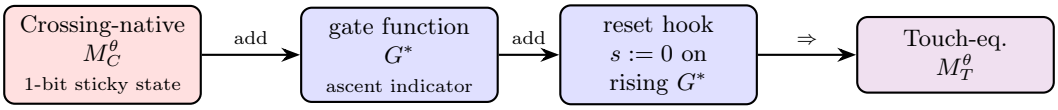


FIGURE 5. From crossing to touch: starting from the crossing-native monitor M_C^θ, two augmentations (a gate function G^* tracking the signal's rises (the consecutive-sample comparison $\tau_{n-1} < \tau_n$), plus a reset hook that clears the state s on each rising edge of G^*) produce a monitor whose validate events coincide with those of the touch-native monitor M_T^θ in type, order, and count (each validate timestamped at the threshold up-crossing, not the excursion peak). Neither augmentation is present in M_C^θ.

10. The augmented monitor as join

The two transformations of Sections 9.1 and 9.2 are symmetric in magnitude: each starts from one of the two minimal monitors and adds exactly one bit of structure to reach the behaviour of the other, but the *kind* of structure added differs by direction. The touch-to-crossing direction (Proposition 9.2) adds a one-bit hysteretic state; the crossing-to-touch direction (Proposition 9.4) adds a 2-window ascent gate. Neither direction is privileged — one bit of structure in each direction, state in one, window/lookahead in the other. The Crossing and Touch automata as the two constrained specialisations of this general automaton are drawn, in their direction-resolved (value-class) form, in Figure 6. We make the symmetry precise by introducing the *augmented monitor*. Its single new power is operational: the tuner may switch each threshold on or off from segment to segment — engaging a wall on the stretches where it wants that wall's events, or sliding it to the absorbing value $\pm\infty$ where it wants the wall silent — so one monitor runs as touch on one stretch and as crossing on another along a single signal. This segment-wise enabling and disabling of thresholds is what generalises the two modalities; that the resulting monitor is, *structurally*, the *join* of M_T and M_C in the capability lattice is the order-theoretic restatement of the same fact. Joining the two modalities is not, however, the same as solving every synthesis problem: threshold gating alone realises only the *separable-contact* sublanguage (Corollary 13.3), and producing the patterns the two modalities cannot jointly express on a fixed signal — on a constant segment, or two same-type events at adjacent ticks — requires *controlling the signal itself* (Appendix D), not merely the thresholds.

Before stating the augmented-monitor definition we extend the static Definition 5.1 predicates to *moving* threshold schedules and to the absorbing values $\pm\infty$. This generalisation is needed because Definition 10.2 below allows $\theta_{\mathrm{u}}, \theta_{\mathrm{d}} : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ to vary with the sample index.

Definition 10.1 (Moving validation and invalidation predicates)**.** Let $\theta : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ be a piecewise-constant threshold schedule. The *moving validation predicate* validate(θ) holds at position n iff

$$\tau_{n-1} < \tau_n \ \wedge\ \tau_{n+1} < \tau_n \ \wedge\ \tau_n \geq \theta(n),$$

i.e. a strict local maximum of τ whose value clears the threshold sample at the SAME position. Symmetrically, the *moving invalidation predicate* invalidate(θ) holds at position n iff

$$\tau_{n-1} > \tau_n \ \wedge\ \tau_{n+1} > \tau_n \ \wedge\ \tau_n \leq \theta(n).$$

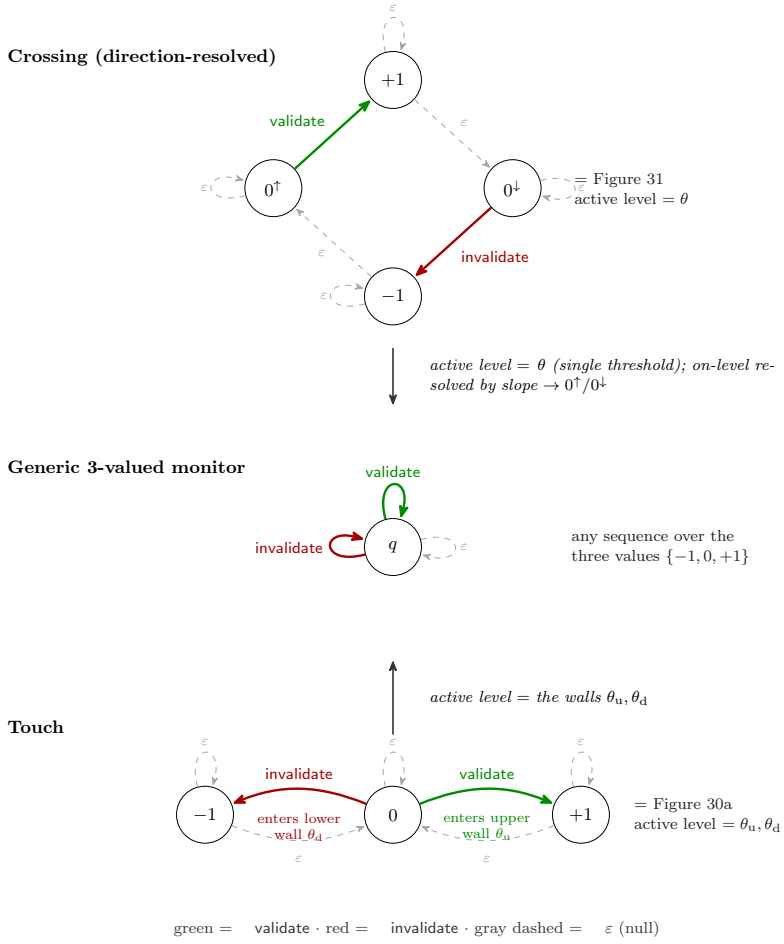


FIGURE 6. CROSSING and TOUCH are two direction-resolved refinements of a single *generic 3-valued monitor*. **Centre:** the generic monitor — one state that emits any sequence over the three verdict values $\{-1, 0, +1\}$ (*validate* $= +1$, *invalidate* $= -1$, $\varepsilon = 0$), with no ordering constraint. Each modality (arrows) is this monitor read with the *active level* fixed, which restricts the verdict sequences and ties each verdict to the signal. **Top:** reading the level as the single threshold θ gives the direction-resolved CROSSING (Figure 31), whose on-level state is split by slope into $0^{\uparrow}/0^{\downarrow}$ (strict *validate*/*invalidate* alternation). **Bottom:** reading the level as the touch walls $\theta_{\mathrm{u}}, \theta_{\mathrm{d}}$ gives TOUCH (Figure 30a). The augmented monitor that joins both modalities — the alternation latch $\{S_{\mathrm{up}}, S_{\mathrm{down}}\}$, $\kappa = (3, 1, 1)$ — is the SCHMITT-internal automaton of Figure 9; the explicit plateau-faithful direction-resolved ternary refining this generic monitor is Figure 11 ($\kappa = (1, 2, 0)$, Section 10). This generic monitor is the *ceiling* of the tuning hierarchy (Proposition 8.1): moving-wall tuning alone reaches the separable-contact sublanguage (Corollary 13.3), and the full free language is attained only when the tuner may also modify the signal (Appendix D).

The absorbing values are interpreted physically as *unreachable walls*: a real-valued signal sampled at finite ticks cannot touch a wall placed at $\pm\infty$. Concretely, BOTH $\theta(n) = +\infty$ AND $\theta(n) = -\infty$ disable $\mathsf{validate}(\theta)$ at position n:

- $\theta(n) = +\infty$: the wall is above every real signal value; the value-clearance test $\tau_n \geq +\infty$ is impossible.
- $\theta(n) = -\infty$: the wall is below every real signal value; while $\tau_n \geq -\infty$ is formally true, the touch semantics requires the signal to physically reach the wall, which never happens in finite ticks for a finite-valued signal. No event is fired.

Symmetric interpretations hold for $\mathsf{invalidate}(\theta)$: both $\theta(n) = +\infty$ and $\theta(n) = -\infty$ disable the predicate, on the same physical-reachability grounds.

The single exception to this disabling convention is the *wall-motion* regime of Appendix A: when the threshold schedule θ is itself moving and the moving wall crosses the signal between positions $n-1$ and n, an event may be fired with type determined by the direction of the relative motion at the crossing point. This is the wall-caused-crossing semantics; the general case treated by the present definition is the static-tick-by-tick semantics under which $\pm\infty$ disables the predicate.

Scope note: absorbing endpoints. The absorbing values $\pm\infty$ are excluded from the firing rules: they are formally admissible parameter values of the value range $V \cup \{-\infty, +\infty\}$ that the rules above explicitly set aside (STRICT on V + disable on $\pm\infty$). The value range thus splits into the active region V and the disabled absorbing endpoints $\pm\infty$, with firing rules defined only on the active region. Downstream uses (Conjecture 10.12 and Theorem 10.14) rely on this convention to keep the parameter family $(V \cup \{\pm\infty\})^2 \times \Pi$ closed under monitor embeddings; the boundary between the active region and the disabled endpoints is the locus of those embeddings.

When θ is constant the moving predicates reduce to the static predicates of Definition 5.1. The sample $\theta(n)$ used in the value-clearance test is the *current* sample at position n; jumps in θ between positions $n-1$ and n do not retroactively alter the predicate evaluated at $n-1$.

Definition 10.2 (Augmented monitor)**.** An *augmented monitor* is a tuple $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$, where $\theta_{\mathrm{u}}, \theta_{\mathrm{d}} : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ are two threshold *schedules* (piecewise-constant functions of the sample index, with finitely many transitions on any finite prefix) and $\pi \in \{\text{ALT}, \text{FREE}\}$ is the alternation schedule. When $\theta_{\mathrm{u}}, \theta_{\mathrm{d}}$ are constant on all of $\mathbb{N}$ we recover the static case of Section 7; when they admit transitions, the monitor realises the threshold/wall motion of Appendix A. We write $\theta_{\mathrm{u}}(n) = +\infty$ for "upper threshold disabled at sample n" (the disabled-gate semantics of Figure 20(a)) and $\theta_{\mathrm{d}}(n) = -\infty$ for the symmetric disabling of the lower threshold. The monitor maintains a state $s \in \{S_{\mathrm{up}}, S_{\mathrm{down}}\}$ initialised to S_{up} and emits, at each position n:

- **Schedule ALT.** If $s = S_{\mathrm{up}}$ and $\mathsf{validate}(\theta_{\mathrm{u}})$ holds, emit $\mathsf{validate}$ and set $s := S_{\mathrm{down}}$. If $s = S_{\mathrm{down}}$ and $\mathsf{invalidate}(\theta_{\mathrm{d}})$ holds, emit $\mathsf{invalidate}$ and set $s := S_{\mathrm{up}}$.
- **Schedule FREE.** The state s is ignored. Every $\mathsf{validate}(\theta_{\mathrm{u}})$ fires a $\mathsf{validate}$ event; symmetrically, every $\mathsf{invalidate}(\theta_{\mathrm{d}})$ fires an $\mathsf{invalidate}$ event.

Remark 10.3 (Coincident-threshold touch is mechanically a level comparator (zero-hysteresis Schmitt))**.** The augmented monitor of Definition 10.2 with $\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = \theta$ and $\pi = \text{ALT}$ is the *mechanical realisation* of the classical *level comparator without hysteresis* — equivalently, the $\Delta \to 0$ degenerate limit of the Schmitt-trigger of Remark 10.4. In the special case

$\theta = 0$ this is the textbook *zero-crossing detector* of signal processing (used, e.g., in PLLs, frequency counters, and zero-crossing demodulators).
Trigger conditions (taking $\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = \theta$, $\pi = \textsc{alt}$, initial state S_{up}):

- validate **at tick** n iff *(pattern)* $\tau_{n-1} < \tau_n > \tau_{n+1}$ (strict local max), *(value)* $\tau_n \geq \theta$, *(gate)* current state $= S_{\mathrm{up}}$.
- invalidate **at tick** n iff *(pattern)* $\tau_{n-1} > \tau_n < \tau_{n+1}$ (strict local min), *(value)* $\tau_n \leq \theta$, *(gate)* current state $= S_{\mathrm{down}}$.

Although the firing rule is internally a *touch* predicate (tangency at a local extremum), the externally observable event stream coincides exactly with the crossing-native monitor M_C^{θ} via the gate-threshold equivalence of Proposition 13.9: every strict local maximum with $\tau_n \geq \theta$ is preceded by an upward transit of θ, and symmetrically for minima. This is the Coincident parameter point of Proposition 10.5 below; see Figure 7(a) for the signal-trace visualisation.

Remark 10.4 (Two-threshold touch is mechanically a Schmitt-trigger)**.** The augmented monitor of Definition 10.2 with $\theta_{\mathrm{u}} > \theta_{\mathrm{d}}$ and $\pi = \textsc{alt}$ is the *mechanical realisation* of the classical Schmitt-trigger encoder. The two distinct touch barriers play the role of the two trigger levels of the electronic Schmitt-trigger, and the state machine $S_{\mathrm{up}} \leftrightarrow S_{\mathrm{down}}$ enforces the snap-action alternation: once S_{up} fires on a maximum tangent to or exceeding θ_{u}, the monitor will not fire again until a minimum tangent to or below θ_{d} is observed. Although the firing rule is internally a touch predicate (the *tangency* event at a local extremum) rather than a level transit, the externally observable event stream is exactly the classical Schmitt-trigger behaviour: one validate per excursion that reaches θ_{u}, one invalidate per excursion that descends to θ_{d}, with the band $(\theta_{\mathrm{d}}, \theta_{\mathrm{u}})$ acting as a hysteresis dead-zone. This is the Schmitt parameter point of Proposition 10.5 below; see Figure 7(b) for the signal-trace visualisation.

The two pictures of this encoder are therefore the same machine seen from opposite sides: the *external* four-state level-loop of Figure 8 ($\kappa = (1, 1, 0)$) and the *internal* two-state extremum-latch automaton of Figure 9 ($\kappa = (3, 1, 1)$) are two realisations of one encoder. The left rail $\{L, M^{\uparrow}\}$ of Figure 8 corresponds to the output-low (pre-validate) state S_{up} and the right rail $\{U, M^{\downarrow}\}$ to the output-high (post-validate) state S_{down}; the two realisations emit the same event sequence (one validate per up-excursion, one invalidate per down-excursion) and differ only in the firing *time* — the external view fires at the threshold crossing, the internal view at the strict extremum, displaced by the excursion's rise (resp. fall) time (Figure 9(b)).

Proposition 10.5 (Three classical encoders as parameter points)**.** *The three classical threshold-encoding families are specific parameter points of the augmented monitor. We name each point by its classical role (small-caps), used consistently throughout the paper:*

Coincident.**:** $\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = \theta$, $\pi = \textsc{alt}$ *— the edge-triggered encoder, validation count* N_X.

Schmitt.**:** $\theta_{\mathrm{u}} = \theta + \Delta$, $\theta_{\mathrm{d}} = \theta - \Delta$ *with hysteresis* $\Delta > 0$, $\pi = \textsc{alt}$ *— the Schmitt-trigger; equivalently the* hysteretic touch *of Section 5 (the stateless touch detector plus a one-bit alternation latch),* $\kappa = (3, 1, 1)$.

Extremum.**:** $\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = -\infty$, $\pi = \textsc{free}$ *— the extremum-based encoder, validation count* N_T.

Proof sketch. (Coincident) Under coincident thresholds and alt, the state transitions exactly once per above-θ excursion; the count is N_X (Proposition 9.2). (Schmitt) With

$\theta_u > \theta_d$ the two-threshold variant is the standard Schmitt-trigger hysteresis. (EXTREMUM) Under FREE with thresholds at $-\infty$, every local maximum fires validate; the count is N_T — the gated-crossing realisation of this touch count is Proposition 9.4, the symmetric companion to the COINCIDENT case above. □

Corollary 10.6 (Classical encoders as a touch-only face of the n=4 parameter cube)**.** *Each of the three classical encoders COINCIDENT, SCHMITT, EXTREMUM of Proposition 10.5 embeds as the parameter point of the quaternary augmented monitor of Definition 10.13 obtained by disabling the central crossing threshold $\theta_{\text{cross}} = +\infty$ and inheriting the same $(\theta_u, \theta_d, \pi)$ coordinates:*

COINCIDENT:: $M^{(4)}(\theta, +\infty, \theta, \textit{ALT})$;

SCHMITT:: $M^{(4)}(\theta + \Delta, +\infty, \theta - \Delta, \textit{ALT})$;

EXTREMUM:: $M^{(4)}(-\infty, +\infty, -\infty, \textit{FREE})$.

In each case the central crossing predicates $\mathsf{crosses}^{\uparrow}(+\infty), \mathsf{crosses}^{\downarrow}(+\infty)$ *are vacuous (per the absorbing-value convention of Definition 10.1 applied to θ_{cross}), so* $\mathsf{validate}_{\text{cross}}$ *and* $\mathsf{invalidate}_{\text{cross}}$ *never fire; the five-valued alphabet $\Sigma^{(4)}$ projects to the ternary Σ via Remark 10.16. The three encoders therefore inhabit the same* touch-only face *$\{\theta_{\text{cross}} \in \{-\infty, +\infty\}\}$ of the four-parameter cube $(\theta_u, \theta_{\text{cross}}, \theta_d, \pi)$.*

Proof. Direct from the proof of Proposition 10.5: with $\theta_{\text{cross}} = +\infty$ the central predicates are disabled, the quaternary monitor reduces to the ternary M of Definition 10.2 by Remark 10.16, and the prior proof applies pointwise to the projected event stream. □

Remark 10.7 (Two new notable points enabled at n=4 beyond the touch-only face)**.** The quaternary parameter cube admits at least two further notable points that are not expressible in the ternary augmented monitor of Definition 10.2:

- CROSSING.: $M^{(4)}(+\infty, \theta, -\infty, \text{ALT})$ — the crossing-native point recovers the crossing-native monitor M_C^{θ} exactly, with sticky-state alternation built into ALT (Theorem 10.14, clause (2)).
- TRIO.: $M^{(4)}(\theta_u, \theta_{\text{cross}}, \theta_d, \text{FREE})$ with all three thresholds finite and distinct — the trio-encoder emits in all four typed verdicts of $\Sigma^{(4)}$ simultaneously, as the disjoint union of M_T at the outer barriers and M_C at the central threshold (Theorem 10.14, "Join" clause). This point is invisible to the touch-only family: it requires the central crossing detector activated independently of the outer touch detectors.

CROSSING inhabits a different face of the cube (θ_u, θ_d disabled, θ_{cross} active), while TRIO occupies the interior (all three thresholds finite) and witnesses the literal touch–crossing join.

10.1. **The general automaton and its constrained specialisations.** The augmented monitor $M(\theta_u, \theta_d, \pi)$ of Definition 10.2 is the *general* ternary automaton of this paper: it holds all tuning knobs free — the two threshold positions θ_u, θ_d and their motion (Appendix A), the alternation schedule $\pi \in \{\text{ALT}, \text{FREE}\}$, the plateau-touch convention (Appendix E, a tuning parameter of the augmented monitor in the same sense as π), together with the semantic-aggregation, signal-modification, and tick-level copy-paste powers of Appendices B to D. Each classical modality automaton is then *the same automaton with specific*

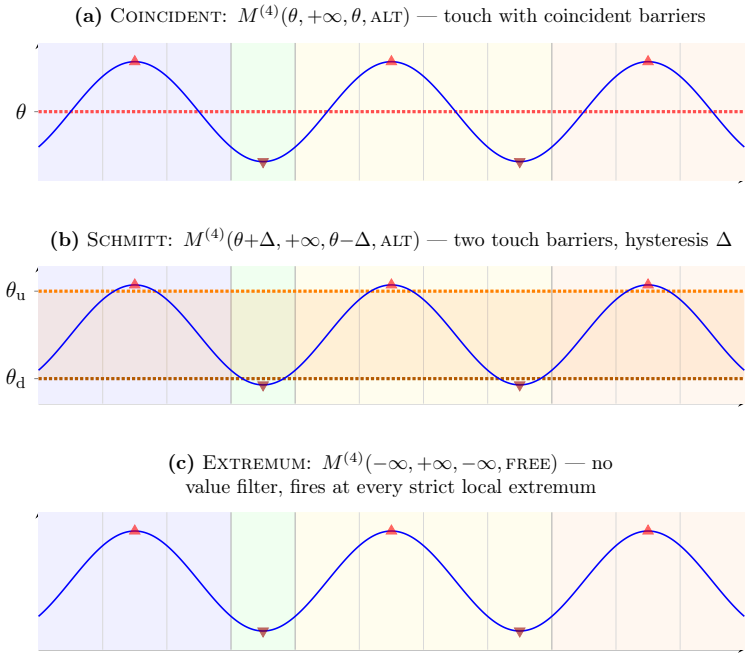


FIGURE 7. Signal traces for the three touch-only classical encoders of Proposition 10.5 on a common sinusoid $\tau(x) = 0.5 + 0.4\cos(2\pi(x-1)/4)$. All three panels share the same three trigger conditions for validate at tick n — *(pattern)* strict local maximum $\tau_{n-1} < \tau_n > \tau_{n+1}$, *(value)* $\tau_n \geq \theta_u$, and *(gate)* the output-low rail $\{L, M^{\uparrow}\}$ (internal latch S_{up}, Figure 9) when $\pi = \text{ALT}$ — and the symmetric three for invalidate on the output-high rail $\{U, M^{\downarrow}\}$ (internal latch S_{down}). The encoders differ only in their *parameter choices*. **(a)** COINCIDENT: single touch barrier at $\theta = 0.5$ (the two barriers coincide, $\theta_u = \theta_d = \theta$), ALT schedule. Each strict local maximum with $\tau_n \geq 0.5$ fires validate if state $= S_{up}$; each strict local minimum with $\tau_n \leq 0.5$ fires invalidate if state $= S_{down}$. Event count N_X=3 coincides exactly with the crossing-native monitor $M_C^{0.5}$ on this signal (gate-threshold equivalence, Proposition 13.9) — this is why COINCIDENT is the mechanical realisation of the zero-hysteresis level comparator (Remark 10.3). **(b)** SCHMITT: two distinct touch barriers ($\theta_u = 0.85$, $\theta_d = 0.15$), hysteresis $\Delta = 0.35$, ALT schedule. Only maxima above θ_u fire validate and only minima below θ_d fire invalidate; the central band (orange) is the hysteresis dead-zone in which small fluctuations are discarded. Mechanical realisation of the classical Schmitt-trigger (Remark 10.4). **(c)** EXTREMUM: thresholds at $-\infty$ disable the value filter (so condition *(value)* becomes vacuously true at every tick), and FREE disables the state gate (condition *(gate)* is removed). Only the *(pattern)* condition remains, so *every* strict local extremum fires regardless of value, with no alternation constraint — any small perturbation of τ that introduces additional extrema would generate additional events here that COINCIDENT/SCHMITT would discard. For the remaining two encoders see Figure 1(a) for the CROSSING signal trace and Figure 13 for TRIO. The discrete-signal-state automata are Figure 8 (external) and Figure 9 (internal).

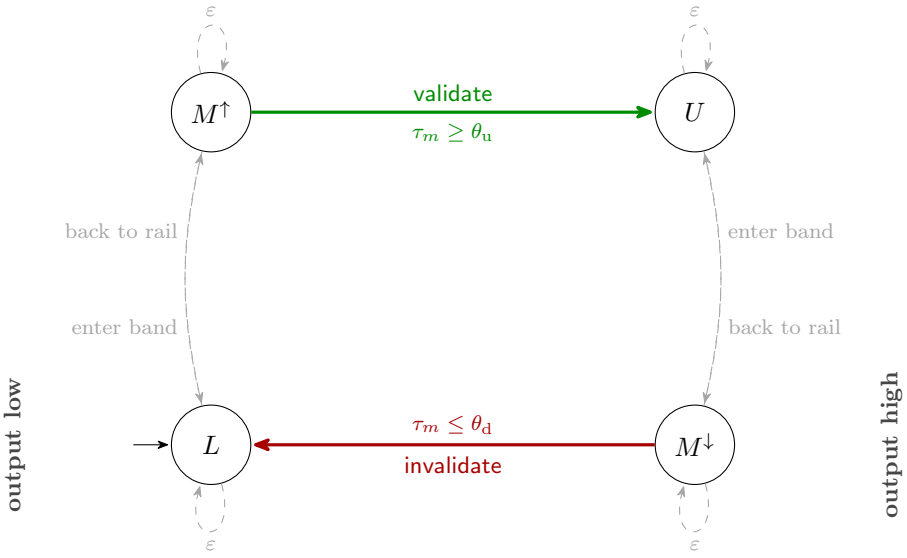
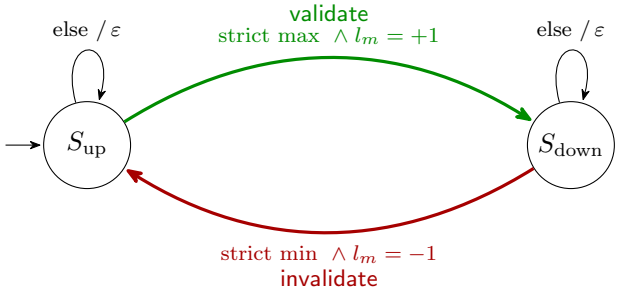


FIGURE 8. Discrete-signal-state SCHMITT automaton as a hysteresis loop — the *external* / crossing realisation, which fires on the threshold crossing, $\kappa = (1, 1, 0)$; the *internal* / monitor realisation is Figure 9. The value-class of the current sample τ_m against the two trigger levels $\theta_{\mathrm{d}} < \theta_{\mathrm{u}}$ (L below, in-band, U above) is combined with a one-bit *latch* — the sticky output bit that gives the trigger its hysteresis: it holds its value until the signal crosses the *whole* band, so chatter inside the band cannot flip it. The two in-band states carry that bit: $M^{\uparrow}$ (the ascending branch, latch still *low*) and $M^{\downarrow}$ (the descending branch, latch still *high*). The left rail $L, M^{\uparrow}$ holds output low, the right rail $U, M^{\downarrow}$ holds output high. Thick coloured edges are the only firing transitions (green validate on entering U from the low rail, red invalidate on entering L from the high rail). The gray dashed edges are silent (ε, no event): the self-loops hold the state (stay below, in-band, or above); *enter band* moves in from a rail ($\theta_{\mathrm{d}} < \tau_m < \theta_{\mathrm{u}}$); and *back to rail* returns to the same rail without flipping the latch ($\tau_m \leq \theta_{\mathrm{d}}$ back to L, $\tau_m \geq \theta_{\mathrm{u}}$ back to U). Setting $\theta_{\mathrm{u}} = \theta_{\mathrm{d}}$ empties the band and collapses the loop to the crossing latch $\{S_{\downarrow}, S_{\uparrow}\}$ of Definition 4.1. This is the *continuous-signal base case*: the signal passes through the band, so there is exactly one validate edge ($M^{\uparrow} \to U$) and one invalidate edge ($M^{\downarrow} \to L$). A *discontinuous* single-tick jump across the whole band ($L \to U$ or $U \to L$) would fire the same event directly; it is excluded here by the continuity assumption (as in Figures 30a and 31). The monitor's value count is read off the figure: it operates at the focus rung with verdict alphabet $\mathbb{V}_3 = \{-1, 0, +1\}$ ($|\mathbb{V}| = 3$, the three value-classes its guards distinguish), and emission alphabet $\{\varepsilon, \mathsf{validate}, \mathsf{invalidate}\}$ ($|\Sigma| = 3$: two distinct firing labels plus ε). The continuous-signal trace is Figure 7(b).

knobs frozen to fixed values: it is a constrained specialisation of $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$, not a separate machine. This is the catalogue reading of Remark 2.8 and Proposition 10.5: the named encoders "are not new automata but parameter points" of the augmented family.



(a) internal monitor-faithful automaton
$\kappa = (3, 1, 1)$: touch predicate $(3, 0, 1)$ + 1-bit alternation latch

(b) one excursion: *external* fires at the crossing, *internal* at the extremum

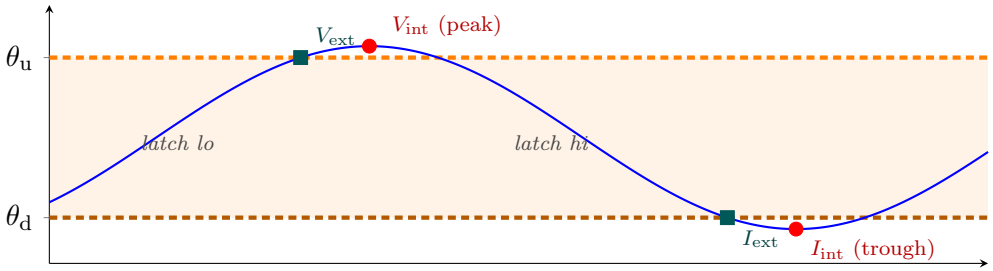


FIGURE 9. Internal, monitor-faithful realisation of the SCHMITT encoder and its mapping to the external loop of Figure 8. **(a)** The internal automaton is exactly the augmented monitor of Definition 10.2 (it fires on the strict extremum at the trigger level): a two-state alternation latch $\{S_{\text{up}}, S_{\text{down}}\}$, initial S_{up}, driven per tick by the integer level $l_m \in \{-1, 0, +1\}$ (-1: $\tau_m \le \theta_{\text{d}}$; 0: $\theta_{\text{d}} < \tau_m < \theta_{\text{u}}$; $+1$: $\tau_m \ge \theta_{\text{u}}$) and the discrete-derivative extremum (strict max: the discrete slope changes $+ \to -$, i.e. the derivative passes through 0; strict min: $- \to +$). It fires validate and flips to S_{down} on a strict maximum with $l_m = +1$, and invalidate back to S_{up} on a strict minimum with $l_m = -1$; every other tick is a silent self-loop (ε). Its complexity is $\kappa = (3, 1, 1)$: the touch predicate $(3, 0, 1)$ plus the one-bit alternation latch. **(b)** The mapping. Both realisations emit the *same* event per excursion — one validate per up-excursion, one invalidate per down-excursion — and differ only in the firing *time* (displaced by the excursion's rise/fall time). The external view (teal squares, $V_{\text{ext}}, I_{\text{ext}}$) fires at the threshold crossing: V_{ext} at the up-crossing of θ_{u}, I_{ext} at the down-crossing of θ_{d}. The internal view (red circles, $V_{\text{int}}, I_{\text{int}}$) fires at the strict extremum: V_{int} at the peak above θ_{u}, I_{int} at the trough below θ_{d}. Under the mapping to Figure 8, the left rail $\{L, M^{\uparrow}\}$ (output low) corresponds to the pre-validate state S_{up} and the right rail $\{U, M^{\downarrow}\}$ (output high) to the post-validate state S_{down}. The monitor's value count is read off the figure: it operates at the focus rung with verdict alphabet $\mathbb{V}_3 = \{-1, 0, +1\}$ ($|\mathbb{V}| = 3$, the three value-classes its guards distinguish), and emission alphabet $\{\varepsilon, \text{validate}, \text{invalidate}\}$ ($|\Sigma| = 3$: two distinct firing labels plus ε). The continuous-signal trace is Figure 7(b).

The reduction *from* each modality *to* the common automaton is therefore the inclusion of a constrained parameter point into the free family; the converse direction — the general automaton dominating every modality — is the join, conjectured in the simulation preorder (Conjecture 10.12) and proved exactly at the quaternary rung (Theorem 10.14).

Table 3 reads off, for each parameter point, which knobs the modality freezes (the constraint) and the resulting capability triple κ of Definition 2.4. The top row records the general automaton with all knobs free.

TABLE 3. The general automaton $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ and its modality specialisations. Each modality is the same augmented monitor with the listed tuning knobs frozen; $\kappa = (w, b, k)$ is the capability triple (Definition 2.4: window w, persistent state bit-width b, anti-causal lag k). The SCHMITT row carries the two realisations of Remark 10.4 (external crossing / internal extremum).

Modality	**Frozen knobs (the constraint)**	$\boldsymbol{\kappa}$
General $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$	all knobs free	$(3, 2, 1)$
COINCIDENT/CROSSING	$\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = \theta$, $\pi =$ ALT	$(1, 1, 0)$
SCHMITT	$\theta_{\mathrm{u}} = \theta{+}\Delta$, $\theta_{\mathrm{d}} = \theta{-}\Delta$, $\pi =$ ALT	$(1, 1, 0)$ ext. / $(3, 1, 1)$ int.
EXTREMUM/TOUCH	$\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = -\infty$ (or $= \theta$), $\pi =$ FREE	$(3, 0, 1)$
DIRRES	plateau convention = DIRRES	$(1, 2, 0)$

The reduction is hub-and-spoke: the general automaton is the hub, each modality a spoke obtained by freezing knobs. TRIO, the quaternary point $M^{(4)}(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \mathrm{FREE})$ of Remark 10.7, is excluded from this ternary hub: it needs the central crossing threshold and so lives one rung up, at $n = 4$ (Theorem 10.14). It is the next rung, not a spoke of the present hub. Figure 10 renders this as a hub-and-spoke specialisation diagram.

Corollary 10.8 (Capability dominance of the general automaton)**.** *The capability triple of the general augmented monitor is* $\kappa(M_{\mathrm{general}}) = (3, 2, 1)$*, the componentwise join of the modality triples of Table 3: the window* $w = 3$ *is the largest, attained by the strict touch predicate of Definition 5.1 (a two-sided extremum test needs* $\tau_{n-1}, \tau_n, \tau_{n+1}$*); the state bit-width* $b = 2$ *is the largest, attained by the DIRRES ternary engaged state* $s \in \{-1, 0, +1\}$ *(*$\lceil \log_2 3 \rceil = 2$*, Appendix E); the anti-causal lag* $k = 1$ *is the largest, from the one-sample lookahead of the extremum test. Every modality's capability triple is dominated componentwise,* $\kappa(M_{\mathrm{modality}}) \leq (3, 2, 1)$*, and each spoke* drops *exactly the components its frozen knobs render inert:*

- *COINCIDENT/CROSSING* $(1, 1, 0)$*: freezing* $\theta_{\mathrm{u}} = \theta_{\mathrm{d}}$ *and firing at the level transit drops the window* $3 \to 1$ *and the lag* $1 \to 0$*; the single alternation latch under* ALT *drops the state* $2 \to 1$*.*
- *SCHMITT* $(1, 1, 0)$ *external: same drop as CROSSING when read as the external hysteresis loop (Figure 8); the internal extremum realisation (Figure 9) restores the touch window/lag, giving* $(3, 1, 1)$*, but never the second state bit.*
- *EXTREMUM/TOUCH* $(3, 0, 1)$*:* $\pi =$ FREE *removes the alternation latch, so the state bit drops* $1 \to 0$ *(indeed* $2 \to 0$ *against the general* $b = 2$*); the touch window* 3 *and lag* 1 *are retained.*

- *DirRes* $(1, 2, 0)$*: firing at a plateau edge keeps the window at* 1 *and the lag at* 0*, while the ternary engaged state raises the state bit to the maximal* $b = 2$ *— the only spoke that attains the join in the state component.*

The three controllable components map onto the three knob axes: the schedule π *sets the alternation state* $b \in \{0, 1\}$*; the plateau convention can raise* b *to* 2 *under DirRes; and the firing locus — level transit versus strict extremum — sets* $(w, k) \in \{(1, 0), (3, 1)\}$*.*

Proof. By reading off κ at each constrained parameter point. The general value $(3, 2, 1)$ is the componentwise maximum over the column of Table 3, each entry of which is the capability triple of Definition 2.4 evaluated on the canonical native automaton of the corresponding parameter point (Proposition 10.5; the DirRes point of Appendix E). Componentwise dominance and the per-component drops are then immediate from the partial order on triples (Definition 2.4). □

Remark 10.9 (The monitor is the join, not a reduction)**.** Proposition 10.5 does not assert touch as primitive over crossing. $M(\theta, \theta, \text{ALT})$ (the Coincident point) recovers crossing; $M(-\infty, -\infty, \text{FREE})$ (the Extremum point) recovers touch. Neither degenerate point is structurally simpler than the other; the augmented monitor exhibits both as parameter choices and admits a continuum of intermediate behaviours. The quaternary extension of Definition 10.13 promotes this from a structural intent (within the touch-only subfamily) to a literal coincidence: the Crossing and Trio points of Remark 10.7 are exact parameter points that realise M_C natively and the touch–crossing join explicitly.

Theorem 10.10 (Ternary-rung closure under threshold tuning; instant-level, one-step lookback, Boolean-combination scope)**.** Scope. *We consider* 3*-valued threshold modalities that are (i) instant-level, (ii) admit at most one-step lookback, and (iii) have firing rule expressible as a Boolean combination of the value-class predicates* $\{v_n{=}{+1}, v_n{=}0, v_n{=}{-1}\}$ *over the ternary alphabet* $\mathbb{V}_3 = \{-1, 0, +1\}$*,* $v_n = \text{sign}(\tau_n - \theta)$*. Methods that add* orthogonal *structure (an output action on the input signal, e.g. the membrane-potential reset of integrate-and-fire neurons; or a temporal-formula layer consuming the event stream, e.g.* LTL_3 *runtime monitors) fall outside this scope.*

Within the stated scope, every *such modality is event-stream equivalent, under an appropriate tuning of the threshold parameters* $(\theta_u, \theta_d, \pi)$ *in the augmented monitor family* M *of Definition 10.2, to either the touch-native monitor* M_T *(Definition 5.3) or the crossing-native monitor* M_C *(Definition 4.1). The closure is realised by the following tuning recipe:*

Crossing branch.: *Take* $\theta_u = \theta_d = \theta$ *and* $\pi = \text{ALT}$*. The monitor* $M(\theta, \theta, \text{ALT})$ *produces the event stream of* M_C^θ *on every signal via the gate-threshold equivalence of Proposition 13.9. This is the Coincident parameter point of Proposition 10.5. Hysteresis variants (*$\theta_u > \theta_d$*,* $\pi = \text{ALT}$*) reduce to crossing-with-hysteresis — still the crossing branch, with a dead-zone band* (θ_d, θ_u) *(Remark 10.4, the Schmitt parameter point).*

Touch branch.: *Take* $\pi = \text{FREE}$ *(any* θ_u, θ_d*, finite or* $-\infty$*). The monitor* $M(\theta_u, \theta_d, \text{FREE})$ *produces the event stream of the touch-native monitor with two barriers — no alternation gate, both detectors always armed. At the limit* $\theta_u = \theta_d = -\infty$ *the value-filter is disabled and every strict local extremum fires; this is the Extremum parameter point of Proposition 10.5.*

Proof. The classification of distinguishable modalities at $|\mathbb{V}| = 3$ is given by Proposition 3.1: the available instant-level predicates over the value-class trace $v_n = \text{sign}(\tau_n - \theta) \in$

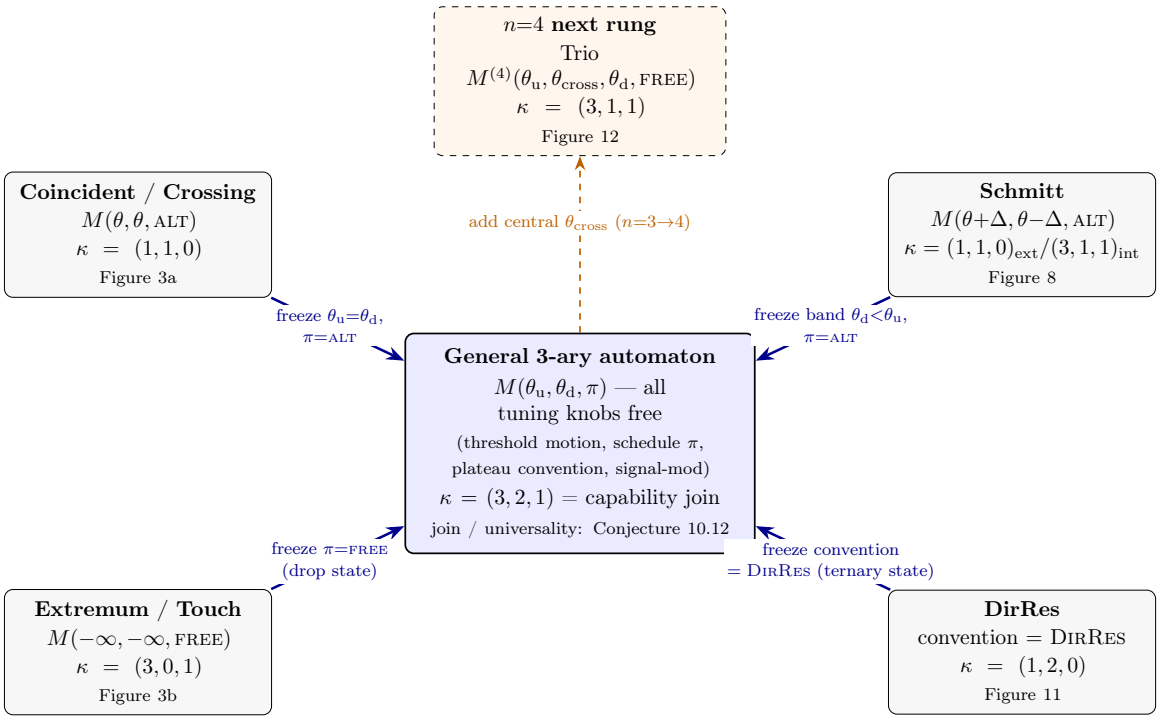


FIGURE 10. Hub-and-spoke specialisation diagram. Each spoke is a named classical *encoder* (COINCIDENT, SCHMITT, EXTREMUM, DIRRES) — a constrained point of the general 3-ary augmented monitor $M(\theta_u, \theta_d, \pi)$ — the hub, with every tuning knob free (Definition 10.2) — recovered by *freezing* knobs (the arrow labels), not a separate machine (Proposition 10.5, Remark 2.8); the arrows therefore run from each spoke *up to* the hub. (Only two of the spokes are the *modalities* of Proposition 3.1 — COINCIDENT/CROSSING and EXTREMUM/TOUCH; SCHMITT and DIRRES are derived realisations of their join.) The capability triple drops componentwise from the hub's $\kappa = (3,2,1)$ to each spoke (Corollary 10.8); this hub value is the componentwise maximum over *all four* spokes — its state bit-width 2 is contributed by DIRRES $(1,2,0)$, not by the two modalities, whose product-join is the smaller $(3,1,1)$ of the TRIO (Theorem 10.14). The hub is also the least upper bound of the two modalities in the simulation preorder (Conjecture 10.12, a conjecture; proven at the quaternary rung in Theorem 10.14). The TRIO (n=4) is the next rung up, reached by adding a central crossing threshold, not a 3-ary specialisation. The two-modality core — CROSSING, TOUCH, and their full-power join as explicit automata — is drawn in their value-class form in Figure 6.

$\{-1, 0, +1\}$ with at most one-step lookback yield exactly two distinguishable kinematic modalities — the transit event of crossing and the attainment event of touch (or equivalently the local-extremum event read from a three-window pattern over the ternary alphabet). Any other 3-valued threshold modality reducible to a Boolean combination of these atomic

predicates is therefore event-stream equivalent (up to verdict relabelling) to one of these two cases.

The augmented monitor M of Definition 10.2 is rich enough to realise both branches simultaneously through parameter tuning: $\pi = \text{FREE}$ disables the state-machine alternation and exposes the touch-native event stream at the chosen (θ_u, θ_d); $\pi = \text{ALT}$ with coincident thresholds $\theta_u = \theta_d = \theta$ enforces the alternation gate that, by Proposition 13.9, yields exactly the crossing-native event stream at threshold θ. The named encoders COINCIDENT, SCHMITT, EXTREMUM of Proposition 10.5 are three specific instances of this closure; any further 3-valued threshold encoder admissible at the ternary rung must, by the modality count of Proposition 3.1, fall into one of the two branches identified above. □

Remark 10.11 (Practical reading of the closure)**.** Theorem 10.10 closes the ternary rung as a *tuning-equivalence class* of monitors: any 3-valued threshold encoder one might propose — including the three classical encoders COINCIDENT, SCHMITT, EXTREMUM of Proposition 10.5, and any intermediate parameter point of the augmented monitor — is a tuning of one of touch or crossing. Genuinely new modalities require an enriched verdict alphabet: the 2×2 lattice at $|\mathbb{V}| = 4$ (Section 3.2) and the literal touch–crossing join $M^{(4)}$ (Definition 10.13) live outside the ternary closure precisely because their event-stream typing $\Sigma^{(4)}$ (5-valued) cannot be expressed at $|\mathbb{V}| = 3$.

Related threshold-detection methods. The augmented monitor of Definition 10.2 subsumes, as parameter points or as the underlying instant-level layer of a composed monitor, several classical threshold-event detection methods documented in the signal-processing, vision-sensor, neuroscience, and runtime-verification literatures. We list the principal correspondences:

- The *Schmitt-trigger* [14] is the SCHMITT parameter point $M(\theta{+}\Delta, \theta{-}\Delta, \text{ALT})$ (Remark 10.4, Proposition 10.5).
- *Zero-crossing detection* (DSP, PLL, frequency counters, folklore since the 1960s; in digital-receiver timing recovery the threshold-crossing methods are one of the classes reviewed in [13]) is the COINCIDENT parameter point at $\theta = 0$: $M(0, 0, \text{ALT})$ (Remark 10.3).
- The *STL atomic predicate* $\square(x \geq \theta)$ [10, 5] is a 2-valued threshold-membership test at the atomic layer; its event lifting (one validate per upward transit of θ) coincides with the COINCIDENT parameter point above (within the lifting framework).
- *Event cameras / Dynamic Vision Sensors (DVS)* [9, 31] emit validate/invalidate (ON/OFF) events when the change in per-pixel log intensity since the pixel's last event reaches a contrast threshold $\pm\theta_C$, after which the pixel resets [31, Sec. 2.4]; for short intervals this approximates a threshold on the temporal derivative. Behaviourally this is the crossing-native monitor $M_C^{\theta_C}$ applied to the log-derivative of the intensity signal — the CROSSING parameter point of the quaternary family (Theorem 10.14, clause (2)) for an idealised pixel; in the physical regime this correspondence is only approximate, subject to the refractory period and analog non-idealities of a real sensor.
- *Integrate-and-fire neurons* [32, 33, 34, 35] fire when the membrane potential $V(t)$ reaches a threshold (in [35], a spike cut-off), then reset. The event *stream* produced is identical to the COINCIDENT stream of M_C^{θ} applied to V, but the reset action $V := V_{\text{reset}}$ is an output *on the input signal*, which the augmented monitor of Definition 10.2 does

not perform; the integrate-and-fire model is therefore, at most, a *behavioural-stream* equivalence rather than a structural parameter point — a correspondence we regard as plausible but do not establish formally here.

- 3-*valued LTL runtime verification* [19, 4] operates at a different *layer* of the monitor stack: the LTL_3 monitor consumes the event stream produced by the augmented monitor and emits its own $\{\top, \bot, ?\}$ verdict over a temporal formula. The two layers compose orthogonally rather than coinciding as parameter points — an architectural lifting rather than a tuning equivalence, which we describe informally rather than prove.
- The *Metric Temporal Logic* atomic predicate [36, 37] attaches real-time bounds to temporal operators over Boolean propositions (STL itself is built on the real-time logic MITL [10]); at the atomic threshold level it coincides with the STL lifting above (at the atomic level only); the temporal envelope is orthogonal.

Within the stated scope of Theorem 10.10 — instant-level, one-step lookback, Boolean combinations of value-class predicates over $\mathbb{V}_3$ — the survey above finds no literature method whose firing rule lies genuinely outside the augmented monitor family. Methods that appear to lie outside (integrate-and-fire, LTL_3 monitors) do so by adding *orthogonal* structure — an output action on the input signal, or a temporal-formula layer consuming the event stream — not by introducing a third instant-level value-class predicate over $\mathbb{V}_3$. This is consistent with, and constitutes positive empirical evidence for, the ternary-rung closure of Theorem 10.10.

The touch-only parameter slice $(\theta_\mathrm{u}, \theta_\mathrm{d}, \pi)$ of the augmented monitor admits a continuum of monitors of which Coincident (on the diagonal $\theta_\mathrm{u} = \theta_\mathrm{d}$), Schmitt (upper triangle $\theta_\mathrm{u} > \theta_\mathrm{d}$) and Extremum (both thresholds at $-\infty$) are notable points; they occupy the touch face of the parameter space ($\theta_\mathrm{cross}{=}+\infty$), with the schedule $\pi \in \{\textsc{alt}, \textsc{free}\}$ a discrete extra axis.

Conjecture 10.12 (Universal property of the augmented monitor)**.** *Let $\mathcal{M}$ be the class of finite-state monitors whose event-emission rule is a Boolean combination of the static-threshold predicate of Definition 4.1 and the strict-extremum predicates of Definition 5.1 with threshold parameters in $V \cup \{-\infty, +\infty\}$, under the moving-threshold operational rule of Definition A.1. Equip $\mathcal{M}$ with the* simulation preorder *$M \preceq M'$ defined by: $M \preceq M'$ iff for every signal Q the event stream $E_M(Q)$ is a subsequence of $E_{M'}(Q)$ at matching event positions and types. Then $M(\theta_\mathrm{u}, \theta_\mathrm{d}, \pi)$ of Definition 10.2 is the* least upper bound *of $M_T^{\theta_\mathrm{u}}$ (touch-native with barrier θ_u) and $M_C^{\theta_\mathrm{d}}$ (crossing-native with threshold θ_d) in $(\mathcal{M}, \preceq)$. Explicitly:*

(1) *Both embeddings hold: $M_T^{\theta_\mathrm{u}} \preceq M(\theta_\mathrm{u}, -\infty, \textsc{free})$ and $M_C^{\theta_\mathrm{d}} \preceq M(+\infty, \theta_\mathrm{d}, \textsc{alt})$.*
(2) *Minimality: for any $M' \in \mathcal{M}$ with $M_T^{\theta_\mathrm{u}} \preceq M'$ and $M_C^{\theta_\mathrm{d}} \preceq M'$, there exists a parameter assignment $(\theta'_\mathrm{u}, \theta'_\mathrm{d}, \pi') \in (V \cup \{-\infty, +\infty\})^2 \times \{\textsc{alt}, \textsc{free}\}$ such that $M(\theta'_\mathrm{u}, \theta'_\mathrm{d}, \pi') \preceq M'$.*

The augmented monitor is therefore the canonical parameter-family realisation of the join $M_T \sqcup M_C$ within the touch-only subfamily; any monitor strictly larger than both modalities in $(\mathcal{M}, \preceq)$ contains a parameter point of the augmented family as a sub-monitor. Both embeddings of clause (1) are formal-but-vacuous when read literally under the subsequence preorder (the right-hand-side monitors emit the empty stream under the convention that $\pm\infty$ disables predicates, per Definition 10.1), so the statement is best read as a structural intent that is realised literally *only at the quaternary rung (Definition 10.13 and Theorem 10.14).*

Proof sketch. (1) Both embeddings are by direct inspection of Definition 10.2: setting $\theta_\mathrm{d} = -\infty$ disables the lower wall and the free schedule disables alternation, so the monitor

reduces to the strict-extremum touch monitor on θ_{u}; the symmetric assignment recovers the crossing monitor on θ_{d} via the gate-threshold equivalence of Proposition 13.9 (Section 9.2). (2) Minimality follows from the fact that the emission rules of Definition 10.2 are exactly the union of the rules of M_T and M_C with no additional combinatorial structure: any monitor that emits at least the events of both must, in particular, emit at the union, which is realised by the parameter point where both walls are active. □

10.2. **The quaternary augmented monitor: a true touch–crossing–touch sandwich.** The augmented monitor of Definition 10.2 is touch-only with state-machine alternation: both walls fire on local extrema, and the crossing modality is recovered only externally via the gate-threshold equivalence of Proposition 13.9. A natural extension at the quaternary rung $|\mathbb{V}| = 4$ promotes one of the three walls to a true *crossing* threshold while keeping the other two as touch barriers. The resulting monitor unifies touch and crossing as parameter points of a single family WITHOUT any external translation. Figure 13 renders the construction: two outer touch barriers bracket a central crossing threshold, and the four event types fire at semantically disjoint signal features.

Definition 10.13 (Quaternary augmented monitor)**.** A *quaternary augmented monitor* is a tuple $M^{(4)}(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \pi)$, where

- $\theta_{\mathrm{u}}, \theta_{\mathrm{d}} : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ are *touch barriers* (upper, lower), used by the touch predicates $\mathsf{validate}, \mathsf{invalidate}$ of Definition 10.1;
- $\theta_{\mathrm{cross}} : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ is a *crossing threshold*, used by the static-threshold transit predicates $\mathsf{crosses}^{\uparrow}, \mathsf{crosses}^{\downarrow}$ defined as: $\mathsf{crosses}^{\uparrow}(\theta_{\mathrm{cross}})$ holds at n iff $\tau_{n-1} < \theta_{\mathrm{cross}}(n) \leq \tau_n$; $\mathsf{crosses}^{\downarrow}(\theta_{\mathrm{cross}})$ holds at n iff $\tau_{n-1} > \theta_{\mathrm{cross}}(n) \geq \tau_n$;
- $\pi \in \{\mathrm{ALT}, \mathrm{FREE}\}$ is the alternation schedule, controlling the touch detectors as in Definition 10.2.

The monitor's emission alphabet is enriched to five values

$$\Sigma^{(4)} := \{\, \varepsilon,\ \mathsf{validate}_{\mathrm{up}},\ \mathsf{validate}_{\mathrm{cross}},\ \mathsf{invalidate}_{\mathrm{cross}},\ \mathsf{invalidate}_{\mathrm{down}} \,\}.$$

The firing rules are:

- $\mathsf{validate}_{\mathrm{up}}$ fires when $\mathsf{validate}(\theta_{\mathrm{u}})$ holds (touch of upper barrier), subject to the alternation gate if $\pi = \mathrm{ALT}$;
- $\mathsf{validate}_{\mathrm{cross}}$ fires when $\mathsf{crosses}^{\uparrow}(\theta_{\mathrm{cross}})$ holds, at the transit tick (crossing of central threshold upward), independently of the alternation gate;
- $\mathsf{invalidate}_{\mathrm{cross}}$ fires when $\mathsf{crosses}^{\downarrow}(\theta_{\mathrm{cross}})$ holds, at the transit tick (crossing of central threshold downward), independently of the alternation gate;
- $\mathsf{invalidate}_{\mathrm{down}}$ fires when $\mathsf{invalidate}(\theta_{\mathrm{d}})$ holds (touch of lower barrier), subject to the alternation gate if $\pi = \mathrm{ALT}$.

Absorbing-value conventions of Definition 10.1 apply to each of the three thresholds independently: $\pm\infty$ disables the corresponding predicate.

When $\theta_{\mathrm{cross}} = \pm\infty$ (crossing threshold disabled) the quaternary monitor degenerates to the ternary $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ of Definition 10.2, projecting the alphabet $\Sigma^{(4)}$ down to the ternary $\{\varepsilon, \mathsf{validate}, \mathsf{invalidate}\}$ by identifying $\mathsf{validate}_{\mathrm{up}} \mapsto \mathsf{validate}$ and $\mathsf{invalidate}_{\mathrm{down}} \mapsto \mathsf{invalidate}$.

Theorem 10.14 (Provable join at the quaternary rung)**.** *Both modalities M_T and M_C are exact parameter points of the quaternary family of Definition 10.13. Explicitly:*

(1) M_T^θ *(the touch-native monitor of Definition 5.3 with barrier* θ*) coincides with the event stream of* $M^{(4)}(\theta, +\infty, \theta, \textsc{free})$ *under the alphabet projection* $\mathsf{validate}_{\mathrm{up}} \mapsto \mathsf{validate}$, $\mathsf{invalidate}_{\mathrm{down}} \mapsto \mathsf{invalidate}$. *Both walls are at* θ*; the crossing threshold is disabled; the free schedule lifts the alternation gate.*

(2) M_C^θ *(the crossing-native monitor of Definition 4.1 with threshold* θ*) coincides with the event stream of* $M^{(4)}(+\infty, \theta, -\infty, \textsc{alt})$ *under the alphabet projection* $\mathsf{validate}_{\mathrm{cross}} \mapsto \mathsf{validate}$, $\mathsf{invalidate}_{\mathrm{cross}} \mapsto \mathsf{invalidate}$ *and the sticky-state alternation built into* M_C*. Both touch barriers are disabled; the crossing threshold equals* θ*; the alternation schedule reproduces the* M_C *sticky-state alternation.*

Both embeddings are at matching event positions and types *(the event-time identity is exact, not modulo shift), under the natural alphabet projections. The parameter point* $M^{(4)}$ *with all three thresholds finite is the disjoint union of* M_T *at the outer barriers and* M_C *at the central threshold — the join of the two modalities as exact parameter points of a single family.*

Proof. (1) At parameters $(\theta, +\infty, \theta, \textsc{free})$ the predicate $\mathsf{crosses}^{\uparrow}(+\infty)$ is impossible (no value $\geq +\infty$), so $\mathsf{validate}_{\mathrm{cross}}$ is never emitted; the symmetric reasoning disables $\mathsf{invalidate}_{\mathrm{cross}}$. The schedule FREE lifts the alternation gate, so $\mathsf{validate}(\theta_{\mathrm{u}}{=}\theta)$ and $\mathsf{invalidate}(\theta_{\mathrm{d}}{=}\theta)$ fire independently at every strict local maximum with value $\geq \theta$ and every strict local minimum with value $\leq \theta$ respectively. These are precisely the firing positions of M_T^θ per Definition 5.3. Under the alphabet projection $\mathsf{validate}_{\mathrm{up}} \mapsto \mathsf{validate}$ and $\mathsf{invalidate}_{\mathrm{down}} \mapsto \mathsf{invalidate}$ the two event streams coincide at every tick.

(2) At parameters $(+\infty, \theta, -\infty, \textsc{alt})$ both touch predicates are disabled (their thresholds are at the absorbing values that make the value-clearance test impossible per Definition 10.1); the only active predicates are $\mathsf{crosses}^{\uparrow}(\theta)$ and $\mathsf{crosses}^{\downarrow}(\theta)$. The state-machine alternation of Definition 10.2 reduces to the sticky-state alternation of the crossing-native monitor of Definition 4.1 once the touch detectors are disabled: $\mathsf{validate}_{\mathrm{cross}}$ at the first upward transit, then $\mathsf{invalidate}_{\mathrm{cross}}$ at the next downward transit, alternating. These are exactly the firing positions of M_C^θ. Under the alphabet projection $\mathsf{validate}_{\mathrm{cross}} \mapsto \mathsf{validate}$ and $\mathsf{invalidate}_{\mathrm{cross}} \mapsto \mathsf{invalidate}$ the two event streams coincide at every tick.

(Join.) At parameters $(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \textsc{free})$ with all three thresholds finite, the four active predicates $\mathsf{validate}(\theta_{\mathrm{u}})$, $\mathsf{crosses}^{\uparrow}(\theta_{\mathrm{cross}})$, $\mathsf{crosses}^{\downarrow}(\theta_{\mathrm{cross}})$, $\mathsf{invalidate}(\theta_{\mathrm{d}})$ fire independently on the signal. The event stream is the disjoint union of the $M_T^{\theta_{\mathrm{u}},\theta_{\mathrm{d}}}$ touch events at the outer barriers and the $M_C^{\theta_{\mathrm{cross}}}$ crossing events at the central threshold, with each event tagged by its source predicate via the five-valued alphabet $\Sigma^{(4)}$. The join interpretation of $M_T \sqcup M_C$ is realised literally at the event level: every M_T event and every M_C event appears in the quaternary monitor's stream with its correct type and at its correct tick. □

Remark 10.15 (CLASP classification lifts pointwise to $\Sigma^{(4)}$)**.** Each of the four event predicates of $\Sigma^{(4)}$ ($\mathsf{validate}(\theta_{\mathrm{u}})$, $\mathsf{crosses}^{\uparrow}(\theta_{\mathrm{cross}})$, $\mathsf{crosses}^{\downarrow}(\theta_{\mathrm{cross}})$, $\mathsf{invalidate}(\theta_{\mathrm{d}})$) is an instant-level finite-witness predicate of the same form classified by Proposition 13.10. The CLASP classification "a single validate emission is a Cosafety witness; absence on every prefix is a Safety property" lifts pointwise to each of the four typed verdicts of $\Sigma^{(4)}$ without modification; the trace-level Mixed reading of Proposition 13.11 likewise extends component-wise. The quaternary extension therefore does not alter the CLASP placement of the augmented monitor — it refines the verdict alphabet without changing the property-theoretic class.

Figure 14 below renders an end-to-end worked instance of the pipeline through the Extremum encoder, threading the augmented monitor of Definition 10.13 from signal to CLASP-cell verdict.

Remark 10.16 ($n = 3$ as the touch-only degeneration of $n = 4$)**.** The augmented monitor of Definition 10.2 is the $\theta_{\text{cross}} \in \{-\infty, +\infty\}$ degeneration of the quaternary monitor of Definition 10.13: disabling the central crossing threshold collapses the five-valued alphabet $\Sigma^{(4)}$ onto the ternary alphabet $\{\varepsilon, \mathsf{validate}, \mathsf{invalidate}\}$ by merging $\mathsf{validate}_{\text{up}}$ and the (now-absent) $\mathsf{validate}_{\text{cross}}$ into $\mathsf{validate}$. The "join" statement of Conjecture 10.12 thus applies to the touch-only subfamily; the genuine touch–crossing join requires the quaternary extension and is the content of Theorem 10.14.

Note (asymmetric projection). The alphabet projection $\Sigma^{(4)} \twoheadrightarrow \Sigma$ is total in the forward direction but *not invertible*: given a ternary stream over Σ alone, the inverse lift to $\Sigma^{(4)}$ is not uniquely determined — the source-predicate tag (which of $\mathsf{validate}(\theta_{\text{u}})$, $\mathsf{crosses}^{\uparrow}(\theta_{\text{cross}})$, etc. fired) is the information that the projection discards. Recovering the quaternary monitor from its ternary projection requires re-instating that tag from the parameter vector $(\theta_{\text{u}}, \theta_{\text{cross}}, \theta_{\text{d}}, \pi)$; the join interpretation of Theorem 10.14 is literal precisely because $\Sigma^{(4)}$ keeps the source tag explicit.

Having closed the construction of the augmented monitor, we pause before the downstream semantics to return to the two rungs deferred from the modality ladder of Section 3 — the degenerate κ-minimal corners of the very monitor family just built.

11. Degenerate rungs: the trivial and binary specialisations

The modality ladder of Section 3 deferred the two *degenerate* rungs at the bottom of the verdict-alphabet spectrum — the trivial rung $|\mathbb{V}|{=}1$ and the binary rung $|\mathbb{V}|{=}2$ — precisely because they are not foundations but *specialisations*. Each is defined by a collapse of the touch and crossing modalities, and each presupposes tuning powers of the monitor beyond the focus rung's static reading. With the two modalities now formally in hand (Sections 4 and 5), together with the capability anatomy and the strict tuning-layer hierarchy $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$ (Section 8 and Proposition 8.1) and the augmented monitor as their join (Section 10), we can place the two rungs where they belong: as the κ-minimal corners of the augmented family, reached by driving the input-modification and wall-motion powers to their degenerate extremes. At these corners the value-class component of the monitor automaton (Remark 2.7) collapses, and what the automaton tracks is no longer the signal's position relative to a threshold but, respectively, the threshold's binary *detection* state ($n = 1$) and the signal's *membership* of two threshold levels ($n = 2$).

11.1. The trivial rung ($|\mathbb{V}| = 1$): degenerate signal–threshold overlap and the no-modality baseline.

The bottom of the ladder is the rung at which the monitor emits a *single* verdict value over its entire lifetime. There is no modality to enumerate because there is no distinguishable predicate: the verdict trace is constant. The operationally interesting question becomes geometric: *on which signal–threshold configurations does the touch–crossing apparatus of Sections 4 and 5 degenerate to a single-valued verdict?*

Figure 15 answers this with the *discontinuous-threshold model*: the signal is constant at the unique encoding level θ, the threshold itself is geometrically in permanent overlap with the signal (it has no other value-position to occupy), and what is binary at this rung

is the threshold's *detection state* – it can be ON (rendezvous-event mechanism enabled) or OFF (rendezvous mechanism disabled). Each OFF $\to$ ON re-activation of detection fires exactly one event of the type fixed by the detection mode, since the geometric overlap is already in place at the moment of re-activation. Three configurations of this binary detection-state exhaust the trivial rung: V-mode detection re-activated each tick realises the trivial-top singleton Σ^ω via $\mathsf{validate}^\omega$; I-mode detection re-activated each tick realises the trivial-bottom singleton $\varnothing$ via $\mathsf{invalidate}^\omega$; detection permanently OFF realises the $\mathsf{L} \cap \mathsf{K}$ non-monitorable cell by *silence* – no event ever fires, and the monitor remains uncommitted in finite time. No mobile threshold and hence no indeterminate verdict (?) ever arises at this rung; the mono-valuedness $|\mathbb{V}|=1$ is a structural consequence of pairing a constant signal with a binary detection state.

Remark 11.1 (Trivial-rung readings and structural absences)**. Gate vs. edge equivalence.** Each ON pulse is, under per-tick reset of the detection control, an OFF $\to$ ON transition; the firing tick coincides in both readings (see Remark 11.4). The interruptions are essential under the edge reading — without them a single entry event would fire and never recur. Under the gate reading they are the schedule of an always-open gate splayed across the tick grid; both encodings realise Σ^ω. **Silence vs. the indeterminate verdict.** In panel (c) $\mathsf{L} \cap \mathsf{K}$ is realised by *silence* (no event ever fires), not by an active indeterminate verdict (?) — the ? verdict requires the mobility of Figure 35(b) to arise, and that mobility is structurally absent here. **Modal absence.** A binary detection state admits no band, so neither continuous-touch (Section 5) nor crossing (Section 4) can fire; the apparatus sits structurally below the threshold at which the modal decomposition of Section 10 becomes operative. Remark 11.2 positions this configuration as the joint signal–threshold degenerate case of a principle developed at the ternary rung (Appendix E); the reader may safely defer that connection.

Remark 11.2 (Bridge to wall-induced touch – deferred)**.** The discontinuous-threshold model of Figure 15 anticipates a principle developed in detail at the ternary rung: events can be generated either by signal motion against a static threshold (signal-caused) or by threshold motion against a static signal (wall-caused), and the latter mechanism is subject to a convention paradox (signal-direction rule vs strict-extremum rule) that motivates the indeterminate verdict (?) at $|\mathbb{V}|=3$. At the trivial rung the threshold has *no* value-position to move to or away from – its only degree of freedom is the binary ON/OFF of its detection state – so the ambiguity-generating motion is structurally absent. Two consequences follow. First, the event-firing predicate collapses to a single deterministic rule: one event per OFF $\to$ ON re-activation, with type fixed by the detection mode. Second, the $\mathsf{L} \cap \mathsf{K}$ cell of the CLASP partition is realised at this rung by *silence* (detection permanently OFF, no event ever fires, monitor uncommitted in finite time), not by the indeterminate verdict (?) that realises it at $|\mathbb{V}|=3$ and above. The mono-valuedness $|\mathbb{V}|=1$ is thus not an extrinsic parameter choice but a structural consequence of removing mobility from both the signal and the threshold. A reader may safely defer the wall-induced-touch connection to Appendix E (where the convention paradox is formalised) and Figure 35 (which visualises it on a constant signal with a mobile wall); we conjecture that the trivial rung is then re-readable as the joint signal–threshold-degenerate special case.

Proposition 11.3 (The trivial rung as the joint specialization of Figures 36 and 37)**.** *The discontinuous-threshold model of Section 11.1 is the simultaneous specialization of two Appendix E-territory constructions:*

(1) *the entry-/exit-/episode-bracketing convention choices of Figure 36 applied to a* plateau extended to the whole tick set $\mathbb{N}$*, i.e. a* permanent on-wall episode*; and*
(2) *the longer-notch wall-motion scenarios of Figure 37, applied under the* at-or-above *state definition B ($s = +1 \Leftrightarrow Q \geq \theta$) and with the wall's value-space mobility* replaced by a binary ON/OFF activation schedule *(no continuous notch trajectory).*

Under this joint specialization the three trivial-rung cells of Figure 15 are realised as the following case-specific reductions of the Figures 36 and 37 conventions:

- Σ^ω (trivial top, V-mode). *Each detection* OFF $\to$ ON *is interpreted as a fresh* entry *into the permanent on-wall episode under the entry-triggered convention of Figure 36(a); with re-activation at every tick, the per-tick V-entries form the stream* $\mathsf{validate}^\omega$*. Equivalently, under Figure 37 state-definition B, each ON pulse is a wall-caused above-crossing (panel (4) of Figure 37) whose direction-of-relative-motion is fixed by the V-mode label.*
- $\varnothing$ (trivial bottom, I-mode). *Symmetric to the V-mode case, with the wall–signal relation flipped (lower-wall episode, descending wall-caused crossing in Figure 37 panel (4) reading), so each re-activation issues* $\mathsf{invalidate}$ *and the stream is* $\mathsf{invalidate}^\omega$*.*
- $\mathsf{L} \cap \mathsf{K}$ (non-monitorable, OFF). *The detection is permanently OFF, no re-activation ever occurs. Equivalently this is Figure 36(b) (exit-triggered) on an episode that has no exit, i.e. the entry event is deferred forever; equivalently Figure 37 panel (2) (signal strictly below the notch level, no contact event) under the at-or-above definition B. No event ever fires; the verdict is uncommitted in finite time; the property lies in* $\mathsf{L} \cap \mathsf{K}$ *by silence.*

The reductions are exhaustive over the three trivial-rung cells and exclusive over the three detection-state configurations. No fourth configuration adds a new cell: the entry-triggered/exit-triggered/episode-bracketing distinction of Figure 36 collapses under the permanent-plateau limit to a single binary cycle (re-entry $\Leftrightarrow$ re-activation), and the A vs B state-definition distinction of Figure 37 collapses because the wall has no value-space position to sit strictly at versus above. This is the operational content of the joint signal–threshold degeneration *claim of Remark 11.2.*

Proof sketch. The proof is a case-by-case reduction. For clause (1) of the trivial-rung correspondence (V*-mode* $\to \Sigma^\omega$): a permanent on-wall episode is the $\to \infty$ limit of a finite plateau of Figure 36; entry-triggered fires once at the start of an episode. The detection ON/OFF cycle splices the permanent episode into a sequence of finite sub-episodes, each of which begins with a fresh entry; the union of entries is the per-tick re-activation pattern, which under any positive-mode (V) fires $\mathsf{validate}^\omega$. Clause (2) is the symmetric mirror at the lower wall. Clause (3) follows by transparently equating "never re-activated" with two equivalent Figures 36 and 37 cases (exit deferred forever; signal strictly below the notch level). The exhaustion claim holds because the trivial rung admits only three distinct constant verdict streams over the singleton-cardinality alphabet $|\mathbb{V}| = 1$ (once a literal is fixed, the stream is determined), and each constant stream is realised by exactly one of the three detection-state configurations. The exclusion claim holds because the three configurations are pairwise distinct on the detection-state trace. □

Remark 11.4 (The $n = 1$ automaton, explicitly)**.** Under the parallel-state-automaton framework of Definition 2.6 and Remark 2.7, the $n = 1$ monitor is the smallest non-trivial automaton in the ladder:

- State set $\mathcal{S}_1 = \{\mathrm{OFF}, \mathrm{ON}\}$.

- Initial state $s_0 = \mathrm{OFF}$.
- Transition rule δ_1 driven by the detection control channel d_m: $\delta_1(\mathrm{OFF}, \text{ON-input}) = \mathrm{ON}$ and dually.
- Event-firing transitions $\mathcal{F}_1$ (per detection mode):

Transition	V-mode	I-mode	OFF-mode	Notes
OFF → ON	validate	invalidate	ε	rising edge fires
ON → OFF	ε	ε	ε	silent
ON → ON	ε	ε	ε	no rising edge
OFF → OFF	ε	ε	ε	no activation

Gate reading (canonical for touch). The table above is the *edge-triggered* reading of the automaton, which fires exactly once per OFF → ON transition. An operationally equivalent *level-triggered gate reading* adopts the predicate "$d_i = \mathrm{ON} \Rightarrow$ fire (V or I per mode)" and treats the $\{\mathrm{OFF}, \mathrm{ON}\}$ state as a per-tick gate on the (tautological) touch contact predicate $\tau_i = \theta$. Under the per-tick reset of d_i adopted in Figure 15, every $d_i = \mathrm{ON}$ sample IS an OFF → ON transition, so the two readings produce identical event streams on the configurations of Proposition 11.3. The gate reading is the natural level-triggered extension of touch semantics to the trivial rung and is the convention used in the PoC notebook `01_n1_trivial_rung`. The V-mode automaton cycled at every tick ($d_m = \mathrm{ON}$ for all even m, OFF for all odd m) fires $\mathsf{validate}^\omega$, realising Σ^ω; cycled at no tick (permanently OFF) fires nothing, realising $\mathsf{L} \cap \mathsf{K}$ by silence (Proposition 11.3).

This automaton is the $n = 1$ specialization of Remark 2.7: the signal automaton component (which at $n \geq 3$ tracks value classes such as $\{-1, 0, +1\}$ or band membership $\{\text{below}, \text{in}, \text{above}\}$) is permanently in the single state 0 at this rung, so the detection control d_m is the only driver of event-firing transitions; the automaton is drawn explicitly in Figure 16.

The trivial rung carries no modality count to enumerate; its function is to establish the geometric baseline above which the remaining rungs activate touch and crossing distinctions. The next rung (Section 11.2) breaks the degeneracy by introducing a non-constant signal at the lowest interesting cardinality, $|\mathbb{V}|$=2.

11.2. **The binary rung ($|\mathbb{V}| = 2$): two-line jumping signal and the degenerate single modality.** *Signal model.* At this rung the input signal is not the continuous sinusoid of the focus rung. We fix *two* thresholds $\theta_0 < \theta_1$ and take $\tau : \mathbb{N} \to \{\theta_0, \theta_1\}$ to be a *piecewise-constant signal* that sits at one of the two threshold levels and may jump to the other at any tick boundary. Geometrically, the signal trajectory is a two-line horizontal-step pattern: the signal is always *exactly at* one of θ_0 or θ_1, and tick boundaries are the only positions at which it can switch. There is no continuous trajectory between the two levels: the signal does not pass through any intermediate value, so there is no transit *across* a threshold in the kinematic sense, and no local-extremum event in the three-window-pattern sense (the signal is constant between jumps, so every position is simultaneously a maximum and a minimum of its local window).

Verdict alphabet. $\mathbb{V}_2 = \{0, 1\}$ with $b_n := [\tau_n = \theta_1]$. The monitor reports which of the two thresholds the signal is currently at. Figure 18 illustrates the signal model.

Proposition 11.5 (The binary rung hosts one modality)**.** *On the two-line jumping signal model with thresholds $\theta_0 < \theta_1$, the available instant-level predicates over $\mathbb{V}_2$ are exactly*

$\{b = 1, b = 0\}$ *(both nontrivial, mutual negations). Any instant-level modality is therefore equivalent up to Boolean equivalence and tautology to the* threshold-membership *modality* $b_n = 1$ *(the signal is currently at the upper threshold). In particular touch and crossing* collapse*: there is no continuous transit* across *either threshold because the signal inhabits the two thresholds only and never any intermediate value, and there is no local-extremum event in the three-window-pattern sense because the signal is constant between jumps.*

Proof. A modality $\mu : V^* \to \{0, 1\}$ that is causal and instant-local on a binary verdict must factor through the single-step predicate b_n. Two binary predicates on a one-bit alphabet differ at most by Boolean equivalence: $\{b = 0\} = \neg\{b = 1\}$, and the other Boolean combinations collapse to $\top$ or $\bot$. Hence the count of distinguishable nontrivial modalities is one.

The classical crossing event "the signal transits a threshold" has no kinematic content here: the trajectory takes the value θ_0 or θ_1 and nothing in between, so the transit *through* the gap (θ_0, θ_1) is never sampled. Only the threshold-jump $b_{n-1} = 0 \to b_n = 1$ is observable, and that is a verdict-stream change rather than a kinematic transit; it is recoverable only from a one-bit lookback, not from the instant-local alphabet itself. The classical touch event "τ_n attains a strict local extremum" is undefined on a constant-between-jumps signal: every plateau position is a (non-strict) local maximum and minimum simultaneously, so the three-window pattern fires nowhere or everywhere depending on convention. Both classical modalities therefore reduce to the threshold-membership predicate. □

The binary rung is structurally degenerate (Figure 17). The two-line jumping signal model and the binary verdict alphabet agree on a single conclusion: one modality, with the touch/crossing distinction invisible. The non-degenerate rungs developed earlier break this degeneracy in two complementary ways: the ternary focus rung (Section 3.1) replaces the two-line jumping signal with a continuous trajectory and reduces the threshold count to one (a single central threshold around which the smooth signal moves); the quaternary rung keeps the continuous trajectory and adds inner and outer thresholds (three thresholds total).

Remark 11.6 (Tuning powers presupposed by the low rungs)**.** The trivial and binary rungs are not merely smaller verdict alphabets: each is the configuration the tuning algorithm reaches by driving a specific subset of its powers — on the input pair (signal, threshold) — to a degenerate extreme. Read against the layer hierarchy of Proposition 8.1, the rungs presuppose:

- *Trivial rung ($n = 1$): two powers.* **(i) A constant input signal** — a degenerate exercise of the Layer-3 input-modification power (Appendix D), the only layer able to set input values, here driven to the extreme that removes all signal mobility. **(ii) Threshold toggling on and off** — the residual freedom of Layer-1 wall motion (Appendix A) once the wall has no value-position to move to or from (Remark 11.2): its only remaining degree of freedom is the binary ON/OFF of the detection control d_m. The value-class component of the automaton is frozen (Remark 11.4), so these two powers — constant input *plus* threshold toggling — are exactly what drives every event at this rung.
- *Binary rung ($n = 2$): one power.* The input signal is fixed to the two-line piecewise-constant jumping signal that inhabits only the two threshold levels (Figure 18) — again the Layer-3 input-modification power, here driven to admit exactly two values and no

intermediate trajectory. No threshold toggling is needed: the membership latch is driven by the input jumps themselves (Proposition 11.5).

Hence the verdict-cardinality ladder $|\mathbb{V}| = 1, 2, 3, \ldots$ and the tuning-power layers $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$ are not independent axes: the low rungs are *reached* by driving the input-modification and wall-motion powers to their degenerate extremes (constant or two-level input; threshold reduced to binary detection). This is why their modalities differ in kind from the ternary rung's — they track detection ($n = 1$) or threshold membership ($n = 2$) rather than the signal's value-class position relative to a threshold ($n \geq 3$). The observation introduces no new construction; it relabels the existing rung models as outputs of named tuning powers.

Remark 11.7 (The automaton ladder as progressive determinacy)**.** Set the rung automata side by side and a second reading of the ladder appears, dual to the degeneration of Remark 11.6. The skeletons barely change: the trivial rung is a two-state detection latch (Figure 16), the binary rung a two-state membership latch (Figure 17), and the ternary crossing monitor is again a two-state latch (Figure 3) — all three at $\kappa = (1, 1, 0)$ — while touch is the stateless three-window detector at $\kappa = (3, 0, 1)$.

$\lvert\mathbb{V}\rvert$	Automaton (κ)	State tracks	Indeterminacy
1	detection latch $(1, 1, 0)$	detection ON/OFF	passive (silence, $\varepsilon \mapsto ?$)
2	membership latch $(1, 1, 0)$	which threshold	modal collapse (T=C)
3	crossing latch $(1, 1, 0)$	value-class sign	*resolved*
	+ touch $(3, 0, 1)$	three-window	(transit $\neq$ attainment)

What changes rung to rung is not the machine but the *value-class resolution of its input*: $|\mathbb{V}|$ rises $1 \to 2 \to 3$ as the tuning algorithm installs one further distinction, and each distinction *reduces indeterminacy* on two fronts. *Verdict indeterminacy:* at the trivial rung the indeterminate cell $\mathsf{L} \cap \mathsf{K}$ is realised *passively*, by silence — the monitor never commits ($\varepsilon \mapsto ?$ under Definition 12.3) — because the value-class component is frozen and only detection ON/OFF carries information. *Modal indeterminacy:* at the binary rung touch and crossing *collapse* (Proposition 11.5), since the signal inhabits only the two threshold levels and never the gap (θ_0, θ_1), so no position exists from which transit and attainment could differ.

The ternary rung admits exactly that gap as the neutral 0-class: the continuous signal now passes *through* the boundary region the binary rung forbade, and the 0-symbol is the structural enabler that distinguishes the transit event (crossing, $-1 \to +1$ through 0) from the attainment event (touch, arrival at 0 without committing) (Proposition 3.1). Hence the touch and crossing automata are not a heavier machine than the low-rung latches but the *same minimal skeleton under a finer input alphabet*: the third value-class — the tuning algorithm's power to admit the once-forbidden gap as a resolved boundary state — is the single added distinction that turns the collapsed, indeterminate single modality of $|\mathbb{V}| \leq 2$ into the two distinguishable modalities of $|\mathbb{V}| = 3$. Read upward the ladder reduces indeterminacy; read downward (Remark 11.6) it is the degeneration obtained by driving the input back onto the threshold. This relabels Propositions 3.1 and 11.5 as the two ends of one resolution axis; it introduces no new construction.

This completes the descent to the bottom of the verdict-alphabet spectrum. The trivial and binary rungs are the degenerate κ-minimal corners of the augmented family (Section 10); the ternary focus rung (Section 3.1) is the smallest rung at which touch and crossing become

distinguishable. We now equip the monitor's output — the event stream — with the algebraic and verdict-semantic structure that any downstream runtime-verification consumer expects.

12. Event stream algebra and LTL3 semantics

The monitor's output is an *event stream*: a function $E : \mathbb{N} \to \Sigma$ with alphabet $\Sigma := \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ (the last symbol representing "no event"). Two natural structures live on the set of event streams: an *algebra* of stream-shaping operators (this section), and a *verdict semantics* that maps streams to LTL3 truth values for higher-level temporal logic (the next subsection). Both structures are needed by downstream analyses that build on top of the augmented monitor, such as temporal-implication calibration.

Definition 12.1 (Event stream operators)**.** The set $\mathcal{E} := \Sigma^{\mathbb{N}}$ carries the following operators:

- *Identity* $\varepsilon^{\mathbb{N}}$: the stream that emits ε at every position.
- *Concatenation* $E_1 \cdot E_2$ at split point $k \in \mathbb{N}$: emit $E_1(n)$ for $n < k$ and $E_2(n-k)$ for $n \geq k$.
- *Window* $E[a, b]$ for $a \leq b \in \mathbb{N}$: emit $E(n)$ for $a \leq n \leq b$ and ε otherwise.
- *Type-projection* $\pi_{\mathsf{validate}}(E)$ and $\pi_{\mathsf{invalidate}}(E)$: keep events of the specified type, replace the others with ε.
- *Type-swap* $\sigma(E)$: exchange $\mathsf{validate}$ and $\mathsf{invalidate}$ everywhere, leaving ε unchanged — the realisation of the sign-convention swap of Appendix E.
- *Parallel composition* $E_1 \parallel E_2$: the stream that emits $E_1(n)$ at positions where $E_2(n) = \varepsilon$, emits $E_2(n)$ at positions where $E_1(n) = \varepsilon$, and emits a collision marker $\perp$ at positions where both are non-ε. Disjoint streams (Σ-supports disjoint) merge cleanly without $\perp$.

$(\mathcal{E}, \cdot, \varepsilon^{\mathbb{N}})$ is a non-commutative monoid; the type-swap σ is an involution of $\mathcal{E}$ that commutes with windowing and concatenation.

Remark 12.2 (Monitor-as-functor)**.** Every monitor M is a function $Q \mapsto E_M(Q) \in \mathcal{E}$ from value-traces to event streams. The augmented monitor of Definition 10.2 is the function-family $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi) \mapsto E_{M(\cdot)}(Q)$ parameterised by a point of the parameter space, valued in $\mathcal{E}$. This makes precise the informal slogan "a monitor is a parametric sampler of an event stream from a value-trace," which we offer as a guiding intuition rather than a formal claim.

12.1. LTL3 verdict semantics.

We read the monitor output through the three-valued semantics of LTL3 [4]: at every position n a proposition has verdict in $\mathbb{B}_3 := \{\top, \perp, ?\}$ (true, false, undetermined) depending on what every trace extension would conclude. Each convention $c \in \mathcal{C}$ of Definition E.4 together with a fixed parameter point of the augmented monitor induces a verdict assignment as follows.

Definition 12.3 (Convention-induced LTL3 verdict)**.** Fix $c \in \mathcal{C}$, a tuple $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$, and a signal Q. Write $E^c_{M(\cdots)}(Q) \in \mathcal{E}$ for the event stream the augmented monitor emits on Q under convention c. The convention-induced verdict $v^c_{M,Q} : \mathbb{N} \to \mathbb{B}_3$ for the atomic

proposition validate is

$$v^c_{M,Q}(n) \;:=\; \begin{cases} \top & \text{if } E^c_M(Q)(n) = \mathsf{validate}, \\ \bot & \text{if } E^c_M(Q)(n) = \mathsf{invalidate}, \\ ? & \text{if } E^c_M(Q)(n) = \varepsilon. \end{cases}$$

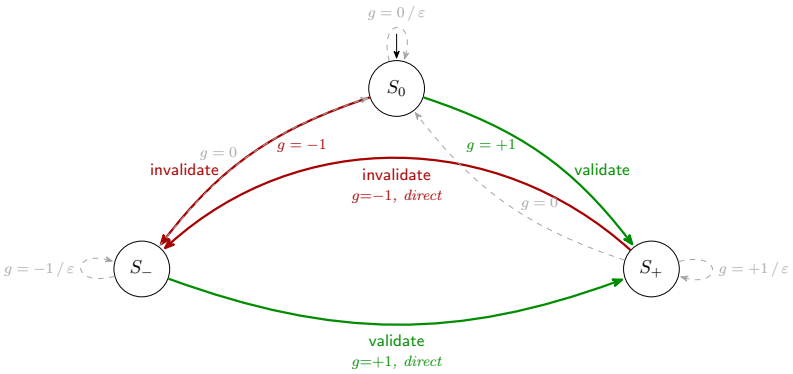
The verdict for invalidate is obtained by exchanging $\top \leftrightarrow \bot$. The convention c enters the verdict map only through $E^c_M(Q)$; thus the mapping

$$v_{(\cdot)} : \mathcal{C} \to \mathbb{B}_3^{\mathbb{N}}$$

sends each convention to a distinct verdict assignment exactly when the conventions disagree on at least one sample of $E_M(Q)$.

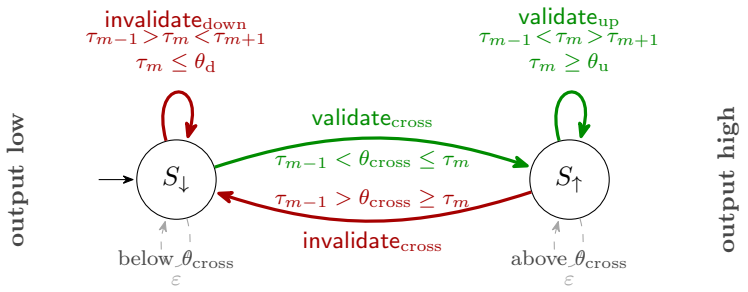
The map $c \mapsto v^c_{M,Q}$ is the formal interface between the augmented monitor and any LTL3-style temporal logic that consumes monitor output. In particular, the five plateau conventions of Appendix E project onto five LTL3 verdict assignments at the same underlying signal and parameter point — verdict differences arise exactly on plateaus and cross-band jumps, as predicted by Figures 34, 36 and 37.

Remark 12.4 (The (V, A) parameter pair of this paper)**.** The augmented monitor of Definition 10.2 and the event stream algebra of Definition 12.1 expose a pair (V, A) where V is the event alphabet and A is the algebra of event-generation primitives over signals. Throughout the paper, V is held fixed at the ternary alphabet $\{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ (cardinality 3), while A varies along the touch/crossing duality of Sections 9.1, 9.2 and 10: Model A uses a single crossing-based primitive over a single threshold parameter $\theta \in \mathbb{R}$, Model B uses two touch-based primitives over two wall parameters $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}) \in \mathbb{R}^2$, and the augmented monitor is the join of the two algebras over the same fixed V. The convention algebra $\mathcal{C} = \mathcal{T}^{|W|} \times \mathcal{S}$ of Definition E.4 is itself an internal axis along which A varies with V unchanged. The dual configuration — V varying at fixed A along a verdict-cardinality ladder — is treated in [3], where A is the standard LTL operator semantics and V ranges over the multi-valued truth domains $\mathbb{B}_n$. We draw this correspondence as an informal organising analogy, not as a formal duality.



$\kappa = (1, 2, 0)$ entry-triggered (causal): window 1, the ternary state $\{S_-, S_0, S_+\}$ costs $\lceil \log_2 3 \rceil = 2$ bits, lag 0. Collapsing S_0 recovers the $(1, 1, 0)$ Crossing latch — plateau-faithfulness costs one extra state bit. The onset-deferred variant back-dates over the whole plateau, so its lag equals the plateau length (unbounded), not a fixed finite triple.

Figure 11. Discrete-signal-state automaton for the *unified ternary direction-resolved* convention of the Crossing and Touch modalities. The input read at each step is the current sign-class $g = \mathrm{sgn}(Q(n) - \mathrm{level}) \in \{-1, 0, +1\}$, where the active level is θ for crossing (Model A) and θ_u or θ_d for touch (Model B). The three states are S_0 (engaged / on-level / undecided, $s_0 = S_0$), S_+ (validate active, $s = +1$) and S_- (invalidate active, $s = -1$). Firing rule: fire on *entering* ± 1, with the event type equal to the sign just entered — exactly one event per on-level episode. The two upper thick edges pass *through* S_0 (a plateau or tangential contact that yields an engaged, undecided moment before the slope resolves); the two bottom thick edges are discontinuous *direct transits* that skip S_0 (a crossing without any on-level sample, e.g. a splice-induced jump). The transition function is total: from each state every input $g \in \{-1, 0, +1\}$ has a defined edge, the silent (gray dashed) self-loops and the $g = 0$ returns to S_0 emitting ε. The drawn machine is *entry-triggered* (causal, $\kappa = (1, 2, 0)$): the ternary state $\{S_-, S_0, S_+\}$ costs $\lceil \log_2 3 \rceil = 2$ state bits, so it carries one bit more than the collapsed two-state Crossing latch, whose $\kappa = (1, 1, 0)$. (The *exit-triggered* and *onset-deferred* timing variants of this automaton are developed later, with the touch conventions, in Appendix E.) This automaton *refines* the static Crossing latch of Figure 3a: collapsing S_0 (no on-level samples, the signal is never tangent to the level) leaves only the two bottom direct-transit edges and recovers the two-state $\{S_\downarrow, S_\uparrow\}$ loop. Here the three states $\{S_-, S_0, S_+\}$ are *exactly* the focus-rung verdict alphabet $\mathbb{V}_3 = \{-1, 0, +1\}$ ($|\mathbb{V}| = 3$): the monitor's value count is the automaton's state count. This identity is specific to the *value-on-state* (Moore) rendering adopted here — one state per verdict class; the monitor's intrinsic memory is the Myhill–Nerode index, which under a transition-labelled (Mealy) emission — as in the stateless touch detector, one control state emitting three values — is independent of $|\mathbb{V}|$. The emission alphabet is $\{\varepsilon, \mathsf{validate}, \mathsf{invalidate}\}$ ($|\Sigma| = 3$); the causation paradox it induces is discussed with the touch conventions (Appendix E).



$\Sigma^{(4)} = \{\varepsilon, \mathsf{validate}_{\text{up}}, \mathsf{validate}_{\text{cross}}, \mathsf{invalidate}_{\text{cross}}, \mathsf{invalidate}_{\text{down}}\}$.
State-based product $\kappa = (3, 1, 1)$ = crossing latch $(1, 1, 0) \sqcup$ touch $(3, 0, 1)$ (max window, sum state-bits, max lag). The free-schedule stateless dual — crossing realised by the 2-window transit predicate — is $\kappa = (3, 0, 1)$.

FIGURE 12. Discrete-signal-state TRIO automaton: the quaternary encoder $M^{(4)}(\theta_{\text{u}}, \theta_{\text{cross}}, \theta_{\text{d}}, \text{FREE})$ of Definition 10.13 drawn as the product of the central-threshold CROSSING latch (Figure 3a, the backbone $S_\downarrow/S_\uparrow$ on θ_{cross}) and the stateless TOUCH detector (Figure 3b, the self-loops). The five-valued alphabet $\Sigma^{(4)}$ tags each firing by its source predicate, coloured by event as in the other automata (green validate, red invalidate): the thick bent edges are the central-threshold transits ($\mathsf{validate}_{\text{cross}}$ on the upward transit $\tau_{m-1} < \theta_{\text{cross}} \le \tau_m$, $\mathsf{invalidate}_{\text{cross}}$ on the downward transit $\tau_{m-1} > \theta_{\text{cross}} \ge \tau_m$), and the thick self-loops are the barrier extrema. *Key:* because $\theta_{\text{d}} < \theta_{\text{cross}} < \theta_{\text{u}}$, while the latch is in $S_\uparrow$ the signal is above θ_{cross} (it up-crossed and has not down-crossed), so only the upper-barrier strict maximum $\mathsf{validate}_{\text{up}}$ ($\tau_m \ge \theta_{\text{u}}$) is geometrically reachable; symmetrically $S_\downarrow$ reaches only the lower-barrier strict minimum $\mathsf{invalidate}_{\text{down}}$ ($\tau_m \le \theta_{\text{d}}$). (A sample tangent to the central threshold, $\tau_m = \theta_{\text{cross}}$, stays in $S_\uparrow$ under the closed convention $\tau_m \ge \theta_{\text{cross}}$ but is filtered out of both touch predicates by $\theta_{\text{d}} < \theta_{\text{cross}} < \theta_{\text{u}}$, so the placement survives the boundary case.) The transition function is total: from each state a central transit fires the crossing event, the reachable barrier extremum fires the touch event, and every other window emits ε (the gray dashed loops) (the silent ε loop also absorbs the strict-equality non-transit $\tau_{m-1} = \theta_{\text{cross}}$, where the strict guard fails and the latch stays put). This is one of two realisations: the state-based product drawn here ($\kappa = (3, 1, 1)$) and, under the free schedule, the stateless 2/3-window dual ($\kappa = (3, 0, 1)$) in which the crossing latch is traded for the 2-window transit predicate (the Myhill–Nerode trade of Definition 2.4). The join is exactly that of Theorem 10.14 — no stronger. At the quaternary rung the value count is larger: verdict alphabet $\mathbb{V}_4 = \{\text{LL}, \text{L}, \text{H}, \text{HH}\}$ ($|\mathbb{V}| = 4$, the four bands the three thresholds $\theta_{\text{d}} < \theta_{\text{cross}} < \theta_{\text{u}}$ cut the value domain into), and emission alphabet $\Sigma^{(4)}$ with $|\Sigma^{(4)}| = 5$ (four distinct firing labels plus ε). The continuous-signal trace is Figure 13.

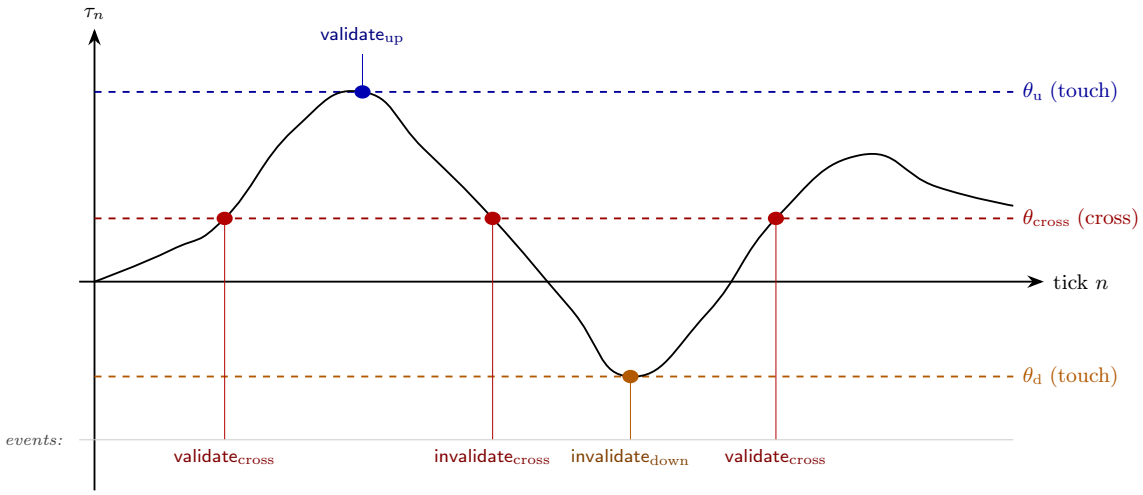


FIGURE 13. The quaternary augmented monitor $M^{(4)}(\theta_{\text{u}}, \theta_{\text{cross}}, \theta_{\text{d}}, \pi)$ as a literal touch–crossing–touch sandwich. Two touch barriers $(\theta_{\text{u}}, \theta_{\text{d}})$ bracket a central crossing threshold θ_{cross}; the four event types fire at semantically disjoint signal features (upper-barrier touch at a local max, central crossing in both directions, lower-barrier touch at a local min), tagged in the five-valued alphabet $\Sigma^{(4)}$. Three parameter degenerations recover the simpler monitors: $\theta_{\text{cross}} = \pm\infty$ recovers the ternary augmented monitor of Definition 10.2 (Remark 10.16); $(\theta, +\infty, \theta, \text{FREE})$ recovers the touch-native monitor M_T^θ; $(+\infty, \theta, -\infty, \text{ALT})$ recovers the crossing-native monitor M_C^θ (Theorem 10.14). The discrete-signal-state automaton is Figure 12.

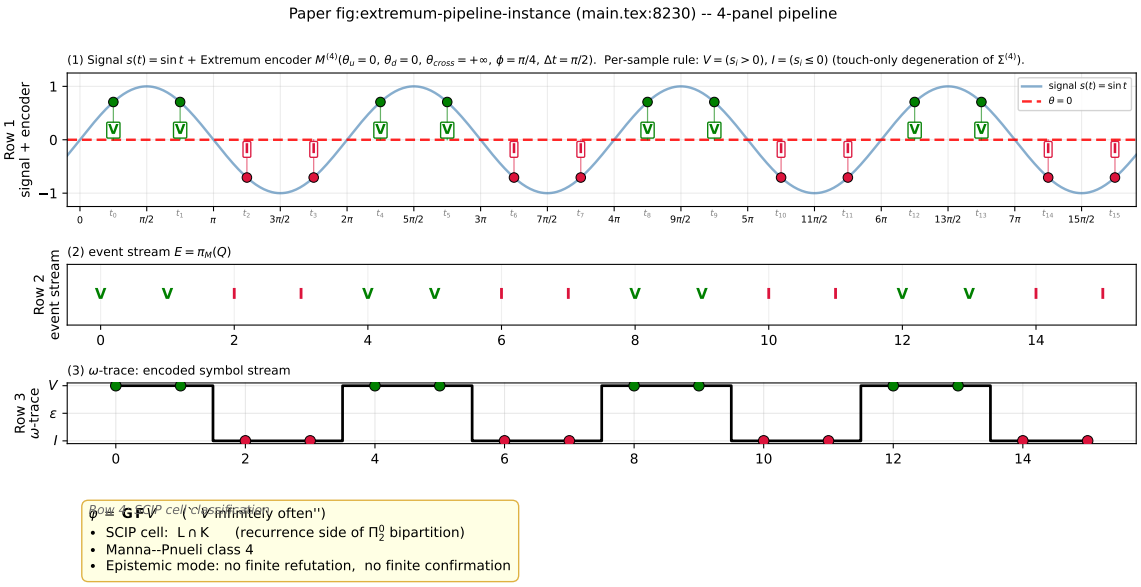
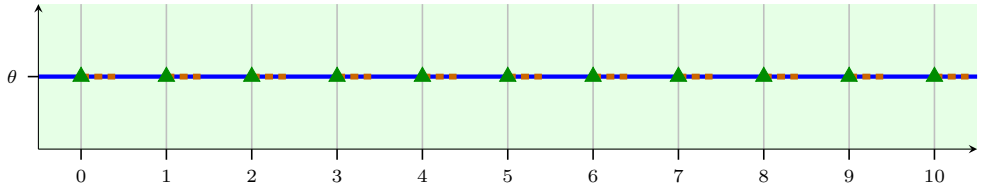
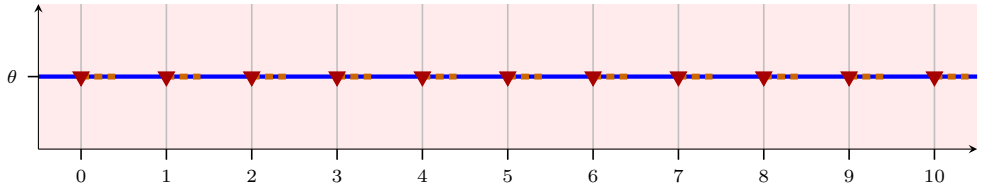


FIGURE 14. End-to-end pipeline instantiation of the EXTREMUM encoder of Corollary 10.6 applied to $s(t) = \sin t$. *Panel 1:* signal and encoder; the EXTREMUM point sets $\theta_{\mathrm{u}} = \theta_{\mathrm{d}} = 0$ and disables the central crossing threshold ($\theta_{\mathrm{cross}} = +\infty$), reducing $M^{(4)}$ to the touch-only ternary monitor of Definition 10.2. The phase offset $\phi \in (0, \pi/2)$ aligns two consecutive samples per positive half-period of $\sin t$. *Panel 2:* the event stream emitted by the encoder over the ternary alphabet $\Sigma = \{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$; here the trajectory does not produce ε since the sampling instants are strictly above or below threshold, so the effective alphabet on this signal is the touch-only degeneration $\{\mathsf{validate}, \mathsf{invalidate}\}$. *Panel 3:* the aggregated ω-trace $\sigma = (\mathsf{validatevalidateinvalidateinvalidate})^{\omega}$. *Panel 4:* CLASP cell classification. The candidate property $\varphi = \mathbf{G}\,\mathbf{F}\,\mathsf{validate}$ inhabits the $\mathsf{L} \cap \mathsf{K}$ cell of the CLASP partition of [3] (Theorem on the 16-cell totality), on the recurrence (Π^0_2) side of the bipartition. Its epistemic mode is twofold-negative: neither a finite refutation nor a finite confirmation is delivered by any prefix of σ. The same template, varied along the parameter space $(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \pi)$, instantiates the other named encoders (Edge / Schmitt / Crossing / Trio) and reaches the corresponding cells of the CLASP partition.

(a) Detection in V-mode: threshold is *interrupted* between activations (*not* moved); each ON pulse opens the gate and fires one validate at the gate-open tick $\to \mathsf{validate}^\omega \to \Sigma^\omega$ (trivial top).



(b) Detection in I-mode: same interruption pattern; each ON pulse opens the gate and fires one invalidate at the gate-open tick $\to \mathsf{invalidate}^\omega \to \varnothing$ (trivial bottom).



(c) Detection permanently OFF: threshold interrupted everywhere, never reactivated; no ON pulse ever placed on the signal, no contact event ever fires $\to \mathsf{L} \cap \mathsf{K}$ (non-monitorable, by silence).

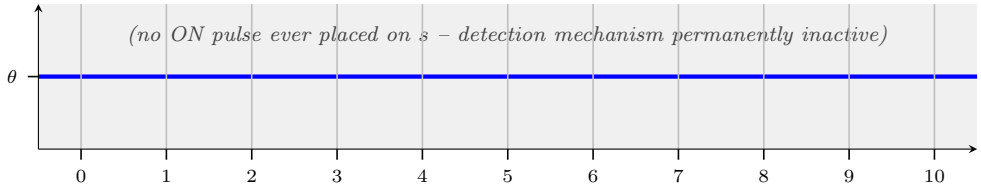


FIGURE 15. **The $|\mathbb{V}|=1$ trivial rung: discontinuous threshold model.** Input: constant $s(n) \equiv \theta$ (solid blue). Threshold has no mobile value-position; its only degree of freedom is the binary detection state, *interrupted* (not moved) between activations. ON pulses (orange segments) sit on the signal at level θ. Panel **(a)** V-mode: 11 pulses fire validate events (green triangles), realising $\mathsf{validate}^\omega \to \Sigma^\omega$. Panel **(b)** I-mode: same schedule fires invalidate (red), realising $\mathsf{invalidate}^\omega \to \varnothing$. Panel **(c)** detection permanently OFF: no pulse, no event, silent trace realising $\mathsf{L} \cap \mathsf{K}$ by silence. See Remark 11.1 for the gate/edge equivalence and the deferral to the ternary degenerate case; this reading is conjectural.

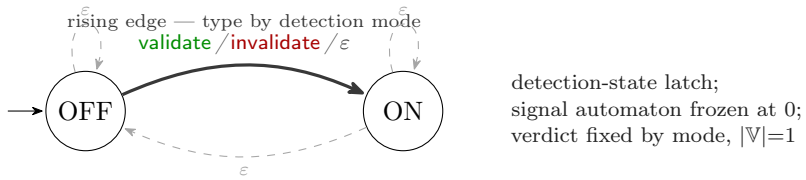


FIGURE 16. The n=1 (trivial-rung) monitor automaton of Remark 11.4: a two-state detection latch $\{\mathrm{OFF}, \mathrm{ON}\}$, initial OFF. Each rising edge OFF$\to$ ON fires exactly one event whose type is fixed by the detection *mode* — validate in V-mode (realising Σ^ω), invalidate in I-mode (realising $\varnothing$), or ε with detection permanently off (realising $\mathsf{L} \cap \mathsf{K}$ by silence). All other transitions are silent (gray dashed). The alphabet is mono-valued, $|\mathbb{V}|$=1: only one event type is ever emitted, the one fixed by the mode.

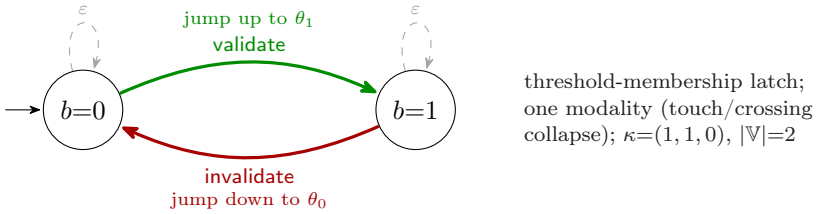


FIGURE 17. The n=2 (binary-rung) monitor automaton: a two-state threshold-membership latch $\{b{=}0, b{=}1\}$ on the two-line jumping signal (Proposition 11.5), where $b_n = [\tau_n = \theta_1]$, initial b=0. The only observable events are the threshold jumps: entering the upper level ($b\colon 0 \to 1$) fires validate, entering the lower level ($b\colon 1 \to 0$) fires invalidate; staying is silent (gray dashed). Touch and crossing *collapse* to this single membership modality — the signal never transits the gap (θ_0, θ_1) (no kinematic crossing) and is constant between jumps (no strict extremum). $\kappa = (1, 1, 0)$, $|\mathbb{V}| = 2$.

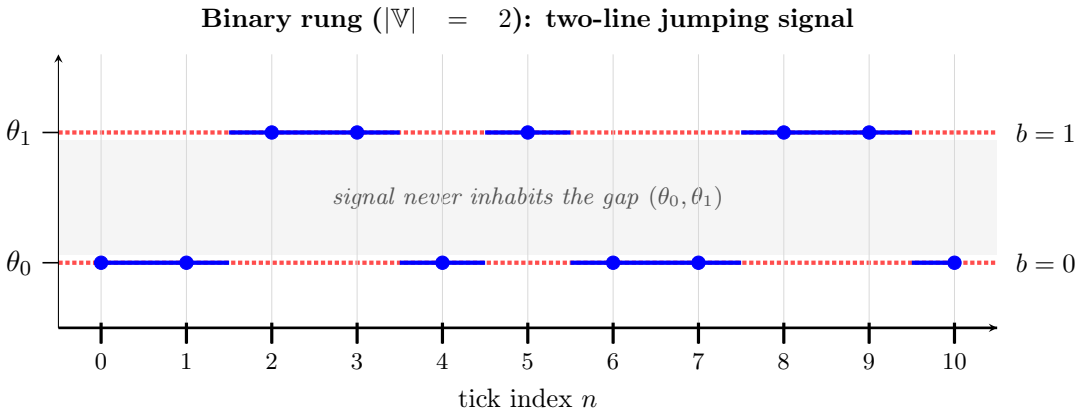


FIGURE 18. Binary rung ($|\mathbb{V}| = 2$) signal model. Two thresholds $\theta_0 < \theta_1$; the signal τ sits at one of the two threshold levels and may jump to the other at any tick boundary. The grey strip in the middle marks the *forbidden gap* (θ_0, θ_1), never inhabited by the signal: there is no continuous trajectory between the two levels and no intermediate sample. The verdict alphabet $\mathbb{V}_2 = \{0, 1\}$ records which threshold the signal is currently at via the predicate $b_n := [\tau_n = \theta_1]$. Touch and crossing collapse to the single *threshold-membership* modality $b_n = 1$ (Proposition 11.5).

13. Synthesis of tuning parameters for a target event pattern

Sections 4 to 10 establish what each modality *can* and *cannot* do. We now invert the perspective: given a fixed input signal Q and a *target event pattern* $E^* = [(t_1, \tau_1), \ldots, (t_n, \tau_n)]$ — where $t_i \in \mathbb{N}$ are sample positions and $\tau_i \in \{\mathsf{validate}, \mathsf{invalidate}\}$ — we ask which tuning parameters of the augmented monitor produce exactly E^* on Q.

This is a model-predictive-control synthesis problem. E^* plays the role of a reference trajectory; Q is the (uncontrollable) plant disturbance; the tuning parameters $\big(\theta_{\mathrm{u}}(t), \theta_{\mathrm{d}}(t), \pi\big)$ are the control variables; the plant model is the augmented monitor of Definition 10.2.

13.1. **Parameter-to-event control map.** Table 4 catalogues which dimension of the event stream each tuning parameter controls. The map is read column-wise: a given target property of E^* — its event times, its alternation pattern, its semantic classification — is achieved by tuning the corresponding parameter.

Tuning parameter	**What it controls**
θ (static)	The level at which the signal is monitored. Determines which excursions of the signal qualify (Model A only).
$\theta(t)$ piecewise (notches)	The *temporal positions* at which events fire. In Model A: a notch placed at t^* crossing $Q(t^*)$ in the right direction produces an event at t^*. In Model B: a wall notch at t^* produces an event only if Q has an extremum at t^*.
Gate $G(t)$	Suppression and reset of the monitor state. Used in Model A to decouple alternation: two validate events without an invalidate between them require gating.
Walls $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}})$	Which local extrema of Q qualify as events (Model B). Lowering θ_{u} enables more maxima as validations; raising θ_{d} enables more minima as invalidations.
Schedule $\pi \in \{\textsc{alt}, \textsc{free}\}$	Whether alternation between validations and invalidations is enforced. Switches the monitor between the Coincident/Schmitt family (alt) and the Extremum family (free).
Band boundaries	The semantic classification of events — which of Pre/X/Body/Post the event is reported under. Has no effect on firing.

Table 4. Mapping from tuning parameters of the augmented monitor to the dimension of control they exert over the event stream. The first three rows control *when* events fire; the next two control *which* events fire; the last controls *how events are classified*.

13.2. **Feasibility of synthesis.**

Definition 13.1 (Separable-contact pattern)**.** Fix a signal $Q : \mathbb{N} \to V$. A target event pattern $E^* = [(t_i, \tau_i)]_{i=1}^{n}$ is a *separable-contact* pattern for the moving-wall touch monitor when each event admits an isolated wall contact that fires it exactly once under the strict convention of Definition 5.1:

(i) *Interior, non-flat tick.* The strict-touch 3-window $(t_i{-}1, t_i, t_i{+}1)$ is defined (an interior tick), and t_i is not inside a constant segment of Q — a flat window $Q(t_i{-}1) = Q(t_i) = Q(t_i{+}1)$ is a *silent zone*, admitting no isolated strict contact.

(ii) *Temporal separability.* No two events of the *same* type fall on adjacent ticks: there is no pair with $t_j = t_i + 1$ and $\tau_i = \tau_j$ (the matching wall cannot both meet Q at one tick and clear it at the next).

Each such event is realised by moving the matching wall to meet $Q(t_i)$ — θ_{u} down to $Q(t_i)$ for validate, θ_{d} up to $Q(t_i)$ for invalidate — at *any* qualifying t_i (extremal or not), the type being fixed by which wall moves, not by the signal's slope.

Proposition 13.2 (Feasibility of synthesis)**.** *Let $Q : \mathbb{N} \to V$ be a signal and $E^* = [(t_i, \tau_i)]_{i=1}^n$ a target event pattern. Adopt the moving-threshold operational rule of Section 9.1: an event fires at a horizontal intersection of the (piecewise) threshold/wall with the signal, with type determined by the signal's direction at the intersection in Model A and by which wall moves into the signal in Model B.*

(1) *For the crossing-native monitor (Model A), E^* is realisable if and only if every event type matches the signal's local direction at the target time: $\tau_i =$ validate requires Q to be ascending at t_i; $\tau_i =$ invalidate requires Q to be descending at t_i.*

(2) *For the touch-native monitor (Model B), with moving walls, E^* is realisable if and only if it is a* separable-contact *pattern (Definition 13.1). Because the type is fixed by which wall moves rather than by the signal's slope, events at* any *tick (extremal or not) are reachable; the only obstructions are constant-segment silent zones and two same-type events at adjacent ticks (see the remark after Corollary 13.3).*

Proof sketch. (1) The signal-direction rule fixes the event type by the sign of $Q(t_i){-}Q(t_i{-}1)$. No threshold manipulation in Model A can override this rule: with a single threshold, the only way to fire an event at t_i is to make θ cross Q at t_i, and the direction of that crossing is dictated by the signal's slope. Hence E^* is feasible iff every τ_i agrees with the local sign of the discrete derivative.

(2) Model B has *two* thresholds, and the type of event is no longer tied to signal direction but to *which wall* moves into the signal. To produce $\tau_i =$ validate at *any* target t_i, set $\theta_{\mathrm{u}}(t_i) = Q(t_i)$ on a brief interval around t_i, while keeping θ_{u} outside Q's range elsewhere; the descending wall meets Q at t_i regardless of the signal's slope there, firing one validate contact. Similarly for invalidate via θ_{d} rising to $Q(t_i)$. The signal's direction is irrelevant. The construction needs the contacts to be *separable* — the wall must clear Q's range between consecutive events — so it fails for a constant segment (no extremum to meet) and for two same-type events at adjacent ticks (the wall would have to be both on Q at t_i and above it at $t_i - 1$). Those targets lie outside wall motion and require modifying the signal itself ($\mathcal{L}_3$ of Proposition 8.1). □

Corollary 13.3 (Asymmetric power, inverted)**.** *For a fixed signal Q, the crossing-native monitor (Model A) is constrained by the signal-direction rule and can realise only target patterns whose event types match the signal's local slope at the target times. The touch-native monitor (Model B), through movable walls, decouples event type from signal direction and realises every* separable-contact *target (events the wall can meet one at a time at any tick, extremal or not; Proposition 13.2(2)). The set of realisable targets is therefore strictly larger for Model B than for Model A under the moving-threshold rule — though still short of the free language: the full set of verdict traces is reached only by also modifying the signal ($\mathcal{L}_3$*

of Proposition 8.1). The asymmetry runs opposite to what a naive reading of the strict-touch semantics of Definition 5.1 would suggest.

Remark 13.4 (Where the generic monitor is attained)**.** The *generic* 3*-valued monitor* at the centre of Figure 6 — one state with a self-loop for each verdict — emits, in the abstract, the *full free language* over $\{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$. Which tuning powers attain it is settled by the layer hierarchy of Proposition 8.1, $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \mathcal{L}_3$. Moving-wall tuning — the only language-extending operation in $\mathcal{L}_1$, the semantic-aggregation tunings being firing-neutral — realises only the *separable-contact* sublanguage of Corollary 13.3: on a fixed signal it cannot fire on a constant segment (a silent zone), nor place two same-type events at adjacent ticks. Tick-level copy-paste ($\mathcal{L}_2$) enlarges the reach but stays within the values Q already contains. The free language — the generic monitor itself — is attained only at $\mathcal{L}_3$, where the tuner is also allowed to *modify the signal* (Appendix D), injecting the extrema that the wall then meets. Within $\mathcal{L}_1$ the synthesis algorithm of Section 13.3 computes, in $O(n+T)$, the wall motion reaching any separable-contact target. The attainment is thus regime-specific: the separable-contact sublanguage is reachable (and proved so in Section 13.3) under moving walls, while the free language is attained only under the signal-modifying regime $\mathcal{L}_3$, which lies outside pure monitor tuning. Under the single-threshold crossing monitor the reachable targets are further constrained to the signal's local slope (Proposition 13.2(1)).

Remark 13.5 (Strict vs. moving semantics)**.** Proposition 13.2 adopts the moving-threshold rule throughout. Under the *strict* touch semantics of Definition 5.1, where touch events fire only at local extrema of Q, Model B is instead more restricted than Model A: targets must coincide with extrema. The two asymmetries — A-constrained-by-direction versus B-constrained-by-extrema — apply under different operational rules and, informally, characterise different regimes of monitor use.

13.3. **Synthesis algorithm and proof-of-concept.** The construction in the proof above lifts to an explicit algorithm that runs in linear time $O(n+T)$ in the target size $n = |E^*|$ and signal length $T = |Q|$ (Proposition 13.6 below). Given (Q, E^*, model), the algorithm either emits a tuning parameter assignment that produces E^* on Q, or — for Model A on a target whose type conflicts with the signal's local slope, or for Model B on a target with a constant-segment (silent-zone) tick or two same-type events at adjacent ticks — emits an INFEASIBLE verdict identifying the offending (t_i, τ_i). The output is compatible with the parametric LaTeX `\def` convention introduced for Figure 22: the synthesis script writes a parameter file that LaTeX can `\input` directly, closing the loop between target specification and rendered figure. The synthesiser is *offline* — it sees all of Q — so it realises the target *timestamps* exactly; the online monitor that later consumes the synthesised tuning still incurs the touch detector's +1 anti-causal decision latency (Section 7) at runtime.

A proof-of-concept implementation is provided in `tools_pocs/monitor_synth/` (Python 3.13, standard library only). It implements both models faithfully — Model A enforces the signal-direction rule of Proposition 13.2(1), Model B the moving-wall separable-contact rule of Proposition 13.2(2) — and ships with a test suite covering: (i) empty targets; (ii) single events (Model A slope-constrained, Model B free at any non-flat interior tick); (iii) extremum-aligned targets (both succeed); (iv) a type-vs-slope mismatch (Model A INFEASIBLE, Model B succeeds — the inverted asymmetry of Corollary 13.3); (v) the genuine Model-B failure modes (constant-segment silent zones, adjacent same-type events,

endpoints); and (vi) random interior targets (Model A usually INFEASIBLE, Model B usually feasible). The asymmetric power of Corollary 13.3 is thus made operational and verifiable: Model B realises the non-extremal wall-caused contacts of Figure 19, while Model A is locked to the signal's slope.

Proposition 13.6 (Synthesis complexity)**.** *For input* (Q, E^*, model) *with* $|E^*| = n$ *target events and a signal of length* $|Q| = T$*, the synthesis algorithm of Section 13.3 runs in time* $O(n+T)$ *and produces a tuning-parameter assignment of size* $O(n)$*:*

- ***Model A.*** *Linear scan through* E^**; at each* (t_i, τ_i) *the algorithm checks the signal-direction rule* $\operatorname{sgn}(Q(t_i) - Q(t_i - 1))$ *against* τ_i *—* $O(1)$ *per event — and either appends a threshold notch* $(t_i, Q(t_i) \mp \varepsilon)$ *to the output tuning, or returns* INFEASIBLE*. Total work* $O(n)$ *on* E^* *plus* $O(T)$ *to precompute the discrete derivative; total* $O(n+T)$*.*
- ***Model B.*** *Linear scan through* E^**; at each* (t_i, τ_i) *the algorithm appends a wall notch on the corresponding wall at level* $Q(t_i)$ *—* $O(1)$ *per event. For a* separable-contact *target (Proposition 13.2(2)) the scan always succeeds in* $O(n)$ *work plus the* $O(T)$ *signal read; for a target on a constant segment, or with two same-type events at adjacent ticks, it returns* INFEASIBLE *at the offending* (t_i, τ_i) *(such targets require signal modification,* $\mathcal{L}_3$ *of Proposition 8.1).*

The output tuning has $O(n)$ *notches; each notch carries* $O(1)$ *scalars (a time index, a target level, optionally a width). No combinatorial search is performed, and the algorithm is on-line: it can stream* E^* *in arrival order and emit notches as they are needed.*

Proof sketch. The construction in the proof of Proposition 13.2 is purely local: each (t_i, τ_i) is realised by an independent threshold/wall notch at t_i, with no inter-event coupling. The total number of notches is therefore $|E^*| = n$, the per-event work is $O(1)$ after an $O(T)$ pre-pass over Q (or $O(1)$ amortised on streamed input), and no global feasibility decision is required: Model A's infeasibility is detected event-locally by the signal-direction check. □

13.4. **Natural event languages and equivalent techniques.** The synthesis problem of Section 13.2 is defined against the *natural* output language of each monitor — the event stream emitted when no tuning intervenes beyond fixing nominal threshold or wall values. Characterising this natural language clarifies what tuning is being asked to escape from.

Remark 13.7 (Natural language of M_C)**.** Fix a threshold θ and a signal Q. Because M_C maintains a one-bit sticky state s that flips at every crossing, every event emission is followed by an event of the *opposite* type before another emission of the same type can occur. The natural language of M_C over the alphabet $\Sigma = \{\mathsf{validate}, \mathsf{invalidate}\}$ is therefore the regular language of *strictly alternating* strings,

$$L_A^{\mathrm{nat}} \;=\; \varepsilon \cup V(IV)^*(I \cup \varepsilon) \cup I(VI)^*(V \cup \varepsilon),$$

recognised by the two-state automaton $\{S_\downarrow, S_\uparrow\}$ with transitions $S_\downarrow \xrightarrow{\mathsf{validate}} S_\uparrow$ and $S_\uparrow \xrightarrow{\mathsf{invalidate}} S_\downarrow$. Crucially, L_A^{nat} is determined by the *monitor*, not by Q.

Remark 13.8 (Natural language of M_T)**.** Fix walls $\theta_{\mathrm{u}}, \theta_{\mathrm{d}}$ and a signal Q. The natural output of M_T is the encoding of the sequence of extrema of Q that satisfy the wall constraints ($Validate$ for a local maximum at value $\geq \theta_{\mathrm{u}}$, $\mathsf{invalidate}$ for a local minimum at value $\leq \theta_{\mathrm{d}}$):

$$L_B^{\mathrm{nat}}(Q) \;=\; \mathrm{enc}\big(\mathrm{extrema}(Q)\big).$$

Unlike L_A^{nat}, this language depends on the *signal*, not on the monitor: a signal with two consecutive maxima (no intervening minimum at value $\leq \theta_{\mathrm{d}}$) yields VV in the natural language of M_T. Strict alternation is a property of the signal's extremum sequence, not an intrinsic constraint of M_T.

Escape from L_A^{nat}. To emit an event stream outside L_A^{nat} — e.g., two consecutive validations without an intervening invalidation — the tuning algorithm must suppress one of the two predicted events of the alternation. Two operationally equivalent techniques are available:

(a) *Threshold deactivation.* Set the gate $G(t)$ to zero on an interval $I \subset \mathbb{R}$. During I no event fires and the state is frozen; when G returns to one the state is reset to 0.
(b) *Threshold relocation beyond signal range.* Set $\theta(t)$, on the same interval I, to a value outside $[\min_{t\in I} Q(t), \max_{t\in I} Q(t)]$ — e.g., a value above $\max Q$. The signal cannot cross the threshold inside I, so the state remains pinned and no event fires.

Proposition 13.9 (Gate-threshold equivalence for M_C)**.** *For every gate schedule $G : \mathbb{N} \to \{0, 1\}$ there exists a piecewise-constant threshold $\theta_G(t)$ such that the event stream of $M_C^{\theta_G}$ on any signal Q coincides with the event stream of the gated crossing monitor with gate G and nominal threshold θ. The converse holds analogously. Techniques (a) and (b) above are therefore operationally equivalent modulo reparameterisation.*

Proof sketch. Given G, define $\theta_G(t) = +\infty$ wherever $G(t) = 0$ (in practice, any value strictly above $\max_t Q(t)$ over the gate-off interval suffices); set $\theta_G(t) = \theta$ elsewhere. Where $\theta_G = +\infty$, no crossing of θ_G by Q is possible, so no event fires — matching the gate-off semantics. Conversely, given a piecewise threshold $\theta(t)$ whose values periodically lie outside Q's range, define $G(t) = 0$ wherever $\theta(t)$ is so positioned. The two event streams coincide. □

Suppression in L_B^{nat}. Symmetrically, in Model B suppression of a naturally occurring invalidation (a touch at the lower wall) admits two operationally equivalent techniques:

(a) *Lower-wall deactivation.* Disable θ_{d} on an interval I — equivalently, set $\theta_{\mathrm{d}}(t) \to -\infty$ during I. The invalidate predicate cannot fire because no value of Q satisfies $Q(t) \leq -\infty$.
(b) *Lower-wall relocation below signal minimum.* Set $\theta_{\mathrm{d}}(t)$ to a value strictly below $\min_{t\in I} Q(t)$ during I. The signal's local minima inside I cannot touch the wall, so invalidate does not fire.

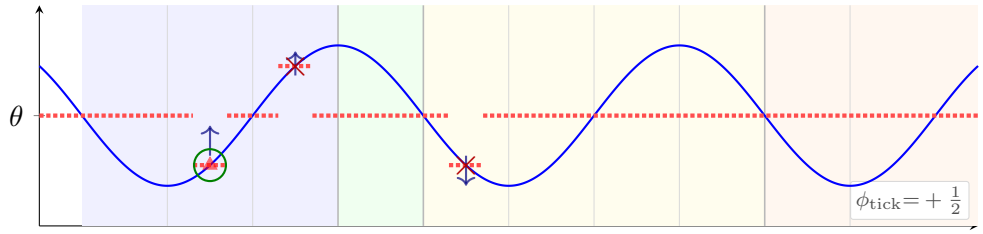
The dual statement holds for suppressing a natural validation via the upper wall. The two techniques produce the same event stream and are operationally equivalent, exactly as in M_C.

Parameter parsimony. The equivalences just stated have a structural consequence: the gate $G(t)$ is *not* a structurally necessary parameter of the augmented monitor of Definition 10.2, provided $\theta_{\mathrm{u}}(t)$ and $\theta_{\mathrm{d}}(t)$ are allowed to range over $\mathbb{R}\cup\{\pm\infty\}$ as piecewise-constant functions. The calibration algebra $\langle T_1, \ldots, T_9, T_{10}, \mathcal{E}\rangle$ introduced in earlier work admits a presentation $\langle T_1, \ldots, T_9, \mathcal{E}\rangle$ in which the T_{10} gating role is absorbed into the threshold/wall schedule. This minimal presentation is closer to the single-parameter family $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ of the augmented monitor.

Figure 19 illustrates a concrete synthesis challenge under the moving-threshold rule: the same target E^* is presented to both models on the same signal Q. Under this rule Model A is infeasible because two of the target events conflict with the signal's local direction; Model B

is feasible because two walls decouple event type from signal direction. Note that the tick frame in Figure 19 is phase-shifted by $\phi_{\text{tick}} = +1/2$ relative to the other figures: tick centres sit at half-integer times so that the target events at $t = 1.5, 2.5, 4.5$ fall *inside* tick windows rather than on tick boundaries. This phase is itself a tuning parameter of the augmented monitor and controls the alignment between event times and tick windows.

(a) Model A: INFEASIBLE — **two targets conflict with the signal-direction rule**



(b) Model B: FEASIBLE **via moving walls — all three targets are realised**

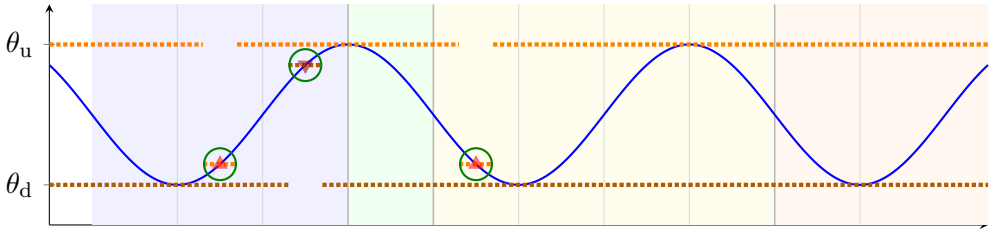


FIGURE 19. Synthesis challenge under the moving-threshold operational rule. Both panels share the same input signal $Q(t) = 0.5 + 0.35\cos(2\pi(t - 3)/4)$ and the same target event pattern $E^* = [(1.5, \mathsf{validate}), (2.5, \mathsf{invalidate}), (4.5, \mathsf{validate})]$. **(a)** Model A is INFEASIBLE. The signal-direction rule forces event type to match the signal's slope at the target time. Small blue arrows indicate the signal's direction at each target: at $t = 1.5$ the signal is ascending so validate is consistent (✓), but at $t = 2.5$ the signal is still ascending — so the target invalidate is impossible (×); similarly at $t = 4.5$ the signal is descending and the target validate is impossible (×). No threshold motion can override the signal-direction constraint. **(b)** Model B is FEASIBLE. With two walls, the event type is determined by *which* wall moves into the signal, not by signal direction. Three wall notches realise E^*: θ_{u} drops to $Q(1.5)$ and $Q(4.5)$ to fire two validate events, while θ_{d} rises to $Q(2.5)$ — even though the signal is ascending there — to fire invalidate. The asymmetric power of Corollary 13.3: Model B's two walls decouple event type from signal direction; Model A's single threshold does not. Both panels use the phase-shifted tick frame $\phi_{\text{tick}} = +1/2$ (tick centres at half-integers), making explicit that the target events sit at *tick centres*: alignment between event times and tick windows is itself part of the tuning, controlled by ϕ_{tick}.

13.5. **CLASP classification of the monitor's predicates.** We position the modalities within the Safety / Cosafety / Liveness / Coliveness partition of LTL properties of [3], which refines the safety–liveness dichotomy [1] and is orthogonal to the Manna–Pnueli hierarchy [12]. This section *consumes* the CLASP partition as an external classification target; it does not re-derive it. The partition, its ω-regular foundations, and the verdict-alphabet ladder it indexes are developed in [3] (self-contained sketches are given in Remarks 13.12 and 13.13). What is new here is solely the *placement* of the augmented-monitor firing predicates inside that fixed partition — the event-generation question studied here, as opposed to the verdict-classification question of [3].

Proposition 13.10 (CLASP class of single events)**.** *The instantaneous predicate "the augmented monitor emits* validate *at position n" is in* Co-safety *for every parameter choice* $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$.

Proof sketch. Detection of $\mathsf{validate}(\theta_{\mathrm{u}})$ requires a 3-window of the value trace plus a comparison to θ_{u}; both are decidable on a finite prefix. Under ALT, the state s is recoverable from the bounded history of previous events, which is itself finite. The predicate is therefore in Co-safety — a finite witness suffices for affirmative resolution. □

Proposition 13.11 (CLASP class of unbounded balanced sequences; infinite-trace)**.** *Under* ALT*, the trace-level predicate "the monitor emits infinitely many* validate *events, each followed by an* invalidate *event" is in* Mixed *— specifically the recurrence cell* $\mathsf{L} \cap \mathsf{K}$ *(*GF*, the* Π^0_2 *side), not the cosafety–liveness cell* $\mathsf{C} \cap \mathsf{L}$*. This classification follows directly from the class definitions of* [3].

Proof sketch (finite-witness ascent). Each validate/invalidate pair is Cosafety witnessable on a finite prefix (Proposition 13.10). The trace-level claim involves a G-quantified condition over the infinite trace and is therefore an *infinite-trace claim* in the sense of the finite/infinite-trace scope of Section 2: no finite prefix determines the infinite suffix tail, and the lift from finite-pair Cosafety to ω-balanced Mixed is justified against the CLASP cell classification of [3]. The argument is: each pair is Co-safety on its finite witness; the conjunction of all such pairs is G-quantified, forcing a Liveness commitment (an infinite sequence of balanced pairs must occur). The prefix-Co-safety witnessing lifts each pair, but the *trace-level* GF recurrence predicate ("infinitely many validate, each followed by an invalidate") is boundary-dense on both sides: it lands in the canonical Mixed cell $\mathsf{L} \cap \mathsf{K}$ (the Π^0_2 recurrence side), not in $\mathsf{C} \cap \mathsf{L}$. Bragetti [3] classifies Φ_{GFp} in $\mathsf{L} \cap \mathsf{K} \setminus (\mathsf{S} \cup \mathsf{C})$, consistent with the $\sin t$ pipeline classification of Figure 14 above. Directly: every finite prefix extends to a satisfying trace (append $(\mathsf{validateinvalidate})^\omega$) and to a violating one (append ε^ω), so the predicate is dense and co-dense, i.e. in $\mathsf{L} \cap \mathsf{K}$; having neither a good nor a bad prefix, it is in neither S nor C. □

Remark 13.12 (Finite-witness lift sketch)**.** The finite-to-infinite reading used above can be summarised in three lines (the summary is ours): (i) split the trace domain V^ω into a finite prefix $V^{\leq T}$ and the infinite suffix (T, ∞) on which the firing predicate is not yet decidable; (ii) verify the property on the prefix using only finite-witness machinery (here: the Co-safety classification of each Validate/Invalidate pair); (iii) lift to the full trace by a G-quantified statement *over the prefix-length parameter* T, which produces the Liveness commitment. Each pair is thus prefix-Co-safety witnessable while the lifted GF recurrence predicate is boundary-dense on both sides; its canonical Mixed location in the CLASP partition is therefore the cell $\mathsf{L} \cap \mathsf{K}$ (the Π^0_2 recurrence side), not $\mathsf{C} \cap \mathsf{L}$.

Remark 13.13 (Self-contained CLASP partition sketch)**.** The four generating classes of the partition are Safety (Safety), Co-safety (Cosafety), Liveness (Liveness), and Coliveness, abbreviated S, C, L, K in [3] (which owns the construction); the four labels read off the (finite-witness-for-violation, finite-witness-for-affirmation) pair: Safety = every violation has a bad prefix; Co-safety = every satisfaction has a good prefix; Liveness = no bad prefix (dense); Coliveness K = no good prefix (empty interior). The intersection cells generated by these four classes form the algebra used here; the cell that this paper labels Mixed is the boundary-dense cell $\mathsf{L} \cap \mathsf{K}$. The lift *mechanism* is a prefix-Co-safety witness on each $\mathsf{validate}/\mathsf{invalidate}$ pair raised by a G-quantification to a Liveness commitment (Proposition 13.11); the resulting trace-level GF recurrence predicate is, however, boundary-dense on both sides. Its *cell identity* is therefore the boundary-dense $\mathsf{L} \cap \mathsf{K}$ (Π^0_2 recurrence side), *not* $\mathsf{C} \cap \mathsf{L}$: in [3] the properties whose boundary is the whole space are exactly those in $\mathsf{L} \cap \mathsf{K}$. We use Mixed as the local name for that placement and do not re-derive its structure. The labelling is the canonical form used in [3]. No claim in this paper requires more than this four-class algebra; the finer Manna–Pnueli [12] hierarchy (with its distinct *Persistence* / *Recurrence* Borel levels, which are orthogonal to the $\mathsf{S}/\mathsf{C}/\mathsf{L}/\mathsf{K}$ algebra) is invoked only as cited related work.

The two propositions place the augmented monitor's specifications within the existing CLASP framework: no new fragment of LTL is required.

The synthesis algorithm of this section takes a *finite* target event pattern E^* and emits a static tuning-parameter assignment that produces E^* on the given signal. A strict generalisation, in which E^* is replaced by an LTL property on the verdict trace $E_M(\sigma) \in V^\omega$ and the tuning parameters evolve as a function of the current parameter point and the trace prefix emitted so far, turns the monitor into an active controller that drives the verdict trace toward satisfaction of the target. This *enforcement-via-parametric-adaptation* mechanism — structurally distinct from the truncation-based enforcement of Schneider's security automata and the edit-based enforcement monitors of [6] — sits outside the scope of the present paper. Its systematic development, and the formal analysis of its convergence guarantees, is left to future work.

14. Robustness, noise, and quantitative margins

The development so far has assumed a noiseless observation $Q : \mathbb{N} \to V$. Real sensor traces carry additive or multiplicative noise, and a useful monitor must remain well-defined under small perturbations. We sketch here how the two modalities differ in robustness, how the augmented monitor inherits a quantitative robustness semantics from the literature of Signal Temporal Logic [5], whose foundational robust-semantics formulation for continuous-time signals is due to Fainekos and Pappas [38], and how the hysteresis construction of the Schmitt trigger [14] appears naturally as a robustness-restoring tuning of the augmented monitor.

Epistemic scope of this section. The robustness margins introduced below are presented as *informal sketches*, not as formally proved quantitative bounds. The linear-vs-quadratic separation of ρ_A vs ρ_B in the neighbourhood of a firing is a generic-case statement that uses standard parabolic local-extremum modelling without a precise genericity constraint. A fully formal treatment with explicit genericity hypotheses, noise-model assumptions, and quantitative bounds is left to future work. The claims in this section are heuristic unless

stated otherwise; the section is intended to position the two modalities in the existing STL robustness landscape, not to supply new quantitative results.

Two natural margins. Each modality admits a canonical *robustness margin* on its native primitive:

$$\rho^{\theta}_{\mathrm{A}}(Q,n) \;:=\; |Q(n)-\theta|, \qquad \rho^{\theta}_{\mathrm{B}}(Q,n) \;:=\; \min\big(\theta_{\mathrm{u}} - Q_{\max}(n),\, Q_{\min}(n) - \theta_{\mathrm{d}}\big),$$

where $Q_{\max}(n) := \max\{Q(n-1), Q(n), Q(n+1)\}$ and $Q_{\min}(n)$ symmetrically. ρ_{A} measures how far the signal is from crossing the threshold at sample n, and ρ_{B} how far the local extremum is from touching either wall. A noise floor of magnitude $\eta < \rho^{\theta}(Q,n)$ cannot flip the event at n in the corresponding modality: the strict inequality $|\eta| < \rho^{\theta}_{\mathrm{A}}(Q,n)$ keeps the sign of $Q-\theta$ stable under perturbation η, and the strict inequality $|\eta| < \rho^{\theta}_{\mathrm{B}}(Q,n)$ keeps every extremum locally bounded away from both walls. In the language of [5], ρ_{A} is the standard STL robustness of the atomic predicate $Q \geq \theta$; ρ_{B} has no STL standard counterpart and is a contribution of the touch modality.

Asymmetric noise sensitivity. The two margins behave differently as Q approaches a firing condition. Near a crossing of θ in Model A, $\rho_{\mathrm{A}} \to 0$ *linearly* in the displacement of Q from the threshold: the monitor is maximally sensitive at exactly the firing time. Near a touch of θ_{u} in Model B, $\rho_{\mathrm{B}} \to 0$ *quadratically* (in the generic case of a strict local maximum, where Q is locally parabolic around the peak): the monitor is less sensitive to noise in a neighbourhood of the firing time. This is the formal counterpart of the engineering intuition that extremum-based detectors are noise-tolerant relative to threshold-crossing detectors.

Hysteresis as a robustness tuning. The Schmitt-trigger remedy for noise-induced multiple firings is the classical *hysteresis* construction: replace the single threshold θ of Model A by two thresholds $\theta^{+} > \theta^{-}$ and require an upward transit through θ^{+} to fire validate and a downward transit through θ^{-} to fire invalidate. Under Definition 10.2 this is the parameter point $\theta_{\mathrm{u}} := \theta^{+}$, $\theta_{\mathrm{d}} := \theta^{-}$, $\pi :=$ ALT, i.e. the SCHMITT parameter point of Proposition 10.5 (non-coincident walls, alternation enforced). The hysteresis gap $h := \theta^{+} - \theta^{-}$ raises the effective Model A margin to $\rho^{\theta^{\pm}}_{\mathrm{A}}(Q,n) \geq h/2$ in the band $[\theta^{-}, \theta^{+}]$, restoring noise immunity at the cost of an event-pair latency proportional to $h/|\dot{Q}|$. The augmented monitor therefore *contains* the hysteresis construction as a single parameter choice; the duality framework recovers the Schmitt trigger as the natural intermediate point between the two minimal modalities.

Robustness under signal modification and copy-paste. The signal-modification operations of Appendix D and the copy-paste operations of Appendix C introduce step discontinuities in Q. Under noise of magnitude η, a step jump ΔQ at a splice boundary is detectable only if $|\Delta Q| > 2\eta$ (so that the post-noise samples on either side remain on the correct side of the wall); below this bound the step is indistinguishable from a noise excursion and the tuning algorithm's intended event may fail to fire. The continuity-preserving copy-paste of Appendix C.2 is therefore not only aesthetically cleaner but also *noise-robust* in this concrete sense: its zero step size is below any positive noise floor. Direction-coherent splices preserve only the sign of the local slope, not the magnitude of any step, so their noise-robustness lies strictly between the discontinuous and the continuity-preserving regimes.

15. Discussion and connections

We close by placing the dual-modality framework in the wider runtime-verification, STL, and automata-theoretic landscape.

Implication monitoring. The augmented monitor framework provides the encoder layer for temporal-implication calibration: a companion analysis [2] partitions bounded observations of temporal implication into eight observation (window) classes and shows that tuning moves the observed trace between them; in the present framework such tuning is motion in the parameter space $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$. Operations that the original tuning algebra treats as separate (threshold shift, hysteresis adjustment, encoding switch from edge to extremum) collapse into parametric paths in this space.

Extremum-seeking loops and the Schmitt tradition. The duality between touch and crossing makes precise an analogy that has remained informal: extremum-seeking loops (touch-native) and Schmitt triggers (crossing-native) are not opposed engineering paradigms but dual parametric specialisations of a single monitor architecture. The transformations of Sections 9.1 and 9.2 are the cost of moving between the two sensing regimes — a cost that is real (an extra bit of state and a finite-state rule, or an ascent detector and a reset hook) but symmetric.

Quantitative monitoring and STL.. Signal Temporal Logic [10] encodes atomic propositions as level comparisons, placing it natively in the crossing regime; the quantitative robustness semantics of STL [5] adds a real-valued margin to each crossing event. The touch regime is closer in spirit to *extremum-seeking control* [7]: it treats local maxima as the elementary unit of information. The augmented monitor framework subsumes both as parameter choices.

Algebraic perspective (informal forward pointer). The framework admits a compact reading in the language of abstract algebra and category theory which we sketch here; a detailed development is left to future work. The paragraphs below are *informal*: they record the categorical reading as a forward pointer rather than as a formal result of this paper. No claim in Sections 4, 5, 10, 10.2, 13, 13.5 and 14 depends on the 2-categorical machinery sketched here; a reader uninterested in the category-theoretic packaging can skip directly to Section 16 without loss of content. The claims in the sub-paragraphs below are heuristic unless stated otherwise.

The monitor as a categorical morphism. Each monitor $M : V^{\mathbb{N}} \to \Sigma^{\mathbb{N}}$ is a causal Mealy function and therefore a morphism in a 2-category **Mon** whose objects are topologised alphabets, whose 1-morphisms are causal bounded-state translators, and whose 2-morphisms are the simulation preorder $M \preceq M'$ of Conjecture 10.12. The join statement of Theorem 10.14 is then literally a categorical *coproduct* (least upper bound) in **Mon**: $M^{(4)} = M_T \sqcup M_C$. The $n = 4$ extension is the *free completion* of $\{M_T, M_C\}$ in the 4-threshold sub-2-category.

Ladder as stratification with increasing algebraic structure. The verdict alphabets along the rung $|\mathbb{V}| = 1, 2, 3, 4$ form a stratification: $\Sigma_1 = \{*\}$ (trivial monoid), $\Sigma_2 \cong \mathbb{Z}/2\mathbb{Z}$ (binary group), Σ_3 is the three-valued Heyting algebra of LTL_3 [4], and Σ_4 is the four-element chain $\bot < \bot_p < \top_p < \top$ of RV-LTL [19]. The information content per tick is $\log_2 n$ bits; the jump $n = 2 \to n = 3$ (+0.58 bit per tick) is precisely the information required to separate the touch and crossing modalities at the focus rung.

Output as element of Cantor space. The event stream $\sigma \in \Sigma^{\omega}$ inhabits the Cantor space (compact, totally disconnected, perfect). Monitor properties $\Phi \subseteq \Sigma^{\omega}$ inherit the Borel

hierarchy classification Σ^0_n, Π^0_n; the CLASP partition of [3] is orthogonal to this classification: it partitions properties by topological character (open/closed, dense/co-dense) rather than by Borel level, and its cell $\mathsf{L} \cap \mathsf{K}$ contains both Π^0_2 (GFp) and Σ^0_2 (FGp) properties. The Manna–Pnueli hierarchy [12] is the Borel-side classification of temporal properties: safety, guarantee, recurrence and persistence correspond to the closed, open, G_δ and F_σ sets.

The duality theorem as capability join. Touch and crossing are not Stone-dual: there is no Boolean negation mapping one to the other. They are instead *capability-dual*: incomparable join-irreducible points in the simulation preorder on **Mon**, whose join exists and is realised by the parameter family $M^{(4)}$. This is structurally analogous to the Joyal–Wraith free completion of incomparable objects in 2-categories.

Cross-disciplinary readings. Peak-detection vs. level-crossing detection is a familiar dichotomy in discrete-time signal processing (see, e.g., [11]). In control theory the two modalities map respectively to extremum-seeking control [7] and switching/hysteresis control (Schmitt [14]). The algebraic runtime-verification framework of Jakšić et al. [24] uses the monoidal structure of runs and the semiring structure of a specification automaton but does not formulate the monitor itself as a categorical morphism. The explicit identification of touch and crossing as capability-dual join-irreducibles in a 2-category of monitors appears to be new here; its full development is left to future work.

Future work. A natural next step is to implement the augmented monitor as a proof-of-concept (validated by the witnessed PoC of Section 16) and extend it with adaptive (time-varying) $\theta_{\mathrm{u}}, \theta_{\mathrm{d}}$, recovering the Robbins–Monro adaptive-threshold scheme. A second direction is to characterise the lattice of intermediate parameter points between Coincident, Schmitt, Extremum, identifying which produce useful non-classical encoders. A third direction is to extend the algebraic perspective above into a full categorical treatment: the 2-categorical structure of **Mon**, the colimit interpretation of the $n \to \infty$ limit of the verdict-alphabet ladder, and the topos-theoretic interpretation of the rising-edge event-firing rule via sheaves over the time-poset.

16. Conclusion

Summary of contributions. Touch and crossing are two basic modalities of validation in threshold monitors. The paper's contributions, each a referenceable claim with its defining result, are the following.

- **C1. Dual-primitive duality.** Touch and crossing are dual primitives: neither is structurally contained in the other, and their capability triples $\kappa(M_C) = (1, 1, 0)$ and $\kappa(M_T) = (3, 0, 1)$ (Definition 2.4) are incomparable. Realised by the touch-as-crossing duality on plateaus (Theorem E.1) and the dual augmentations of Sections 9.1 and 9.2.
- **C2. Pointwise count inequality.** The crossing and touch validation counts satisfy $N_X \leq N_T$ on every signal, with a characterised equality case (Proposition 6.1).
- **C3. Continuous-to-discrete sampling correspondence.** Continuous-signal monitoring (STL-native crossing) and the discrete runtime-verification event stream are two samplings of one signal; the touch/crossing duality is the invariant of the map and the sub-tick events are its explicit excluded interval (Proposition 2.10).
- **C4. Augmented monitor and classical encoders.** The augmented monitor $M(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi)$ (Definition 10.2) is the n=3 touch-only family; the three classical

encoders (edge detector, Schmitt trigger, extremum detector) are its named parameter points COINCIDENT, SCHMITT, EXTREMUM (Proposition 10.5), and the ternary rung is closed under threshold tuning (Theorem 10.10).

C5. Quaternary join. The quaternary monitor $M^{(4)}(\theta_{\mathrm{u}}, \theta_{\mathrm{cross}}, \theta_{\mathrm{d}}, \pi)$ (Definition 10.13) realises M_T and M_C as exact parameter points, settling the touch–crossing join as a literal parameter coincidence at $|\mathbb{V}|$=4 (Theorem 10.14), with the classical encoders recovered on the n=4 cube (Corollary 10.6).

C6. CLASP classification. The validation predicate is Cosafety at the instant level (Proposition 13.10); under ALT alternation the unbounded balanced trace-property is Mixed (Proposition 13.11).

A seventh claim is conjectural and kept separate from the established contributions above:

K1. Universal-join conjecture. The augmented monitor is the universal join (least upper bound in the simulation preorder) of M_T and M_C (Conjecture 10.12).

The modality landscape extends beyond the focus rung $|\mathbb{V}| = 3$. We located touch and crossing on a *modality ladder* indexed by the verdict-alphabet cardinality $|\mathbb{V}| \in \{1, 2, 3, 4\}$ (Section 3): the trivial rung is the modality-free baseline (constant signal with a binary detection state, no touch/crossing decomposition); the binary rung is degenerate (touch and crossing collapse to threshold-membership); the ternary rung is the touch/crossing duality (the focus of this paper); and the quaternary rung hosts a 2×2 modality lattice along orthogonal touch/crossing and weak/strong axes, the two degenerate rungs being developed in Section 11. This cardinality view is analogous to the verdict-alphabet ladder of [3], though it answers a different question at the same $|\mathbb{V}|$ index — which kinematic event modalities a monitor can express, rather than which verdicts a monitor for a given property can reach.

Implications for runtime verification. The framework supplies a vocabulary in which encoder choices become explicit parameters of a single monitor architecture, rather than competing dogmas. Concretely:

- Engineers debating "edge-based vs. extremum-based" monitoring now have a join construction that subsumes both as parameter points, plus a count inequality that quantifies the discrimination.
- The plateau-touch convention algebra $|\mathcal{C}_B| = 72$ (Definition E.4) catalogues the design space of touch-event firing rules. Choosing a convention is no longer a hidden assumption; it is a parameter triple $(C, \theta_{\mathrm{u}}, \theta_{\mathrm{d}})$.
- The synthesis algorithm (Section 13.3) closes the loop from a target finite event pattern back to a tuning-parameter assignment in linear time, opening the door to online parametric adaptation as an enforcement mechanism complementary to the edit-based enforcement monitors of Falcone–Fernández–Mounier [6].

Limitations. The development is restricted to discrete-time real-valued signals on $\mathbb{N}$ with totally ordered values; the plateau-touch conventions assume strict-extremum semantics and would require revisiting for genuinely continuous-time signals. The robustness analysis (Section 14) gives canonical margins on the native primitives but does not develop a full noise model. The quaternary rung is presented in Section 3.2 as an enumeration; a full deep dive on its 2×2 lattice (count inequalities, transformation laws, augmented-monitor generalisation) is left to future work.

Future work. Six directions stand out:

(1) *Quaternary deep dive.* Develop the 2×2 modality lattice at $|\mathbb{V}| = 4$ at the same level of formal detail as the $|\mathbb{V}| = 3$ rung: count inequalities, transformation theorems, augmented-monitor generalisation under three thresholds.
(2) *Noise and quantitative monitoring.* Couple the robustness margins ρ_A (linear) and ρ_B (quadratic) to a STL-style quantitative semantics; study how additive noise on the signal affects modality count and event-time stability.
(3) *Enforcement-via-adaptation.* Generalise the finite-pattern synthesis algorithm of Section 13.3 to LTL trace properties via online parametric adaptation, and develop convergence guarantees under bounded-noise signals.
(4) *Higher-arity ladders.* Investigate $|\mathbb{V}| > 4$ rungs — in particular STL robustness ($|\mathbb{V}| = \infty$) — and the saturation of the modality count beyond $n = 4$.
(5) *Calibration orbits.* Characterise the observational window classes arising in temporal-implication calibration as the orbit of the augmented monitor's parameter space under the tuning taxonomy of Appendix B.
(6) *Modality selection as a meta-property (least tuning effort).* Because touch and crossing coincide as exact parameter points of $M^{(4)}$ at $|\mathbb{V}| = 4$ (Theorem 10.14), modality choice is a matter of tuning *effort*, not expressive power. We propose, as a heuristic, the principle: *among the modalities realising the target event set on the signal, deploy the one of least tuning effort.* The interchangeability holds under a tuning algorithm acting on the *thresholds*; modality interchangeability is *not* asserted when the tuner pre-conditions the *signal* itself (that case is excluded). This opens a richer monitor-parametrisation space classifying monitors quantitatively (effort / cost) and qualitatively (modality, verdict cardinality n); a full treatment of this space is left as an open problem. [3] treats the verdict-side selection (Q1 monitorability, Q2 which n); monitorability is not determined by the CLASP cell [4, Lemma 3.8].

Reproducibility and artifact availability

Paper artifact. The LaTeX source is a single file `articles/dual_modalities/main.tex` maintained under the author's repository. Most figures are tikz / pgfplots code embedded directly in the source; two multi-panel figures (the hero touch/crossing comparison and the extremum-pipeline instance) are included as external PDF files under `figures/`, rendered from a matplotlib generator and regenerable via the helper script `figures/regenerate.py`. Reproduction therefore requires the two PDFs (shipped alongside the source) or the generator script; all remaining figures need no external files. Build instructions: three passes of `pdflatex` on a TeX Live 2024 or MiKTeX 24.1 installation with the standard packages `amsmath`, `amssymb`, `amsthm`, `mathtools`, `enumitem`, `array`, `booktabs`, `microtype`, `graphicx`, `hyperref`, `cleveref`, `placeins`, `tikz`, and `pgfplots` (with `compat=1.18`). No external bibliography manager is needed; the bibliography is in `thebibliography` inline.

Data and code. This is a theoretical paper. No external datasets or executable code are required to verify the claims; all propositions, theorems and corollaries are checked by direct inspection of the proofs.

Citation. A `CITATION.cff` file in the paper directory provides metadata for citing this work in the Citation File Format.

Appendix A. Threshold and wall motion

The static thresholds $\theta, \theta_{\mathrm{u}}, \theta_{\mathrm{d}}$ of Definitions 4.1 and 5.3 are the static restriction of Definition 10.2's threshold schedules $\theta_{\mathrm{u}}(n), \theta_{\mathrm{d}}(n)$ to constant functions. Allowing those schedules to admit transitions yields the *moving-threshold operational rule*, the semantic basis of every tuning operation in Appendices A and B and of the synthesis results of Proposition 13.2.

Definition A.1 (Moving-threshold operational rule)**.** Let $Q : \mathbb{N} \to V$ be a signal and $\theta : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ a piecewise-constant threshold schedule (Model A); for Model B replace θ by the pair $\theta_{\mathrm{u}}, \theta_{\mathrm{d}}$ with the same form. Define the relation indicator $r_\theta(n) := \operatorname{sgn}(Q(n) - \theta(n))$ (with the convention $\operatorname{sgn}(0) = 0$). Under the moving-threshold rule, the monitor emits an event at sample n iff $r_\theta(n) \neq r_\theta(n-1)$ — a *flip* of the relation between Q and θ — with type determined by the signal-direction rule of Definition 4.2 for Model A, or by the wall on which the flip occurs for Model B. Equivalently, an event fires whenever signal and threshold cross each other, regardless of whether the crossing was caused by motion of Q (signal-caused), motion of θ (wall-caused), or both. Static-threshold semantics (Definitions 4.1 and 5.3) is the special case of this rule in which θ is constant and only Q-motion can flip r_θ.

Definition A.2 (Moving-touch predicate (forward sketch))**.** Let $Q : \mathbb{N} \to V$ be a signal and $\theta_{\mathrm{u}}, \theta_{\mathrm{d}} : \mathbb{N} \to V \cup \{-\infty, +\infty\}$ piecewise-constant wall schedules. The *moving-touch validation predicate* $\mathsf{validate}^*(\theta_{\mathrm{u}})$ holds at position n iff one of the following *contact conditions* is met:

- *Signal-caused touch.* τ_n is a strict local maximum of Q and $\tau_n \geq \theta_{\mathrm{u}}(n)$, with θ_{u} constant on $\{n-1, n, n+1\}$.
- *Wall-caused touch (entry-resolved).* $\tau_n = \theta_{\mathrm{u}}(n)$ and $\theta_{\mathrm{u}}(n-1) > \tau_{n-1}$ (the wall has moved down onto the signal at n, from above); under the DirRes *entry* resolver of Definition E.4 the predicate fires at n (the entry edge of the wall-meets-signal contact interval).

The symmetric moving-touch invalidation predicate $\mathsf{invalidate}^*(\theta_{\mathrm{d}})$ holds at strict local minima reaching $\theta_{\mathrm{d}}(n)$, or at wall upward motion onto a constant signal, with the symmetric entry/exit resolvers. The moving-touch monitor $M_T^{*,\theta_{\mathrm{u}}(\cdot),\theta_{\mathrm{d}}(\cdot)}$ emits an event at every tick at which one of these predicates holds, with tie-breaking by the convention chosen in Definition E.4. The static touch monitor of Definition 5.3 is the special case in which both walls are constant and only *signal-caused* touches contribute (wall-caused touches are vacuous because the wall never moves onto the signal).

Definition A.2 is the operational basis of every *wall-induced* event in this paper (e.g. Figure 35 (b), the plateau-on-wall panels of Figure 37, and the synthesis of Proposition 13.2 (2)); the corresponding technical predicate used by the augmented monitor of Definition 10.2 and downstream is given in Definition 10.1. The decoupling between event type and signal direction stated in Corollary 13.3 is the consequence of the wall-caused branch: in Model A no analogous branch exists because there is only one threshold.

Proposition A.3 (Realisability inequality under moving-threshold)**.** *Let $Q : \mathbb{N} \to V$ be a fixed signal with at least one ascending sample and one descending sample. For $X \in \{\mathrm{A}, \mathrm{B}\}$, let $\mathcal{R}^X(Q) \subseteq \mathbb{N} \times \{\mathsf{validate}, \mathsf{invalidate}\}$ be the set of (time, event-type) pairs realisable*

on Q under the moving-threshold rule of Definition A.1 with arbitrary piecewise-constant threshold/wall schedules in model X. Then

$$\mathcal{R}^{\mathrm{A}}(Q) \subsetneq \mathcal{R}^{\mathrm{B}}(Q),$$

with the inclusion strict for every non-monotonic Q.

Proof sketch. By Definition 4.2, Model A fires validate at n only if $\operatorname{sgn}(Q(n)-Q(n-1)) = +1$ and invalidate only if $\operatorname{sgn}(\cdot) = -1$. Therefore $\mathcal{R}^{\mathrm{A}}(Q) \subseteq \{(n, \mathsf{validate}) : Q(n) > Q(n-1)\} \cup \{(n, \mathsf{invalidate}) : Q(n) < Q(n-1)\}$. Model B's moving-wall rule of Proposition 13.2 realises both event types at *any* n irrespective of the sign of $Q(n)-Q(n-1)$. Hence $\mathcal{R}^{\mathrm{A}}(Q) \subseteq \mathcal{R}^{\mathrm{B}}(Q)$, and the strict containment follows because the set $\{(n, \mathsf{validate}) : Q(n) < Q(n-1)\}$ (and its symmetric I-on-ascent counterpart) lies in $\mathcal{R}^{\mathrm{B}}(Q) \setminus \mathcal{R}^{\mathrm{A}}(Q)$ as soon as Q admits at least one descending sample (resp. ascending sample). □

The pointwise count inequality of Proposition 6.1 therefore holds under static-threshold semantics but *reverses* under the moving-threshold rule: Model B is strictly more expressive than Model A when the algorithm is permitted to move the walls/threshold. This reversal is made precise by the realisability inclusion $\mathcal{R}^{\mathrm{A}}(Q) \subsetneq \mathcal{R}^{\mathrm{B}}(Q)$ of Proposition A.3. The two inequalities together characterise the duality: Model B is weaker than Model A under static thresholds (fewer events because extrema are rarer than crossings) and strictly stronger under moving thresholds (independent walls vs. direction-bound single threshold). Both regimes are simultaneously available on the augmented monitor, and the choice between them is the content of the tuning algorithm.

Figure 20 previews the two transformations that Sections 9.1 and 9.2 formalise. Both panels show the same goal — produce a stream of three validate events with *zero* invalidate events, an asymmetric pattern that neither minimal monitor produces natively. Panel (a) achieves it on the crossing side by *gating* the threshold: between every validate the threshold is temporarily disabled (gray segments), so the signal can descend without firing the corresponding invalidate. Panel (b) achieves it on the touch side by *shaping the signal envelope*: the sinusoid's lower amplitude is raised so that its minima sit above θ_{d} and never touch the lower wall, removing every invalidate.

A second mechanism, complementary to the interruption of Figure 20, is to *move* the threshold (or walls) *away* from the signal in a target window. Where interruption disables the firing rule outright on an interval, movement-away keeps the rule active but ensures the threshold and the signal never meet — so no flip, no touch, no event. Figure 21 illustrates the technique on both modalities, on the same unmodified signal as Figure 1.

A third mechanism is to move the threshold(s) *toward* the signal: instead of stepping away to suppress events, step into the signal's path to *generate* them. In Model A this raises or lowers θ across the current signal level, causing an instantaneous state flip and therefore an event *without requiring the signal to change direction at that moment* — though the signal-direction rule still constrains the type of event that can fire. Figure 22 illustrates this mechanism on the crossing modality; its analogue in Model B is bounded by the touch rule and is discussed in the caption of that figure.

Remark A.4 (Generality of the notch technique)**.** The example of Figure 22 instantiates a fully general capability. Under the tuning model considered up to this point, a validate event can be placed at *any* time t^* along an ascending segment of Q by inserting a notch that raises θ to a level $\theta^* < Q(t^*)$ at $t \lesssim t^*$ (so Q horizontally crosses θ^* at $t = t^*$ during its rise) and closes the notch before the descending phase. Symmetrically, an invalidate event

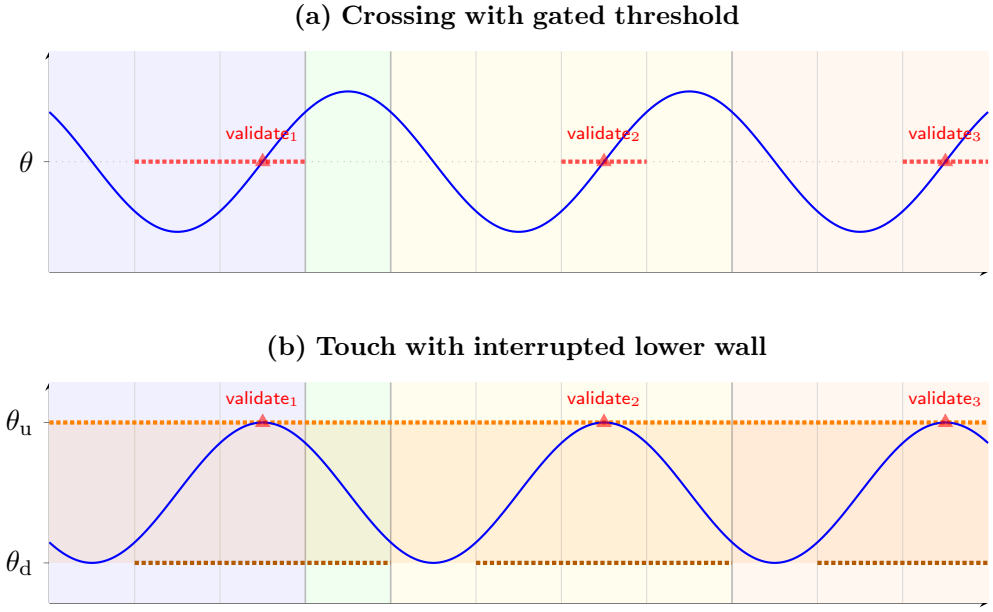


FIGURE 20. Two ways to produce an asymmetric event stream (three validations, no invalidations), one per modality, on *the same unmodified signal*. **(a)** *Gated crossing:* the single threshold is temporarily disabled (gray dotted segments) between consecutive validate events so the signal can descend below the threshold level without firing invalidate. **(b)** *Touch with interrupted lower wall:* the upper wall θ_{u} is always active so each maximum of Q fires validate, while the lower wall θ_{d} is *interrupted* on three short intervals around the signal's minima ($t = 0, 4, 8$), so the natural invalidate touches do not fire. In both panels the signal Q is unchanged with respect to Figure 1; the tuning algorithm acts only on the threshold and walls.

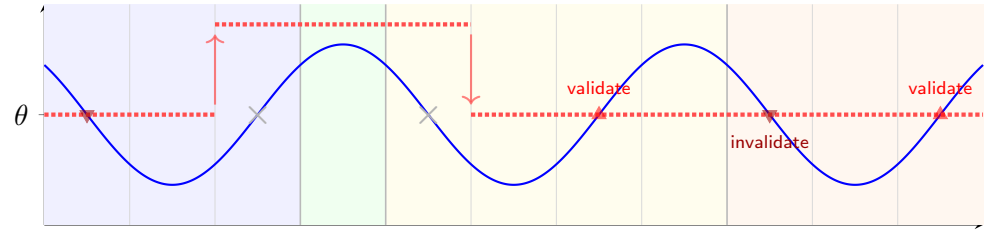
can be placed at any time along a descending segment by lowering θ to $\theta^* > Q(t^*)$. Multiple non-overlapping notches on the same ascending (resp. descending) phase generate multiple validations (resp. invalidations). The only constraint is the signal-direction rule of Figure 19: notch-induced event type is forced by the local slope of Q. The touch-modality analogue is set aside — the two-wall model ties natural firings to extrema of Q, so wall motion alone either enables specific extrema (movement *into* the signal on a reduced-amplitude Q) or suppresses them (movement *away*).

APPENDIX B. SEMANTIC-AGGREGATION TUNINGS

B.1. **Signal parameters, tick anatomy, and band-boundary tuning.** Before turning to the higher-level transformations between modalities (Sections 9.1 and 9.2), we make the parametric family of input signals explicit. Throughout this paper the running illustration uses a discrete-time sinusoidal signal

$$Q(t) \;=\; \mu \;+\; A\,\cos\!\left(\frac{2\pi}{T}\,(t-\varphi)\right), \qquad t \in \mathbb{N}, \tag{B.1}$$

(a) Model A: θ lifted away ⇒ no crossing in the window



(b) Model B: walls retracted away ⇒ specific extrema fail to touch

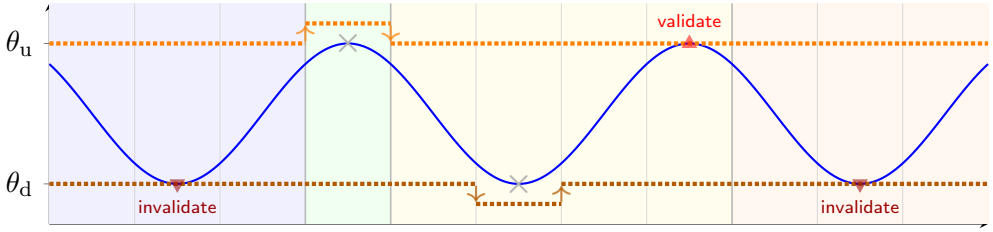


FIGURE 21. Threshold/wall *movement away* from the signal as a suppression technique. The signal Q is unchanged with respect to Figure 1; the tuning algorithm acts only by lifting the threshold (Model A) or retracting the walls (Model B) inside a target window. **(a)** Model A: θ is lifted from 0.5 to 0.95 on $t \in [1.5, 4.5]$. Inside this interval the signal never reaches the threshold, so the natural validate at $t = 2$ and invalidate at $t = 4$ are both suppressed. Movement-away is therefore a *collateral* suppression: every event whose flip would have occurred in the window is removed. **(b)** Model B: θ_{u} is lifted to 0.95 on $[2.5, 3.5]$ around the maximum at $t = 3$, and θ_{d} is dropped to 0.05 on $[4.5, 5.5]$ around the minimum at $t = 5$. Each extremum is selectively detouched: V at $t = 3$ and I at $t = 5$ are suppressed while the other extrema fire normally. Compared to Model A, Model B's suppression is finer-grained — each wall acts on one event-type independently, so suppressing V at one maximum does not affect the surrounding I events.

controlled by four scalar signal parameters (*DC offset* μ, *amplitude* A, *period* T, *phase* φ) and three sampling-grid parameters that determine *how* Q is observed: the *tick-frame phase* ϕ_{tick} (the sub-tick alignment of the integer time axis, introduced in Figure 19; we use a separate symbol from the schedule π of the augmented monitor in Definition 10.2 to avoid notational collision), the *tick width* w (here $w = 1$ under the canonical sampling rate $f_s = 1$ sample per unit time), and the *band-boundary vector* $(b_{\mathrm{PX}}, b_{\mathrm{XB}}, b_{\mathrm{BP}})$ that partitions the time axis into Pre/X/Body/Post. The Nyquist constraint $T \geq 2/f_s = 2$ guarantees that the discrete samples capture the sinusoid without aliasing; throughout the figures we use $T = 4$, well above the Nyquist limit. All seven knobs are tuning parameters of the augmented monitor: they reshape *when* the signal sits inside a tick window and *where* the resulting events are classified, but *none of them changes the signal's sample values* $Q(t)$.

Figure 23 renders this parametric layer in three panels. Panel (a) annotates the canonical sinusoid with each of the four signal parameters (μ, A, T, φ), marks the discrete samples

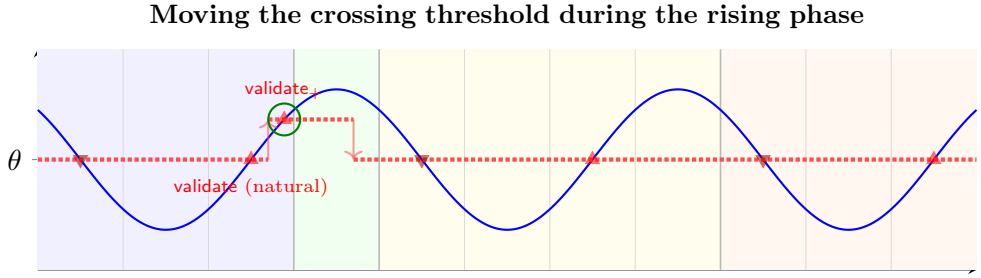


FIGURE 22. Threshold-notch tuning on the crossing modality: movement *toward* the signal as a generation mechanism. Nominal $\theta = 0.5$ (dashed horizontal); vertical arrows mark the notch motion at $t = 2.2$ and $t = 3.2$ but do not themselves fire events. Natural validate at $t = 2$; the raised threshold $\theta = 0.7$ is crossed horizontally by the still-ascending sinusoid at $t \approx 2.39$, firing a second validation $\mathsf{validate}_+$ purely from wall motion. Notch closes at $t = 3.2$ above both signal levels (no event). Subsequent natural events at $t = 4, 6, \ldots$ proceed unchanged. See Remark A.4 for the generality of the technique.

$Q(0), Q(1), \ldots, Q(10)$, highlights one tick window of width $w = 1$, and labels the four bands. Panels (b) and (c) then exercise the band-boundary knob: with the original partition (panel (b), $b_{\mathrm{BP}} = 7.5$) the invalidate event at $t = 8$ falls in Post; after the window-extension operation T_3^+ of the tuning algebra (Appendix B) moves the boundary to $b_{\mathrm{BP}} = 8.5$ (panel (c)), the same event — at the same time, on the same signal, under the same firing rule — is reclassified as belonging to Body.

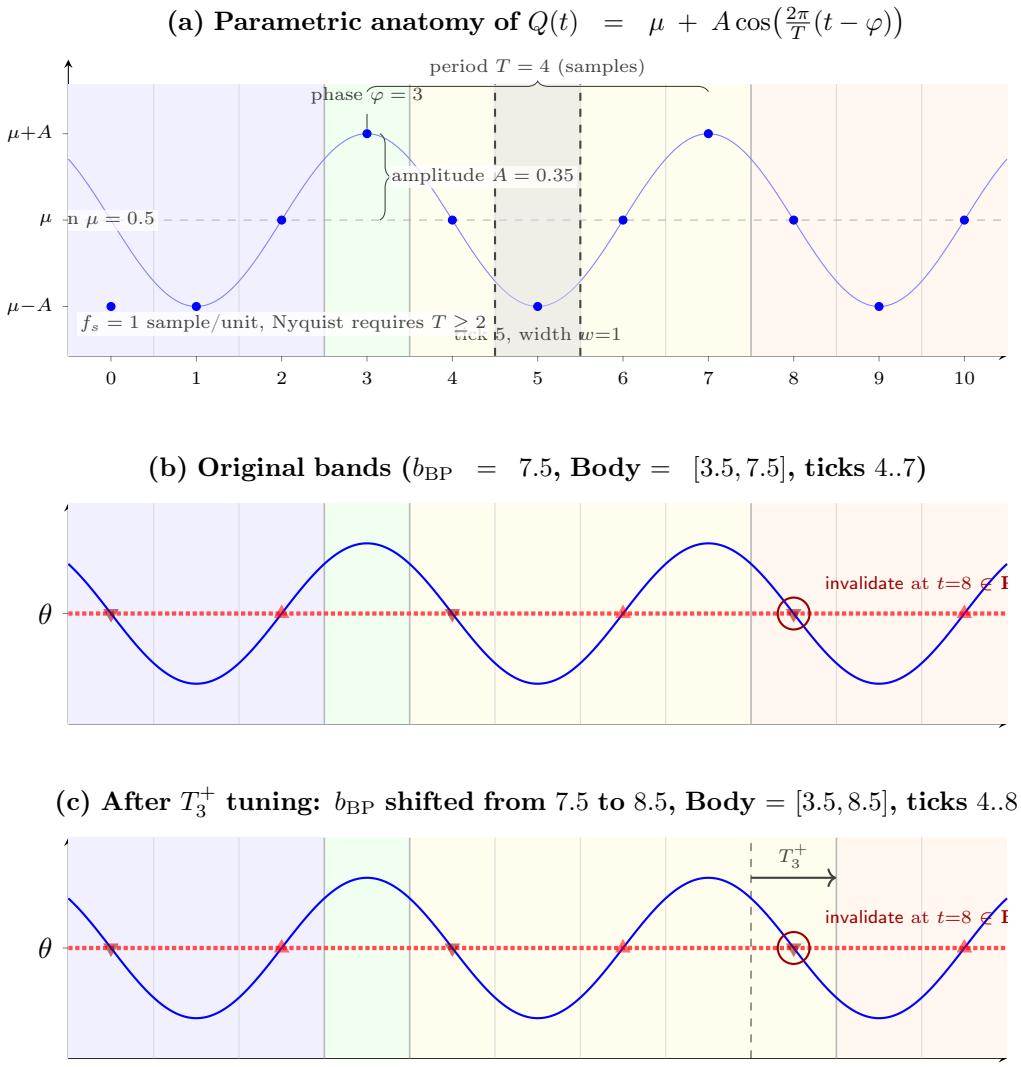


FIGURE 23. Parametric layer of the running illustration and one band-boundary tuning. **(a)** Anatomy of $Q(t)$ (Equation (B.1)): the four signal parameters μ, A, T, φ are labelled in place on the canonical $\mu = 0.5, A = 0.35, T = 4, \varphi = 3$ instance. The blue envelope is the continuous-time sinusoid; the dark blue dots are the discrete samples $Q(0), \ldots, Q(10)$ that the monitor actually observes (sampling rate $f_s = 1$; Nyquist requires $T \geq 2$). The gray strip highlights one tick window of width $w = 1$. The four bands Pre/X/Body/Post are labelled at the top. **(b)** With the original band-boundary vector $(b_{\mathrm{PX}}, b_{\mathrm{XB}}, b_{\mathrm{BP}}) = (2.5, 3.5, 7.5)$, the invalidate event at $t = 8$ (inside tick 8) lies in Post. **(c)** A single-knob retuning by T_3^+ shifts b_{BP} from 7.5 to 8.5 (the old position is shown as a dashed ghost line and the operation as an arrow labelled T_3^+). The same event now lies in the extended Body band. No sample value has changed; only one knob of the parameter family of Equation (B.1) plus its grid has moved. Operations of this kind can move any band boundary, reassigning any event between adjacent bands.

B.2. **Tuning tick widths.** Beyond shifting the band boundaries $(b_{\mathrm{PX}}, b_{\mathrm{XB}}, b_{\mathrm{BP}})$, the tuning algorithm acts on a finer-grained knob of the same parametric layer: the individual tick widths w_i. Up to this point the eleven tick windows have been treated as a uniform partition with $w_i = 1$ for every i. This is a convenience, not a constraint. The tick partition $b_0^{\mathrm{tick}} < b_1^{\mathrm{tick}} < \cdots < b_N^{\mathrm{tick}}$ is itself a tuning parameter: each boundary b_i^{tick} can be moved *independently*, and the resulting tick widths $w_i = b_i^{\mathrm{tick}} - b_{i-1}^{\mathrm{tick}}$ can be shortened or lengthened at will. Bringing two adjacent boundaries closer compresses a region of the time axis into finer granularity (more ticks per signal cycle, useful when several events crowd a short interval); pushing them apart coarsens it (one wider tick that aggregates several signal samples and any events within them). The sampling rate f_s remains fixed throughout — tick width is a *semantic* aggregation parameter on top of the sampling grid, not a resampling.

Figure 24 makes this freedom explicit on the same signal in three retune steps: starting from the canonical uniform partition ($w_i = 1$ for every i), a single boundary b_i^{tick} is pushed by *increasingly large amounts*, and the affected tick w_i shrinks while its neighbour w_{i+1} grows by the same amount. Crucially, there is no upper or lower bound on how far a boundary can move: the tuning algorithm can make a tick last *as long as it wants* — from a near-zero sliver collapsed onto the band boundary to a window that absorbs several signal cycles — and the only constraint is the ordering $b_{i-1}^{\mathrm{tick}} < b_i^{\mathrm{tick}} < b_{i+1}^{\mathrm{tick}}$, i.e. that ticks do not cross. The signal Q, the threshold, the firing rule, and the underlying sample times are all identical between the three panels; only the tick-width vector (w_i) has been retuned.

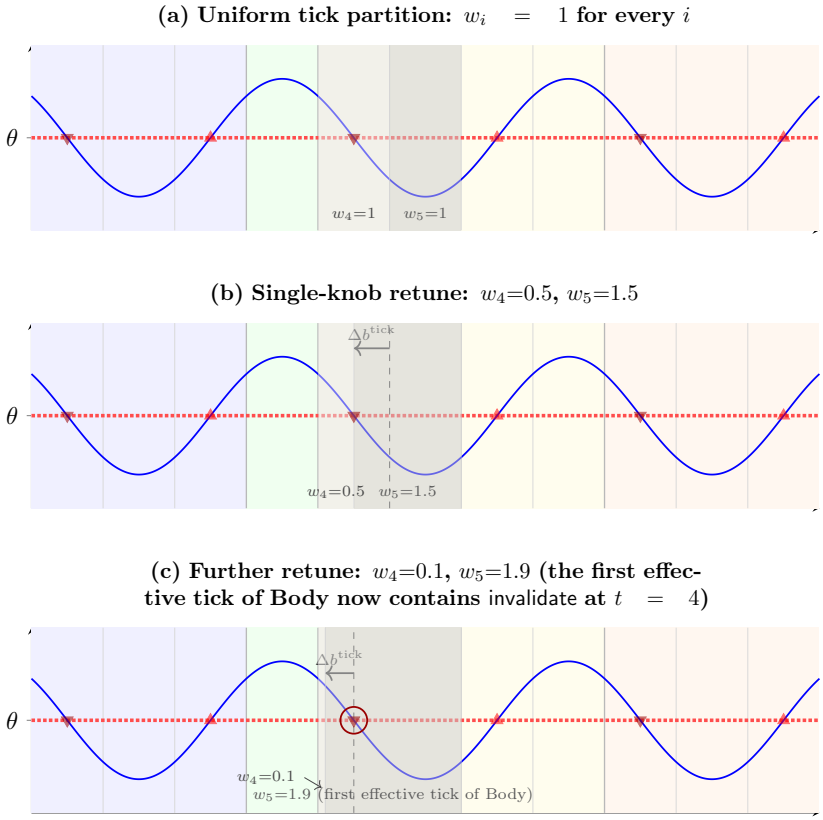


FIGURE 24. Tuning tick widths, in three retune steps. The tick partition is itself a parameter of the monitor: each boundary b_i^{tick} can be moved independently, with *no upper or lower bound on the displacement* — the tuning algorithm can make any individual tick last as long as it wants, shortening one tick and lengthening its neighbour by the same amount. **(a)** Canonical uniform partition: all tick windows have width $w = 1$. Tick 4 (light gray) and tick 5 (dark gray) are highlighted for reference. **(b)** First retune: the boundary between tick 4 and tick 5 is moved from $t = 4.5$ to $t = 4.0$ (the old position is shown as a dashed ghost line and the move as the arrow labelled Δb^{tick}). Tick 4 is compressed to $w_4 = 0.5$ (window $[3.5, 4.0]$); tick 5 is expanded to $w_5 = 1.5$ (window $[4.0, 5.5]$). **(c)** Further retune (continuous with (b)): the same boundary is pushed from $t = 4.0$ to $t = 3.6$. Tick 4 becomes a near-zero sliver at the X/Body band boundary ($w_4 = 0.1$, window $[3.5, 3.6]$); tick 5 grows to $w_5 = 1.9$ (window $[3.6, 5.5]$). The invalidate event at $t = 4$ — which in panel (a) fell inside tick 4 and in panel (b) already sat in tick 5 — is now squarely inside tick 5, which has become the *first effective tick of the Body band*. Crucially the tick count per band is unchanged across the three panels: Body still contains exactly the four ticks $4, 5, 6, 7$. Only their widths have been retuned. No sample value has changed and no firing rule has changed; this is purely a semantic retuning of the aggregation grid, freely interleavable with the band-boundary moves of Figure 23 and the threshold motions of Figures 20 to 22.

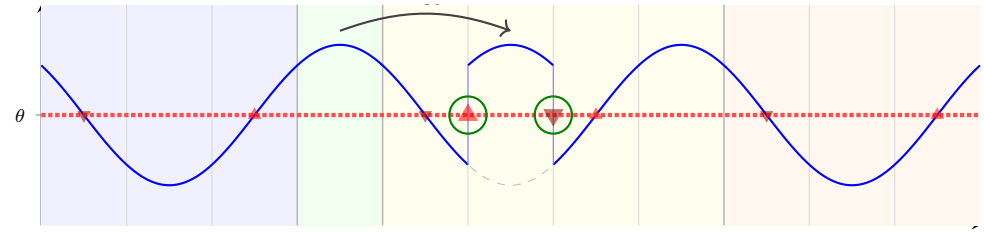
Appendix C. Tick-level signal copy-paste

Between the parameter retunings of Section 7 (which leave Q unchanged) and the free signal modifications of the next section (which can synthesise arbitrary Q), there is an intermediate power: the tuning algorithm can *copy* the content of any tick window of Q into any other tick window, *overwriting* the destination with the source pattern. No sample value is fabricated — every value present in the output already appeared somewhere in the canonical record — but the order in which segments are presented to the monitor is no longer the canonical one. This is *partial mangling*: strictly weaker than fabricating Q from scratch, strictly stronger than the parameter tunings of Section 7, and useful precisely when the canonical record already contains a pattern that the algorithm wishes to deploy at a different position in time. Operationally the algorithm *ignores* the destination tick's original content and *splices in* an observed segment of Q taken from elsewhere; the segments available to the splice are exactly those that the canonical Q makes available, so the construction is grounded in observation rather than invention.

The copy-paste power has two regimes that we treat separately, because they expose the modalities to different artefacts. In the *general regime* (Appendix C.1), the values of Q at the boundaries of the source and destination ticks need not agree, and the splice introduces *discontinuities* (instantaneous jumps) at the destination tick's boundaries — jumps that themselves act like threshold crossings in Model A while being invisible to Model B's touch rule. In the *continuity-preserving regime* (Appendix C.2), the source and destination boundary values match, so the splice produces no jumps; only the *interior* of the destination tick is substituted, and the two modalities react only to changes in the *extremum* of that tick. Both regimes are partial mangling — only previously observed segments are used — but they shift events in qualitatively different ways.

C.1. **General copy-paste (with boundary discontinuities).** Figure 25 illustrates a single such splice on the canonical sinusoid $Q(t) = 0.5 + 0.35\cos(2\pi(t-3)/4)$ of Figure 23. The content of the maximum tick 3 (centred at the global peak $Q = 0.85$) is copied into the minimum tick 5 (centred at $Q = 0.15$), overwriting the natural minimum with a transplanted maximum. Sample values at the tick boundaries $t = 4.5$ and $t = 5.5$ jump discontinuously: at $t = 4.5$ the signal steps up from the canonical $Q(4.5) = 0.253$ to the transplanted segment's starting value 0.747, and at $t = 5.5$ it steps back down by the same amount. These step discontinuities are *themselves* threshold crossings under Model A and hence fire events; under Model B they are not local extrema and do not touch the walls, but the transplanted peak at $t = 5$ now reaches θ_{u} and fires a V touch where the canonical signal would have fired an I. The asymmetric consequences for the two modalities are characteristic of the copy-paste regime: Model A reacts to the *boundary jumps* introduced by the splice, Model B reacts to the *kind of extremum* that the splice has substituted at the centre of the destination tick.

(a) Model A: canonical Q with tick 3 copied into tick 5 — splice boundaries fire V and I events



(b) Model B on the same spliced signal: the transplanted max at $t = 5$ fires a V where the canonical min would have fired an I

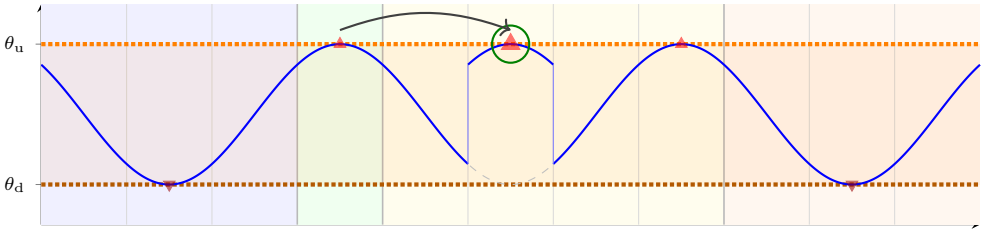


FIGURE 25. Tick-level signal copy-paste: the content of tick 3 (the canonical maximum, $Q(3) = 0.85$) is spliced into the position of tick 5 (the canonical minimum, $Q(5) = 0.15$). The canonical sinusoid (dashed gray ghost) is shown for reference; the spliced signal (solid blue) coincides with the canonical outside $[4.5, 5.5]$ and reproduces the maximum-tick shape inside. Sample values jump by 0.494 at each splice boundary. **(a)** Model A: the canonical descending crossing at $t = 4$ and the ascending crossing at $t = 6$ are unaffected (their ticks are not spliced). The two splice-boundary jumps cross θ at $t = 4.5$ (upward $\Rightarrow V$) and at $t = 5.5$ (downward $\Rightarrow I$) — two extra events fire purely because of the splice. **(b)** Model B: the canonical invalidate at $t = 5$ is replaced by a validate at $t = 5$ — the transplanted peak touches θ_u where the canonical trough would have touched θ_d. The boundary jumps are not local extrema and do not register in touch modality. The asymmetry is structural: Model A sees the splice boundaries, Model B sees the substituted extremum at the tick centre.

C.2. **Continuity-preserving copy-paste.** The discontinuities of Appendix C.1 were not intrinsic to the copy-paste idea — they arose because the canonical sinusoid takes *different* values at the boundaries of different ticks. If the base signal is chosen so that all tick boundaries share a common value $Q = \theta$, then any tick can be spliced into any other position without introducing a jump. The simplest such base is the half-cosine landscape

$$Q_v(t) = \theta + a_i \cdot \cos\big(\pi(t - i)\big) \qquad \text{for } t \in [i - \tfrac{1}{2}, i + \tfrac{1}{2}], \tag{C.1}$$

which assigns one independent amplitude $a_i \in [-A, A]$ to each tick i and is identically equal to θ at every half-integer boundary (since $\cos(\pm\pi/2) = 0$). A continuity-preserving splice replaces a_i at the destination tick with a_j read off the source tick; the rest of Q_v is unchanged and the signal remains continuous everywhere.

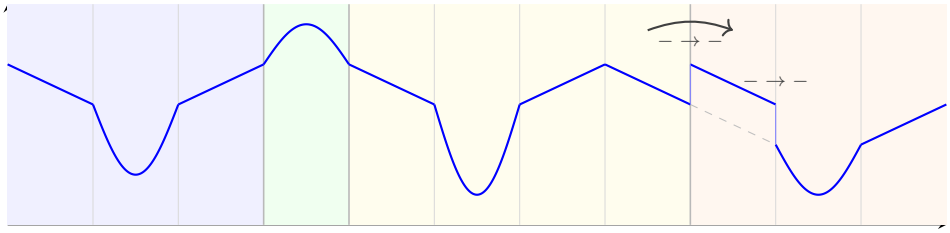
The uniform-boundary condition $Q(t_{i\pm1/2}) = \theta$ that we have used for Q_{v} is the *simplest* sufficient condition for continuity — every tick is interchangeable with every other. It can be *relaxed* to a strictly weaker requirement: rather than demanding a common boundary value across all ticks, only the source and destination ticks (and the destination's immediate neighbours) need to have *matching* boundary values. The collection of available splices then forms a *puzzle*: tick j can be spliced into the position of tick i iff $Q(t_{j-1/2}) = Q(t_{i-1/2})$ and $Q(t_{j+1/2}) = Q(t_{i+1/2})$. Two pieces fit only when their edges match, and any subset of ticks whose boundaries form a closed matching graph admits arbitrary continuity-preserving rearrangements among themselves. The uniform-boundary landscape Q_{v} is the maximally flexible such graph (a complete graph on all 11 ticks); more idiosyncratic signals can still admit non-trivial continuity-preserving splices on smaller compatible subgraphs.

A further relaxation drops the requirement that boundary *values* agree, and asks only that boundary *directions* agree. Concretely, tick j is *direction-coherent* as a splice into the position of tick i iff $\operatorname{sgn}\dot{Q}(t^-_{i-1/2}) = \operatorname{sgn}\dot{Q}_{\mathrm{src}}(t^+_{j-1/2})$ and $\operatorname{sgn}\dot{Q}_{\mathrm{src}}(t^-_{j+1/2}) = \operatorname{sgn}\dot{Q}(t^+_{i+1/2})$ — the spliced tick must start with the same slope sign as the previous tick's final slope, and end with the same slope sign as the next tick's initial slope. Direction-coherent splices still admit value discontinuities at the boundaries, but the discontinuities are now *co-directional* with the local flow: the signal steps up where it was already rising, steps down where it was already falling, and never reverses against the surrounding direction. This is strictly weaker than the value-matching puzzle regime (every value-matching splice is direction-coherent, but not conversely) and is the most permissive condition under which the spliced signal still *looks like* a continuation of its neighbourhood, even though it is not formally continuous.

C.3. **Direction-coherent copy-paste.** Figure 26 contrasts the two sub-regimes on a common base signal Q_{dc} whose ticks have heterogeneous shapes and boundary values (linear descending/ascending ramps and half-sine min/max bumps, with consistent boundary continuity). Panel **(a)** splices a descending ramp from a higher-valued region into a lower-valued descending tick: slope signs match at both boundaries (descending → descending) but boundary *values* differ, so the splice introduces visible value jumps at $t = 7.5$ and $t = 8.5$ — a *direction-coherent splice with jumps*. Panel **(b)** splices a shallow min-bump into a position currently occupied by a deeper min-bump: slope signs match (descending → ascending) *and* boundary values match (both ticks are flanked by the same neighbours), so no jump occurs — only the depth of the trough changes. This is a *direction-coherent splice without jumps*, simultaneously satisfying the value-matching puzzle condition.

Figure 27 demonstrates one such splice on a varied Q_{v} with amplitudes $(a_0, \ldots, a_{10}) = (0.15, -0.35, 0.25, 0.35, -0.15, -0.35, 0.20, 0.35, -0.10, -0.35, 0.10)$: the full-amplitude maximum of tick 3 ($a_3 = +0.35$) is spliced into the position of tick 4 (originally a small minimum, $a_4 = -0.15$). Because the splice preserves $Q_{\mathrm{v}} = \theta$ at $t = 3.5$ and $t = 4.5$, the two modalities react only to the substituted interior: in Model A the natural crossing at the (formerly) sign-change boundary $t = 3.5$ is suppressed and a new crossing appears at $t = 4.5$ (the splice has shifted, not added, the boundary event); in Model B a new validate touch fires at $t = 4$ because the transplanted peak now reaches θ_{u}. Compared with Figure 25, no spurious events are generated by step jumps — the splice acts cleanly on the *extremum* of the destination tick and on the *type* of boundary crossing on either side.

(a) Direction-coherent splice *with* jumps: tick 7 (desc. 0.75 → 0.55) spliced into tick 8 position (desc. 0.55 → 0.35)



(b) Direction-coherent splice *without* jumps: tick 1 (min-bump, depth 0.35) spliced into tick 5 position (min-bump, depth 0.45)

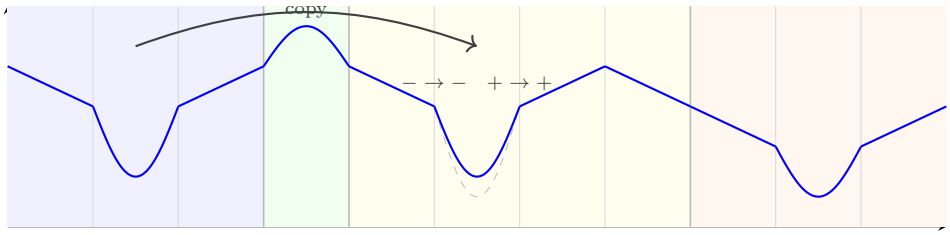
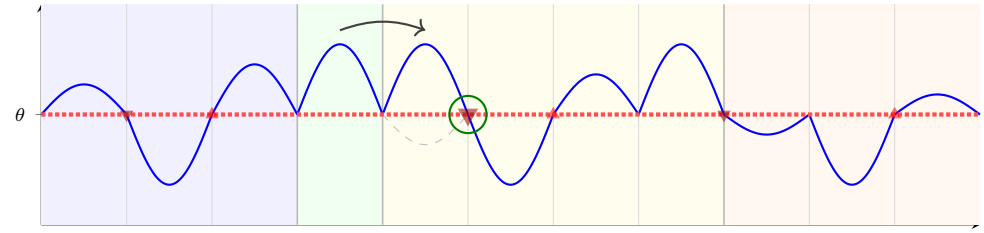


FIGURE 26. Direction-coherent copy-paste in its two sub-regimes on a custom base signal Q_{dc} (ghost gray dashed). The base is constructed from heterogeneous tick types (linear ramps and half-sine min/max bumps) with non-uniform boundary values, but with full canonical continuity. **(a)** A descending ramp from a higher-valued region of Q_{dc} (tick 7, ramp $0.75 \to 0.55$) is spliced into the position of tick 8 (originally a descending ramp $0.55 \to 0.35$). The slope signs at the splice boundaries match in both directions (descending → descending at $t = 7.5$ and at $t = 8.5$), but the boundary values disagree, so the spliced signal jumps by 0.20 at each boundary. The jumps are direction-coherent (they do not reverse the local slope sign) but *not* continuity-preserving. **(b)** A shallow min-bump (tick 1, depth 0.35) is spliced into the position of tick 5 (originally a deeper min-bump, depth 0.45). Boundary values match (both bumps return to 0.55), so no jump occurs; only the depth of the trough changes. This splice is simultaneously direction-coherent and value-matching (puzzle). The figure makes precise the strict ordering *uniform-boundary* $\subset$ *value-matching* $\subset$ *direction-coherent* $\subset$ *unconstrained* copy-paste: each weaker regime admits all splices of the stronger regimes plus additional ones.

(a) Model A on Q_v with continuity-preserving splice (tick 3 → tick 4): event I shifts from $t = 3.5$ to $t = 4.5$



(b) Model B on the same splice: a new V touch fires at $t = 4$ where a small bump was

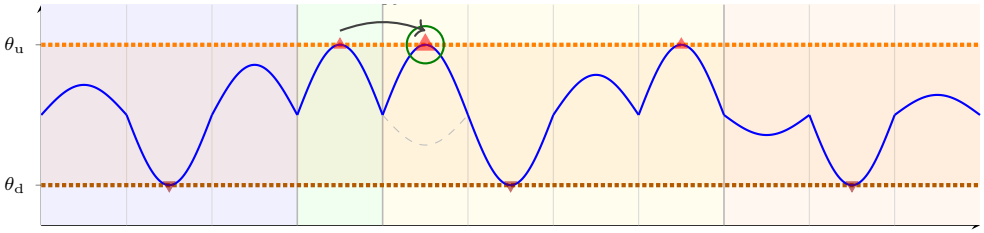


FIGURE 27. Continuity-preserving copy-paste on the half-cosine landscape Q_v of Equation (C.1) with amplitudes $(0.15, -0.35, 0.25, 0.35, -0.15, -0.35, 0.20, 0.35, -0.10, -0.35, 0.10)$. The full-amplitude maximum of tick 3 ($a_3 = +0.35$) is spliced into the position of tick 4 (originally $a_4 = -0.15$). Because $Q_v = \theta$ at every half-integer boundary, the splice introduces *no* step discontinuity. **(a)** Model A: the canonical descending crossing at $t = 3.5$ (present in the ghost signal) is suppressed because both sides of the boundary now sit above θ; a new descending crossing appears at $t = 4.5$, where the spliced maximum meets the original full minimum of tick 5. The total event count is unchanged — the splice has *shifted* the boundary crossing by one tick rather than adding spurious ones, in contrast with the discontinuous splice of Figure 25. **(b)** Model B: the transplanted peak at $t = 4$ now reaches $\theta_u = 0.85$ and fires a new validate touch where the original $a_4 = -0.15$ bump was too small to touch either wall. The five extremum events that fire on the canonical Q_v remain unchanged. Both panels demonstrate that, under the continuity-preserving regime, copy-paste acts cleanly on the *interior* of the destination tick: Model A sees the change as a shift of the boundary type, Model B sees it as a change in the amplitude of the central extremum.

Appendix D. Modifying the input signal

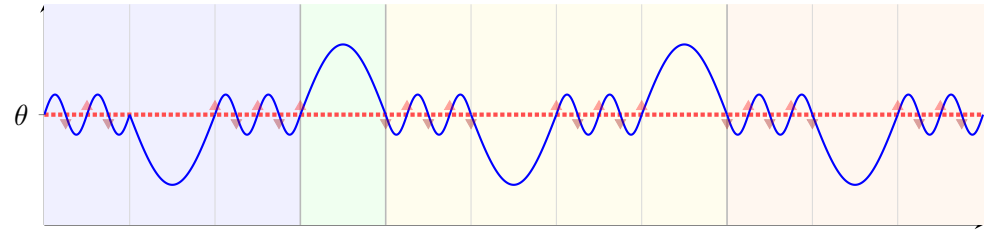
The retunings of Section 7 all act *inside* the monitor — on thresholds, walls, gates, band boundaries, tick widths. The copy-paste of Appendix C already breaks the monitor/signal boundary, but only by rearranging observed segments. A complementary class of operations acts *outside* the monitor, on the input signal Q itself, modulating its amplitude and frequency on a per-tick basis. This is a strictly stronger capability: it changes which crossings or touches occur at all, not just how they are classified. We open the discussion with the canonical illustration of variable amplitude-frequency modulation, familiar from the analog-to-digital conversion and discrete-time signal-processing literature [11] where it isolates the contributions of bandwidth and dynamic range.

Figure 28 shows the same modulated signal under both modalities. The signal is piecewise sinusoidal, alternating two regimes on the tick grid:

- **Oscillation ticks** (here ticks $0, 2, 4, 6, 8, 10$): $Q(t) = \theta + a\sin(4\pi t)$ with low amplitude $a = 0.1$ and $4\pi t$ chosen so $Q = \theta$ at every half-integer. Two full cycles of the carrier fit inside one tick, producing three interior zero-crossings of θ per tick (so the burst contains four V/I events in $[t_{\text{left}}, t_{\text{right}}]$).
- **Extremum ticks** (here ticks $1, 3, 5, 7, 9$, alternating min and max): $Q(t) = \theta \pm A\cos(\pi(t - t_c))$ with full amplitude $A = 0.35$. The signal smoothly reaches one extremum at the tick centre and returns to θ at the tick boundaries.

The sign of the osc amplitude is flipped on osc ticks that sit between two falling extrema (here ticks 4 and 8) so the boundary slope matches the neighbouring extremum and the signal has no cusps. The first tick is an osc tick with the positive-sign carrier, so Q starts above θ, oscillates rapidly, and returns to θ exactly at the boundary with the next tick (a minimum), as required.

(a) Model A: rapid V/I bursts in osc ticks, single V / I at extremum ticks



(b) Model B: osc bursts are silent (between the walls); only the extrema fire

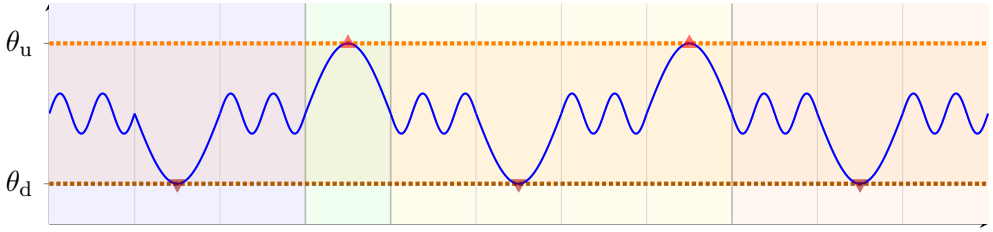


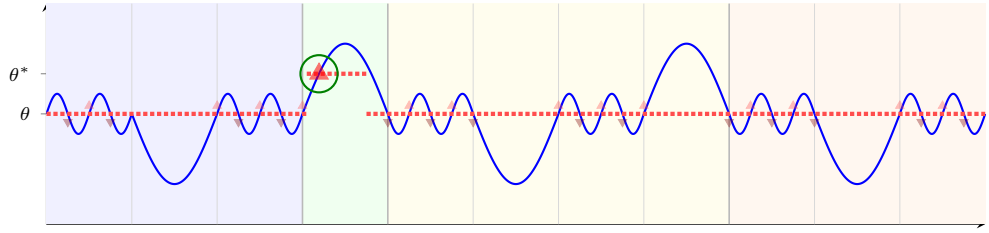
FIGURE 28. Amplitude-frequency modulation as the first signal modification. The same modulated Q is monitored under both modalities. In each tick, the signal is either a high-frequency low-amplitude burst around the threshold (osc ticks $0, 2, 4, 6, 8, 10$) or a slow half-cosine that reaches an extremum at the tick centre (extremum ticks $1, 3, 5, 7, 9$, alternating min and max), and the two regimes share the value $Q = \theta$ at every tick boundary so the overall signal is continuous. **(a)** Model A (single threshold $\theta = 0.5$): each osc tick produces a burst of rapid validations and invalidations (four interior zero crossings of the carrier plus the boundary transits), while each extremum tick fires a single V at a maximum or I at a minimum. **(b)** Model B (walls $\theta_u = 0.85$, $\theta_d = 0.15$): the osc bursts have amplitude 0.1 and remain strictly between the walls — they are silent — while only the extrema touch the walls and fire. This is the modulated analogue of the well-known illustration that distinguishes the contributions of bandwidth (Model A counts oscillations) from dynamic range (Model B counts excursions to the limits).

D.1. **Combined signal and threshold tuning.** The amplitude-frequency modulation of Figure 28 shows what signal modification *alone* can do, and the threshold-notch tuning of Figure 22 shows what threshold-segment movement *alone* can do. Taken *together* they form a precision instrument with a clean two-step recipe. Signal modification *positions the local maxima and minima of Q wherever they are wanted* — any time and (up to amplitude) any value; threshold-segment movement *lowers the threshold (or upper wall) onto exactly that local maximum, or raises it to the local minimum, at the desired instant.* The two steps are independent: the modification step shapes the landscape, the threshold step lands on the chosen peak. The composition eliminates every limitation that either knob has in isolation — on the modulated signal of Figure 28, the notch technique of Figure 22 can therefore place an extra event at any chosen time t^*, and it does so *symmetrically* in both modalities. In

Model A the notch raises the threshold across an ascending sub-segment of the modulated signal, firing one extra validate at the intersection time; in Model B the analogue notch lowers the upper wall onto a local maximum of an oscillation burst that was silent in Figure 28(b), forcing a validate touch where none existed before. Given complete control over Q, the two-knob combination is arbitrarily precise. This combined regime realises the top layer $\mathcal{L}_3$ of Proposition 8.1: signal modification supplies the Layer-3 expressivity that strictly contains the threshold-motion-only Layer 1 and the copy-paste-only Layer 2, so the extra events placed here are exactly the streams in $\mathcal{L}_3 \setminus \mathcal{L}_2$.

Figure 29 illustrates the recipe on the same modulated signal of Figure 28. In panel **(a)** (Model A) the threshold is notched up to $\theta^* = 0.7$ on $t \in [2.55, 3.25]$, a window that spans most of the maximum tick 3. The signal, ascending from $\theta = 0.5$ at $t = 2.5$ toward its peak 0.85 at $t = 3$, crosses θ^* at $t \approx 2.694$ — firing one *extra* validate event precisely there. All other events of Figure 28(a) fire unchanged. In panel **(b)** (Model B) the upper wall is notched down to $\theta^*_{\mathrm{u}} = 0.6$ on $t \in [1.7, 2.45]$, a window that spans most of the osc tick 2 and contains the single local maximum at $t = 2.125$ where $Q = 0.6$. The notched wall meets the signal exactly at this local max, firing a validate touch in a tick that was silent in Figure 28(b). The five natural extremum events of Figure 28(b) fire unchanged. By choosing the notch position and level, the same recipe places a validate or invalidate at any t^* in either model.

(a) Model A on the modulated signal of Figure 28: θ-notch fires an extra V at $t^* \approx 2.694$



(b) Model B on the same modulated signal: wall notch on θ_u fires a V inside a silent osc burst at $t^* = 2.125$

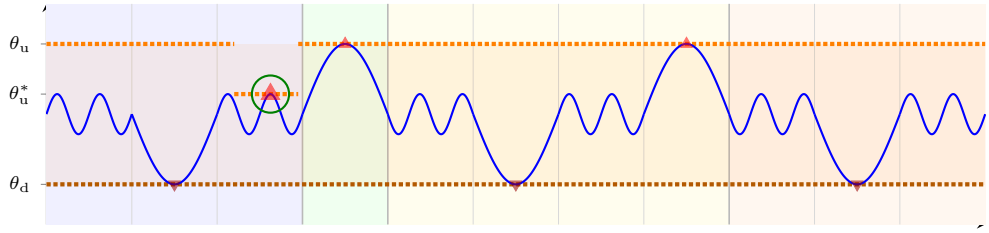


FIGURE 29. Combined signal modification and threshold-segment movement on the modulated signal of Figure 28. Both panels show that the notch technique of Figure 22 carries over to the modulated signal of Appendix D *symmetrically* for the two modalities: an extra event can be placed at any chosen time by a single notch. **(a)** Model A: the threshold is notched up from $\theta = 0.5$ to $\theta^* = 0.7$ on $t \in [2.55, 3.25]$, an interval that spans most of the maximum tick 3 (Q ascends from 0.5 at $t = 2.5$ toward 0.85 at $t = 3$). The signal crosses θ^* at $t^* \approx 2.694$ (the unique time where $Q(t) = 0.7$ on this ascent), firing one *extra* validate event (highlighted in green). The 27 natural events of Figure 28(a) fire unchanged. **(b)** Model B: the upper wall is notched down from $\theta_u = 0.85$ to $\theta_u^* = 0.6$ on $t \in [1.7, 2.45]$, an interval that spans most of the osc tick 2 and contains the single local maximum at $t = 2.125$ where $Q = 0.6$ (the earlier local maximum at $t = 1.625$ is left outside the notch). The notched wall meets the signal at this local max, firing a validate touch in a tick that was *silent* in Figure 28(b). The five natural extremum events fire unchanged. By choosing the notch position, level, and modality-appropriate direction (up-notch in Model A on an ascending subsegment; down-notch on θ_u in Model B at a local max), the recipe places a validate or an invalidate at any t^* with arbitrary precision.

APPENDIX E. THE CONVENTION ALGEBRA OF TOUCH EVENTS

The tuning layers of Appendices A, C and D reshaped *which* events a monitor fires; one orthogonal degree of freedom remains — *how* a touch event is timed when the signal lingers at the wall. We now refine a definitional choice that has been latent since the introduction of the touch modality in Section 5: until this section every result and every figure has silently adopted the STRICT convention (touch fires only on strict local extrema). The

convention becomes operationally distinguishable in one specific scenario — a tick window whose content is a flat horizontal segment at *exactly* the level of one of the walls, say $Q(t) \equiv \theta_{\mathrm{u}}$ on an interval of width $L > 0$ — a configuration which the signal-modification operations of Appendix D make constructively available. How many validate events should the touch monitor fire on such a plateau? Five answers are defensible, each corresponding to a distinct semantic convention for "touch". We name the conventions STRICT, WEAK, SAMPLE(f_s), EDGE, and DIRRES $\in \{\mathsf{entry}, \mathsf{exit}\}$[2], and write $C^{\theta}_{Q}(n) := \mathbf{1}\big[Q(n) = \theta_{\mathrm{u}}\big]$ for the at-wall indicator on Q relative to θ_{u} (the analogues for θ_{d} and for invalidate are obtained by flipping signs). A plateau is a maximal interval $[a, b] \subset \mathbb{N}$ with $C^{\theta}_{Q}(n) = 1$ for all $n \in [a, b]$. Each convention is then the following finite-state rule on plateaus.

(C1) STRICT-EXTREMUM: fire validate at $n \in [a, b]$ iff $Q(n-1) < Q(n) > Q(n+1)$ (strict local max). On any plateau with $b > a$, no such n exists; the plateau fires 0 events.

(C2) WEAK-EXTREMUM: fire validate at every $n \in [a, b]$. Plateau fires $b - a + 1$ events (uncountably many in the continuum limit).

(C3) SAMPLE(f_s): fire validate at every sample of the plateau under sampling rate f_s. For our discrete signals with $f_s = 1$ this coincides with WEAK; the rate index is kept to mark the dependency on f_s for finer/coarser grids.

(C4) EDGE-TRIGGERED: fire validate at $n = a$ (entry) and at $n = b$ (exit). Plateau fires exactly 2 events when $b > a$, exactly 1 when $a = b$.

(C5) DIRRES: fire exactly one validate per plateau, at $n = a$ under the entry sub-variant and at $n = b$ under the exit sub-variant. Type at the firing time is set by the wall (validate on θ_{u}, invalidate on θ_{d}); the plateau itself is the ternary state $s = +1$ of the conventions of Appendix E.

Each rule extends to invalidate on θ_{d} by reflection. The five conventions are formally distinct event-stream functors $Q \mapsto E$ that agree on point-like extrema ($a = b$) and diverge exactly on plateaus of positive width.

- **Strict-extremum convention (zero events).** The touch modality of Section 5 fires validate when Q attains a *strict* local maximum at the wall: $Q(t-1) < Q(t)$ and $Q(t+1) < Q(t)$ with $Q(t) = \theta_{\mathrm{u}}$. A horizontal plateau has no strict local maximum — every interior point of the plateau equals its neighbours — and so fires *no* event. This is the convention adopted throughout the paper; it preserves the count inequality of Section 6 and keeps the event stream well-defined as a countable sequence on any finite signal.
- **Weak-extremum convention (uncountably many events).** If one admits every point of the plateau as a degenerate (non-strict) maximum, the event multiplicity is uncountable: a continuum of touches fires at every t in the interior of the plateau. The convention is intuitively appealing — the signal is visibly "kissing" the wall — but produces an ill-defined countable event stream and immediately breaks the count inequality.
- **Sample-rate-bounded convention ($\lfloor L f_s \rfloor + 1$ events).** At the monitor's discrete sampling rate f_s, the plateau is observed at the $N = \lfloor L f_s \rfloor + 1$ samples t_k with $Q(t_k) = \theta_{\mathrm{u}}$. Treating each such sample as a touch yields a finite count that grows linearly with the sampling rate and the plateau width. With our canonical $f_s = 1$ sample per unit

[2]The name EDGE here refers to the plateau-*entry* firing rule, not to the edge-triggered reading of the n=1 automaton in Remark 11.4. Touch remains level-triggered per-tick under all five conventions; EDGE simply restricts firing to the first plateau sample.

and a plateau spanning exactly one tick ($L = 1$), the result is two events at the boundary samples. The count is well-defined but *depends on the sampling rate*: refining the grid multiplies the event stream, which is undesirable for a calibration-stable monitor.

- **Edge-triggered convention (exactly two events).** The simplest finite resolution treats only the transitions *onto* and *off* of the plateau as events: one validate at the rising transit where Q first attains the wall, one at the falling transit where Q leaves it (or one, if the plateau begins or ends at the domain boundary). Under this convention a plateau is a single "compound extremum" bracketed by two flips, and the event count is independent of the sampling rate.
- **Direction-resolved convention (one event per plateau, type chosen by the restart direction).** A plateau is treated as a *decision-deferred* state of the monitor: while Q sits at the wall (or at θ, for Model A) the monitor is "poised but undecided"; the type of event is settled only at the moment Q leaves the plateau, by the sign of the departure slope. In Model B with Q on θ_{u} the rule fires a single validate when Q next leaves the wall — whether going back down (a peak-like resolution) or continuing above (the wall "shielded" the signal as it climbed through). On θ_{d} the symmetric rule gives a single invalidate. In Model A the plateau on θ also produces exactly one event at the departure, of type validate if $\dot{Q}$ becomes positive after the plateau and invalidate if it becomes negative. Each plateau, regardless of width, contributes exactly one entry to the event stream. The convention admits a clean *ternary interpretation* of the monitor's instantaneous state:

$$s(t) \in \{-1,\, 0,\, +1\} \qquad \text{with} \qquad \begin{aligned} s = +1 &\;\Leftrightarrow\; \textsf{validate active},\\ s = -1 &\;\Leftrightarrow\; \textsf{invalidate active},\\ s = 0 &\;\Leftrightarrow\; \text{plateau / no event yet.} \end{aligned}$$

  Points strictly on the threshold or wall are read as $s = 0$: the sensor is engaged but undecided, and the next non-degenerate motion of Q resolves the value to ± 1. This convention unifies the behaviour of the two modalities under a single rule ("plateau-on-the-active-level resolves to one event by departure direction") and matches the operational intuition of Schmitt-trigger-style sensors that delay a verdict until the input recovers a definite slope.

  In Model B the ternary state space $\{-1, 0, +1\}$ separates two refinements of the direction-resolved convention that differ only in *timing*, the type being already pinned by the wall that the signal has touched:

  – *Entry-triggered.* The validate fires at the moment of the $0 \to +1$ transition (the first contact of Q with θ_{u}); the invalidate fires at the $0 \to -1$ transition. The event *announces* the start of the $+1$ / -1 episode and the monitor remains silent for the rest of the plateau.

  – *Exit-triggered.* The validate fires at the moment Q leaves $+1$ — either because it returns to 0 (back into the inter-wall band) or because it jumps to -1 through a value discontinuity introduced by a splice (Appendix C). The event *closes* the $+1$ episode and the monitor commits only after the episode has demonstrably ended. Symmetrically on -1.

  The two refinements coincide on point-like extrema (entry = exit when the plateau has zero width) and differ only on extended plateaus and on cross-band discontinuities. At a $+1 \to -1$ jump introduced by a splice, the entry-triggered convention fires only the new

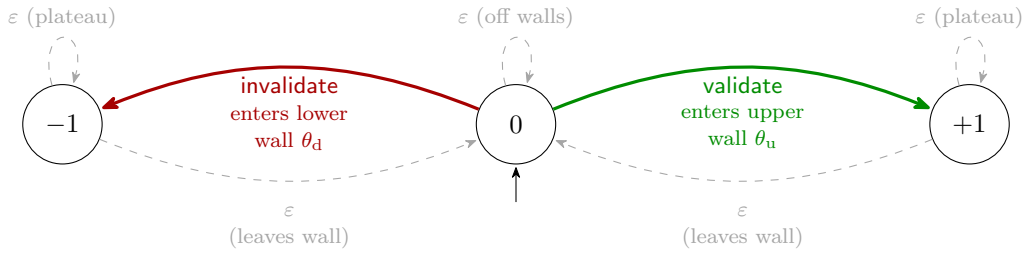
invalidate (the validate was already issued on entry to +1 in the past); the exit-triggered convention fires both events at the same instant — one validate closing the abandoned +1 episode and one invalidate opening the new −1 episode. The choice between the two refinements is itself a tuning parameter of the monitor: a *leading-edge* sensor that wants to react as early as possible adopts entry-triggered; a *trailing-edge* sensor that wants to commit only after evidence has accumulated adopts exit-triggered.

The entry- and exit-triggered DirRes conventions, and the direction-resolved crossing, admit explicit discrete-value-class automata. Figures 30a and 30b draw the two touch refinements and Figure 31 the direction-resolved crossing; the silent (ε, also written λ) transitions carry the signal's wall/threshold dynamics with no emitted event. In all three the states are integer value-classes, the emission alphabet is $\{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ (the two firing labels plus the null event ε), and the transition function is total under the continuous-signal assumption (every value-class move from every state is drawn, no direct wall-to-wall jump). As drawn these are *nondeterministic* ε-NFAs whose silent moves are genuine ε transitions. They are monitors, not partial acceptors: every value-class state is accepting, so determinisation introduces a rejecting state only where an event order is geometrically unreachable — as the dead state $\perp$ does for the crossing (Figure 33). Their deterministic (DFA) refinements are given below.

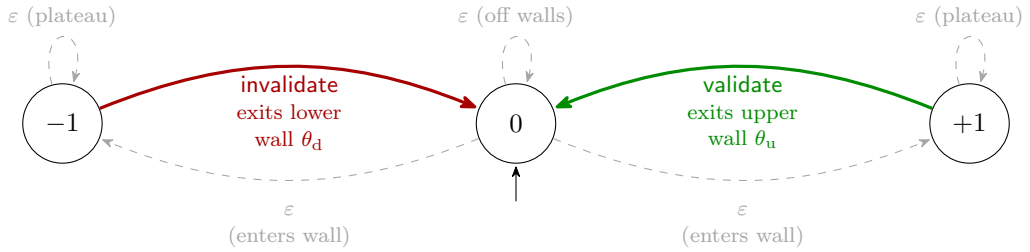
Determinisation. Treating the silent ε moves as ε-transitions and applying the subset construction over the event alphabet $\{\mathsf{validate}, \mathsf{invalidate}\}$ yields the deterministic automata of Figures 32a, 32b and 33. The construction abstracts away the signal-reading geometry — the value-class states merge through their ε-reachability — so each DFA recognises only the *event-language*. Both touch conventions recognise the *same* language, the unconstrained $\{\mathsf{validate}, \mathsf{invalidate}\}^*$: the exit convention already collapses to a single accepting state, and although the entry convention's subset construction produces a two-state automaton (q_V/q_I, recording which wall was last entered), both of its states accept and are Nerode-equivalent, so it minimises to that very same single state. The entry/exit distinction — and the last-entered-wall memory — is thus a *transducer* (Mealy-output) feature, invisible to the event-language: as language acceptors the two touch automata *coincide*. Only the crossing retains genuine structure — its alternation $(\mathsf{validate}\,\mathsf{invalidate})^*$ survives both determinisation and minimisation. (As *transducers reading the value-class input* the original automata are already deterministic; the collapse reflects the loss of that input structure under the ε-NFA abstraction.)

The two direction-resolved automata above (touch and crossing) are one construction read at different active levels; the unified ternary automaton that subsumes them, Figure 11, is drawn earlier alongside the capability join, and all three are collected in Figure 6.

Throughout this paper we adopt the **strict-extremum** convention. It is the simplest definition that preserves the count inequality, keeps the event stream finite on any finite signal, and matches the dominant practice in the touch-detector literature (peak detectors, extremum seekers). The other four conventions are available as *design choices* for monitors that need them — they are part of the tuning vocabulary of the touch modality, not properties of the signal — and the choice itself is a tuning parameter of the augmented monitor in the same sense as the schedule π of Section 10. The *direction-resolved* convention is particularly attractive when one wants both modalities to obey a single uniform rule (one event per plateau, type set by departure slope) and to surface a clean ternary state space $\{-1, 0, +1\}$; the same alphabet is the natural input to downstream temporal-logic monitors.



(A) Entry-triggered: fires on *entering* a wall. This is the expressive value-on-state (Moore) touch automaton whose minimal (Mealy) form is the touch corner of Figure 3.



(B) Exit-triggered: fires on *leaving* a wall.

FIGURE 30. TOUCH automaton, the two DIRRES plateau refinements (Appendix E). In both, the states are the integer value-classes (0 = off the walls, +1 = on the upper wall θ_u, −1 = on the lower wall θ_d), the emission alphabet is $\{\mathsf{validate}, \mathsf{invalidate}, \varepsilon\}$ with ε the null transition, and the continuous-signal base case holds (no direct $-1 \leftrightarrow +1$ wall-to-wall jump). The two conventions differ only in the firing edge: **(a)** the entry-triggered convention fires on *entering* a wall (validate on entering the upper wall, invalidate on entering the lower wall), and the wall-leaving, plateau, and off-wall moves are ε; **(b)** the exit-triggered convention fires on *leaving* a wall (validate on leaving the upper wall, invalidate on leaving the lower wall), and the wall-entering and plateau moves are ε.

Corollary: constant segments are silent zones. The strict-extremum convention for Model B and the signal-direction rule of Figure 22 for Model A together imply that a tick whose content is reduced to a flat segment at *any* level c — whether or not c coincides with a wall — fires no event under any threshold or wall motion that the tuning algorithm may deploy. In Model B the constant signal has no extremum, so no wall notch can create one without modifying Q itself. In Model A the constant signal has zero slope, so no event type matches under the signal-direction rule: a moving threshold would otherwise fire V on a relative ascending crossing and I on a relative descending one, but the rule forbids both unless $\dot{Q}$ has the matching sign at the crossing, and on a plateau $\dot{Q} \equiv 0$. Constant segments are therefore *silent zones* for both modalities under our strict conventions: events can be inserted into such a tick *only* by signal modification that locally restores monotonic motion at the desired times — exactly the operation of Appendix D.1 for which signal mod and threshold mod are jointly necessary. The silent-zone property is what makes a constant tick the cleanest "erase" primitive of the copy-paste construction (Appendix C): pasting a

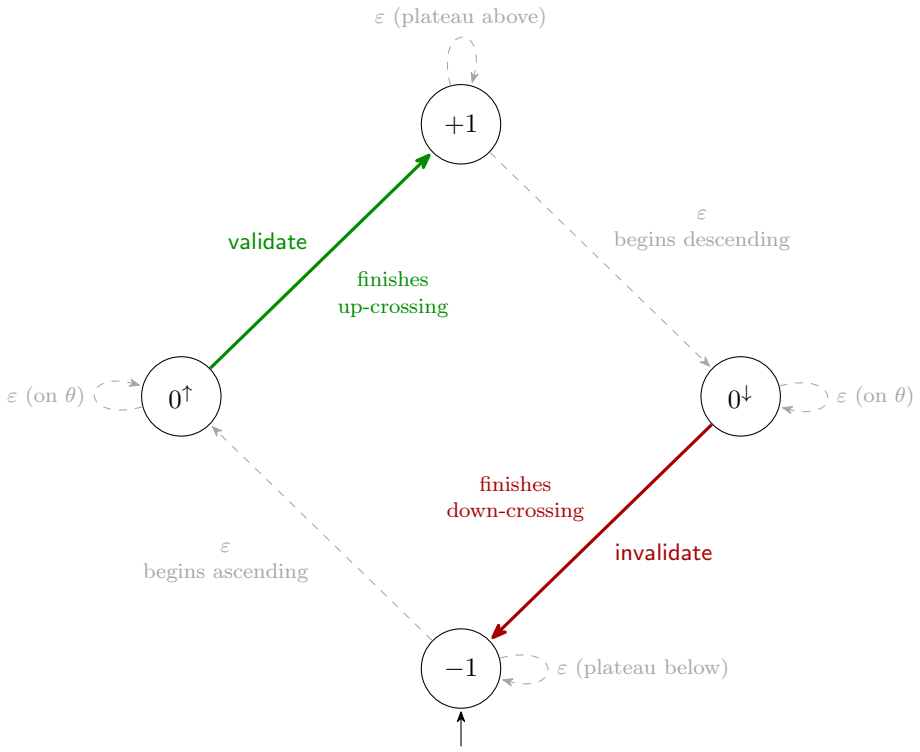
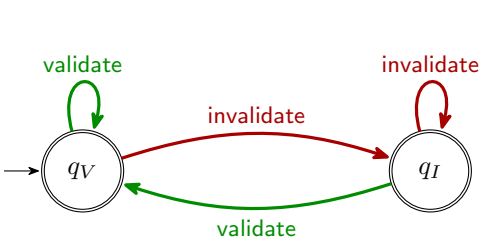
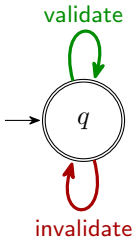


FIGURE 31. CROSSING automaton, *direction-resolved*, entry-triggered. The on-threshold state is split by direction into $0^{\uparrow}$ (ascending) and $0^{\downarrow}$ (descending), resolving the plateau-on-threshold case (Definition 4.2). The superscript arrow is the sign of the discrete slope on the threshold (rising vs. falling). Validate on finishing an up-crossing (entering $+1$), invalidate on finishing a down-crossing (entering -1); the ε (null) transitions are the entries into the on-threshold state and the plateaus. The cycle is $-1 \xrightarrow{\varepsilon} 0^{\uparrow} \xrightarrow{\text{validate}} +1 \xrightarrow{\varepsilon} 0^{\downarrow} \xrightarrow{\text{invalidate}} -1$. Continuous-signal base case. This is the expressive value-on-state (Moore) crossing automaton; its $|Q| = 4$ *refines* the three value-classes $\mathbb{V}_3$ by slope-splitting the on-level class 0 into $0^{\uparrow}/0^{\downarrow}$, so the Moore one-state-per-class reading holds with that refinement ($|Q| = 4 \neq |\mathbb{V}| = 3$, unlike the strict $|Q| = |\mathbb{V}| = 3$ identity of Figure 11). Its minimal (Mealy) form is the crossing corner of Figure 3.

flat segment into a tick erases every event that would otherwise have fired there, without introducing any new ones, irrespective of the surrounding threshold trajectory. Figure 34 renders the property visually: threshold and wall notches that are placed *inside* a constant tick fail to produce any event in either modality, while identical motions placed in a non-constant region of Q would have done so.



(A) Entry convention: two-state memory $q_V = \{+1, 0\}$, $q_I = \{-1, 0\}$.



(B) Exit convention: single state $q = \{0, +1, -1\}$.

FIGURE 32. Subset-construction DFAs of the two touch conventions (Figures 30a and 30b). Both recognise the unconstrained event-language $\{\mathsf{validate}, \mathsf{invalidate}\}^*$ and minimise to the single accepting state of panel **(b)**. **(a)** The entry convention's subset construction produces a two-state automaton recording which wall was last entered; both states accept and are Nerode-equivalent, so the q_V/q_I memory is a Mealy-output distinction, not a language one. **(b)** The exit convention collapses directly to one state accepting every event sequence (the ε-closure of the start state already contains both walls).

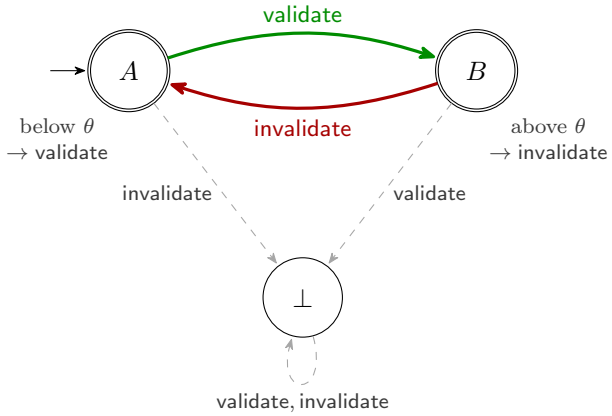
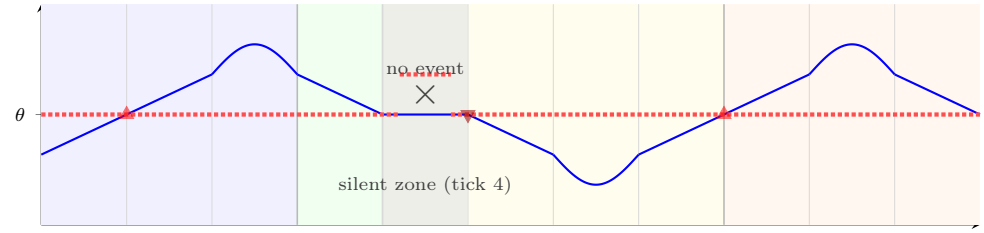


FIGURE 33. Subset-construction DFA of Figure 31. The four signal states collapse to two live classes: the start $A = \{-1, 0^{\uparrow}\}$ (below the threshold, or on it ascending — the only firing that can come next is a validate, completing an up-crossing) and $B = \{+1, 0^{\downarrow}\}$ (above the threshold, or on it descending — the only firing that can come next is an invalidate, completing a down-crossing). The two events must therefore strictly *alternate*, beginning with validate: the recognised language is the prefix-closure of (validate invalidate)* (every alive prefix accepts, so both A and B are accepting). An out-of-order event — a second validate before any invalidate (from B), or an invalidate while still below (from A) — is geometrically impossible and falls into the non-accepting dead state $\perp$. Unlike the touch DFAs, which accept the unconstrained $\{\mathsf{validate}, \mathsf{invalidate}\}^*$, the crossing's alternation is genuine structure and survives both determinisation and minimisation.

(a) Model A: tick 4 held constant at θ is a silent zone — a θ-notch on $[3.7, 4.3]$ fires no event



(b) Model B: the upper wall notched onto the constant tick fires no V touch (no strict max)

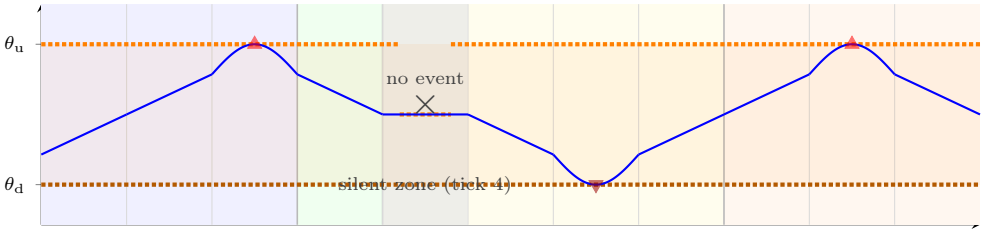


FIGURE 34. Constant segments are silent zones in both modalities. The base signal (solid blue) is piecewise linear with mixed ramps and half-sine bumps elsewhere, and held constant at $\theta = 0.5$ on tick 4 (shaded slate). **(a)** Model A: the threshold is notched up from θ to 0.7 on $t \in [3.7, 4.3]$, a window entirely inside the silent tick. Both the raise at $t = 3.7$ and the drop at $t = 4.3$ would have flipped the sign of $(Q - \theta)$ on a non-constant signal; on the plateau $\dot{Q} = 0$ and the signal-direction rule forbids both V (asc. required) and I (desc. required) — so the notch fires no event. **(b)** Model B: the upper wall is notched down from $\theta_u = 0.85$ to 0.5 on the same interval, so the wall sits directly on the plateau. Under the strict-extremum convention the constant signal has no strict local maximum and no V touch fires. Identical notches placed inside non-constant regions of Q would generate events (as in Figures 22 and 29); silence is a property of *where* the notch is placed, not of the notch itself. The red triangles in each panel mark the events that DO fire on this same signal outside the silent zone — V at ascending crossings / strict maxima and I at descending crossings / strict minima — demonstrating by contrast that the gray $\times$ at tick 4 is a deliberate absence.

Symmetric case: static wall, moving signal. The discussion above has framed the silent-zone property from one side — the wall (or the threshold) is the moving element, the signal is held constant. The touch rule is in fact *symmetric*: a validate fires whenever Q has a strict local maximum at the wall level θ_{u}, regardless of whether the rendezvous was caused by lowering the wall onto a stationary signal or by raising the signal up to a stationary wall. Three sub-cases exhaust the static-wall scenario, each of which applies the same strict-extremum convention:

- **Signal peaks at the wall.** Q rises toward θ_{u}, attains it at a strict local maximum $Q(t^*) = \theta_{\mathrm{u}}$ with $Q(t^* \pm 1) < \theta_{\mathrm{u}}$, then descends. Exactly one validate touch fires at t^*. This is the canonical Model B event.
- **Signal passes through the wall ascending.** Q rises monotonically across θ_{u} and continues above it. The signal is at the wall level only at the single transit time, but that time is *not* a local maximum of Q — the signal keeps rising — so no validate touch fires. Model A would register an ascending crossing here; Model B is blind to it.
- **Signal reaches the wall and plateaus.** Q rises to θ_{u} and stays there on a flat segment. The plateau has no strict local maximum (every interior point equals its neighbours), so once again no validate touch fires. This recovers the silent-zone analysis above with the roles of the wall and the signal exchanged.

The symmetric analysis for θ_{d} and invalidate is obtained by flipping the sign of $Q - \theta$. The asymmetry between the modalities persists across the static-wall and moving-wall framings: Model A sees *every* signal–threshold rendezvous as a crossing (subject to the signal-direction rule), Model B sees only those rendezvous that coincide with a strict extremum of Q.

Why is the no-event verdict defensible in Model B?. The silent-zone analysis is intuitive in Model A: the signal-direction rule of Figure 22 ties event types directly to the sign of $\dot{Q}$, and $\dot{Q} \equiv 0$ on a plateau gives a clean physical reason for the conclusion. Model B's silence stands on a structurally different footing. The touch rule does not refer to $\dot{Q}$ at all — it refers to the *shape* of Q in a neighbourhood of the wall level, and asks whether that shape exhibits a strict local extremum at the wall. A plateau presents no such extremum, regardless of how many walls the monitor maintains. Two consequences are worth emphasising:

- *Adding a second wall does not break the silence.* The two walls of Model B operate independently: θ_{u} detects strict local maxima at its level, θ_{d} detects strict local minima at its level. A constant signal lacks both kinds of extremum simultaneously, so *neither* wall can fire even though there are twice as many "targets" as in Model A. Doubling the wall count does not buy the modality any sensitivity on plateaus — the constraint is at the signal side, not at the wall side.
- *The alternative conventions are local relaxations, not new modalities.* If the strict rule feels too strict for a given application — for instance, in a sensor that should "count" the time spent at the wall — the five conventions of Appendix E (strict, weak, sample-bounded, edge-triggered, direction-resolved) are available without changing the underlying definition of Model B. The *edge-triggered* convention is the natural choice when one wants exactly two events per plateau (one onto, one off): it converts the silent plateau into a pair of bracketing flips while leaving the count inequality of Section 6 intact, because it promotes the plateau's boundary transits to events rather than its interior.

The conclusion is therefore stable across reasonable choices: under the strict convention adopted in this paper, Model B is silent on any constant tick — and the silence is a property of the touch rule itself, not an accident of having a single wall. If a designer needs the constant tick to fire events, the right intervention is to *change the touch convention* (or to modify Q locally), not to add more walls.

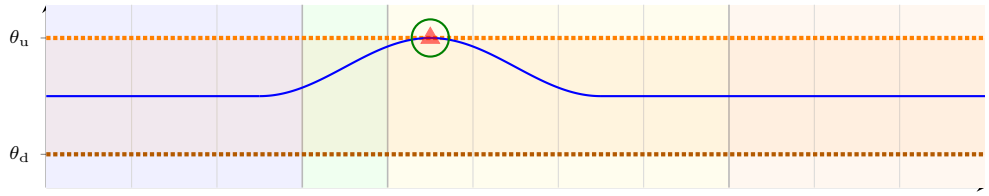
Signal-caused vs. wall-caused state transitions: the causation paradox. The direction-resolved convention with its ternary state $s(t) \in \{-1, 0, +1\}$ generates events on transitions of s, but the transitions can be caused by either signal motion (with walls fixed) or wall motion (with the signal fixed) — and the convention as stated cannot tell them apart. Figure 35 makes this indistinguishability visible. Panel (a) shows a *signal-caused* touch: Q rises into a strict peak at θ_{u} and descends, with both walls static. Panel (b) shows a *wall-induced* "touch": Q is constant at level $c < \theta_{\mathrm{u}}$, and the upper wall is notched down to c over an interval, sitting on the constant signal before returning. In the ternary state both panels produce the same trajectory $0 \to +1 \to 0$, and direction-resolved entry-triggered fires one V in each. Under strict-extremum, however, only panel (a) fires (the strict maximum at the peak), and panel (b) is silent (no strict extremum on a constant signal). The two panels visually pose the question of whether wall motion alone is a legitimate event cause: "yes" adopts direction-resolved as stated; "no" requires the rule to be restricted to signal-caused transitions, restoring the silent-zone property.

Entry vs. exit timing on extended plateaus and cross-band jumps. Figure 36 renders the entry-triggered and exit-triggered refinements of the direction-resolved convention on a Model B signal that contains both a straightforward plateau-on-wall and a splice-induced cross-band discontinuity. Panels (a) and (b) show the same plateau-on-θ_{u} on $[2, 5]$ with events marked under the two timings — entry-triggered fires V at $t = 2$ (left edge), exit-triggered at $t = 5$ (right edge). Panel (c) introduces a copy-paste discontinuity at $t = 7$ that jumps the signal from θ_{u} directly to θ_{d} (Appendix C.1): entry-triggered fires only the new I at $t = 7$ (the V for the $+1$ episode was already issued at its entry $t = 2$), while exit-triggered fires only the closing V at $t = 7$ (the I will fire later, when Q leaves θ_{d}). The two refinements thus *never* fire the same event stream when extended plateaus or cross-band jumps are present.

Longer notches than plateau: four cardinal cases. A subtler scenario arises when the wall notch is wider in time than the signal's plateau on it. The resulting event count depends both on the convention chosen and on the precise relationship between Q and the notched wall in the parts of the notch *outside* the plateau. Figure 37 arranges four cardinal subcases on a common base. In each panel the upper wall is notched down to 0.5 on the wide interval $[1.5, 6.5]$; what varies across panels is the trajectory of Q inside that interval and the resulting state behaviour.

Touch as crossing: the plateau-on-wall is a temporally-extended crossing pair. Panel **(1)** of Figure 37 admits a particularly clean reading that bridges the two modalities. The plateau of Q on the notched wall is interpreted as the touch-modality *equivalent* of two crossing events of Model A — one ascending crossing at the entry edge of the plateau ($t = 3$) and one descending crossing at the exit edge ($t = 5$). The plateau itself, on which $Q \equiv \theta_{\mathrm{u}}^{*}$ for a positive duration, is the touch modality's "engaged" state: a temporal extension of the

(a) Signal-caused touch: Q rises to a strict peak at θ_u; walls static. All conventions fire $1\,V$ at $t = 4$.



(b) Wall-induced touch: Q constant; the upper wall is notched down onto Q over $[2.5, 5.5]$. Strict: 0 events; direction-resolved: $1\,V$.

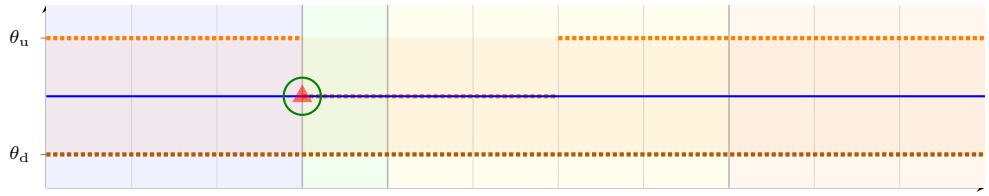


FIGURE 35. The causation paradox of direction-resolved touch. The two panels produce the same ternary trajectory $0 \to +1 \to 0$ but for structurally different reasons. **(a)** *Signal-caused.* Q rises into a strict peak at θ_u at $t = 4$ and returns; the walls are static. All five conventions of Appendix E agree on exactly one validate event at $t = 4$ — the peak is a strict local maximum touching the wall. **(b)** *Wall-induced.* Q is constant at 0.5; the upper wall θ_u is notched down from 0.85 to 0.5 on $[2.5, 5.5]$ and returns. The wall meets the signal not because Q moved but because θ_u moved. Under strict-extremum the event count is 0 (no strict extremum on a plateau). Under direction-resolved entry-triggered the count is $1\,V$ (V = validate, I = invalidate) at $t = 2.5$ (or $t = 5.5$ under exit-triggered), driven purely by wall motion. The two readings differ on whether the tuning algorithm should be permitted to fire events through wall motion alone. The underlying automaton is Figure 11.

instantaneous point at which a Model A signal would have crossed the threshold from below (entry) or from above (exit). Formally:

$$\underbrace{Q(t^-_{\text{entry}}) < \theta^*_u \text{ and } Q(t^+_{\text{entry}}) = \theta^*_u}_{\text{ascending arrival on the wall}} \overset{\text{Model B}}{\longleftrightarrow} \underbrace{Q(t^-) < \theta^*_u \text{ and } Q(t^+) > \theta^*_u}_{\text{Model A ascending crossing}},$$

$$\underbrace{Q(t^-_{\text{exit}}) = \theta^*_u \text{ and } Q(t^+_{\text{exit}}) < \theta^*_u}_{\text{descending departure from the wall}} \overset{\text{Model B}}{\longleftrightarrow} \underbrace{Q(t^-) > \theta^*_u \text{ and } Q(t^+) < \theta^*_u}_{\text{Model A descending crossing}}.$$

Both modalities agree on the *count* (one V at the entry, one I at the exit) and on the *type* assigned by the direction-determined convention; they differ only on the *temporal interpretation* of what sits between the two events. Model A sees the wall level as an instantaneous threshold, so the entry and exit coincide at a single point — the plateau collapses to zero duration. Model B sees the wall level as an extended contact, so the plateau is a positive-duration "engaged" episode separating the two virtual crossings. As the plateau width

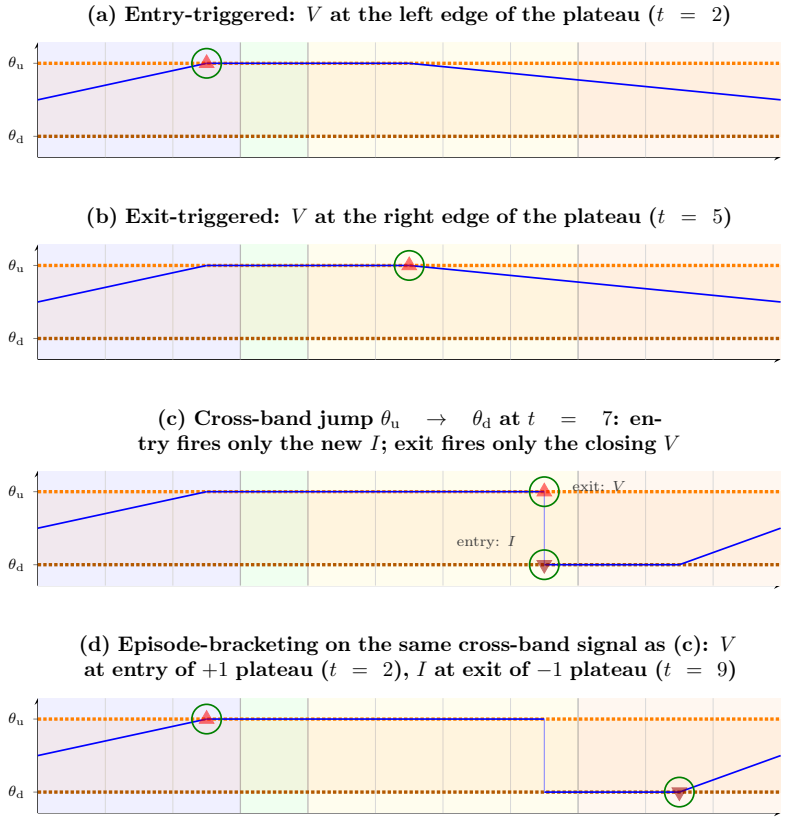


FIGURE 36. Entry-triggered vs. exit-triggered direction-resolved touch in Model B, with edge-triggered as the third comparison case. All three panels share the same upper wall $\theta_u = 0.85$ and lower wall $\theta_d = 0.15$. **(a)** A plateau on θ_u on $[2, 5]$: entry-triggered fires V ($V = \mathsf{validate}$, $I = \mathsf{invalidate}$) at $t = 2$ (the $0 \to +1$ transition). **(b)** The same plateau, exit-triggered: V fires at $t = 5$ (the $+1 \to 0$ transition). **(c)** A splice-induced cross-band discontinuity: Q sits at θ_u on $[2, 7]$, jumps to θ_d at $t = 7$, and sits there on $[7, 9]$. Entry-triggered fires only the new I at $t = 7$ (the V for the prior $+1$ episode was already issued on entry at $t = 2$); exit-triggered fires only the closing V at $t = 7$ (the I for the new -1 episode will fire later, on its exit at $t = 9$). The two conventions never fire the same instantaneous event at a cross-band jump. **(d)** An *episode-bracketing* convention applied to the same cross-band signal of panel (c). V fires at the entry of the $+1$ plateau ($t = 2$) and I fires at the exit of the -1 plateau ($t = 9$). The cross-band jump at $t = 7$ fires no event of its own — it is treated as an internal transition between the two halves of a single bracketed excursion. This is the natural reading when the algorithm wants events to mark only the *opening* of the upper-wall episode and the *closing* of the lower-wall episode, leaving the intermediate kinetics silent. Compared with the entry-triggered and exit-triggered conventions of panel (c), which both fire a single event *at* the jump, the episode-bracketing convention treats the cross-band jump as a continuation rather than as a flip. The underlying automaton is Figure 11.

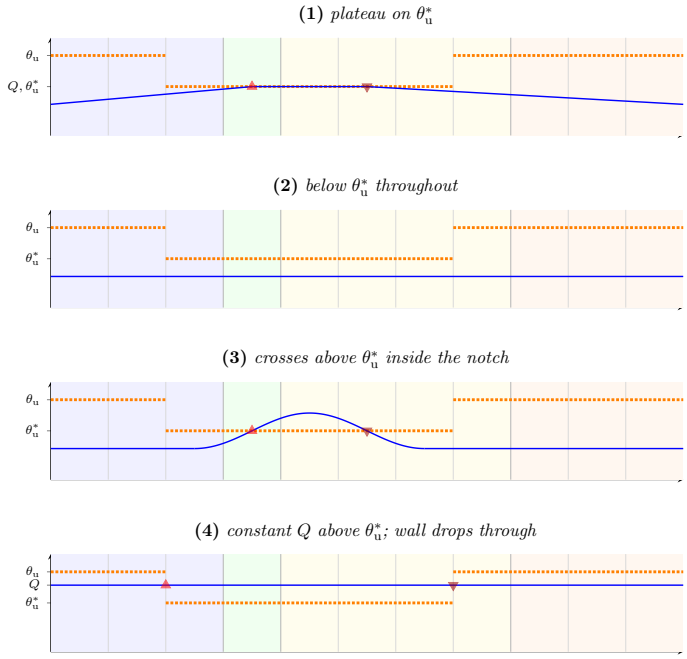


FIGURE 37. Longer-notch scenarios: four cardinal cases for an upper wall notched down to $\theta_u^* = 0.5$ on the wide interval $[1.5, 6.5]$, exposing how the event count depends on the definition of the ternary state $s = +1$ and on which transitions trigger. Writing A for the strict at-wall definition $s = +1 \Leftrightarrow Q = \theta_u$, B for the at-or-above definition $s = +1 \Leftrightarrow Q \geq \theta_u$, and *T1* for signal-caused-only trigger, the four panels contrast as follows. **(1)** Q rises continuously to a plateau at θ_u^* on $[3, 5]$ and descends back inside the notch. Under strict-extremum the plateau has no strict maximum, so 0 events fire. Under any of the other four conventions one event fires at each plateau edge with type set by the direction of crossing: V at the ascending entry $t = 3$, I at the descending exit $t = 5$. **(2)** Q remains strictly below θ_u^* throughout: zero events under all conventions. **(3)** Q rises across θ_u^* at $t = 3$, peaks at 0.7 at $t = 4$, and returns by $t = 5$, all inside the notch. The two signal-caused crossings of the notched wall at $t = 3$ (ascending) and $t = 5$ (descending) are the only events. Under A (strict at-wall) with the extremum requirement the contacts are point-like and no event fires; under B (at-or-above) one event fires at each crossing with type determined by direction: V at $t = 3$, I at $t = 5$. **(4)** Q is constant at a level above θ_u^* while the wall drops through it: under A no events (the wall does not cause contact at a strict extremum); under B the wall crossing through the signal at the notch edges fires two events with type set by the direction of the *relative* crossing — at $t = 1.5$ the wall falls below Q so the relation flips from $Q < \theta_u$ to $Q \geq \theta_u$ (V), and at $t = 6.5$ the wall rises back above Q so the relation flips back to $Q < \theta_u$ (I). Cases (3) and (4) isolate the choice between A and B: at-or-above lets the tuning algorithm fire events on *above-crossings*, even when those crossings are wall-caused, whereas strict at-wall confines events to exact signal-wall coincidences at extrema.

shrinks, Model B's reading degenerates onto Model A's: the plateau collapses to a point and the two virtual crossings become the canonical Model A crossing pair at a single time. Conversely, every Model A crossing can be "inflated" into a Model B plateau by widening the window during which Q sits at the wall level — exactly the regime explored in this section. Conventions (4) (edge-triggered) and (5) (direction-resolved) of Appendix E are the two natural ways to make this duality concrete: edge-triggered keeps the entry and exit as two distinct events, direction-resolved selects one of them as the canonical representative of the whole plateau. This is the elementary bridge between Model A's *point-crossing* semantics and Model B's *plateau-engagement* semantics, and it generalises to every other case of Figure 37 by transporting the same duality from above to below or from upper to lower wall.

Theorem E.1 (Touch-as-crossing duality on plateaus; case-by-case)**.** *Let $Q : \mathbb{N} \to V$ be a signal with a plateau $P = [a, b] \subset \mathbb{N}$ on a wall θ_{u}, i.e. $Q(n) = \theta_{\mathrm{u}}$ for all $n \in P$. Let the* bracket signs *of P be the ordered pair $(s_-, s_+) \in \{-, +\}^2$ with $s_- = \mathrm{sgn}(Q(a-1) - \theta_{\mathrm{u}})$ and $s_+ = \mathrm{sgn}(Q(b+1) - \theta_{\mathrm{u}})$; we treat the boundary case $a = 0$ (left bracket undefined) or b at the trace end (right bracket undefined) by convention* DirRes*, which selects the entry-event direction from the present side.*

For each bracket-sign pair, let E_P^{B} be the event multiset that the touch-native monitor of Definition 5.3 fires on P under the Edge*-convention (C4) of Appendix E, and let E_P^{A} be the event multiset that the crossing-native monitor of Definition 4.1, read under the signal-direction rule of Definition 4.2 (operative throughout Appendix A, so a tangential sample $\tau_n = \theta_{\mathrm{u}}$ fires no event), produces on the* inflated *signal $\tilde{Q}$ defined as $\tilde{Q}(n) = Q(n)$ for $n \notin (a, b)$ and, on the interior, $\tilde{Q}(n) = \theta_{\mathrm{u}} - \mathrm{sgn}(s_- + s_+)\,\delta$ (any $\delta > 0$) for $n \in (a, b)$ — the interior is displaced to the side* opposite *the brackets (above when both brackets lie below the wall, below when both lie above, and left on the wall when the brackets straddle it), so that each wall-contact edge of P becomes a genuine crossing of θ_{u} in $\tilde{Q}$. Then E_P^{B} and E_P^{A} agree as* type-ordered *multisets — same events, same types, same order, same count — with each timestamp determined only* up to position within the on-wall stretch $[a, b{+}1]$*: the* Edge *convention timestamps each event at the wall-contact edge it belongs to, whereas the inflated crossing of $\tilde{Q}$ may place it at the opposite edge of that stretch (the two coincide on a length-*1 *plateau). The timestamps stated below are the* Edge*-convention representatives; in each of the four cases:*

- $(-,-)$**::** *plateau bracketed below on both sides — the canonical case originally treated in the informal statement. Events:* {validate *at* a, invalidate *at* $b+1$}.
- $(+,+)$**::** *plateau bracketed above on both sides (signal descends* to *the wall and re-ascends). Events:* {invalidate *at* a, validate *at* $b+1$} *— the symmetric pair, type-swapped relative to the $(-,-)$ case.*
- $(-,+)$**::** *plateau ascends across the wall (signal transits from below to above at the entry edge and leaves above at the exit edge). Events:* {validate *at* a} *(entry only); the exit is not a flip and produces no event under C4.*
- $(+,-)$**::** *plateau descends across the wall. Events:* {invalidate *at* $b+1$} *(exit only).*

The single-step collapse at $b = a$ is a separately stipulated *boundary case (not a metric limit): a length-1 plateau $P = [a, a]$ under bracket signs $(-,-)$ produces the canonical Model-A* validate/invalidate *pair both timestamped at a; under $(-,+)$ or $(+,-)$ produces a single Model-A ascending/descending crossing at a; and under $(+,+)$ produces* no *event in either model — the touch monitor sees a strict minimum* at *the upper wall and fires nothing, while*

the crossing monitor fires nothing because the single on-wall sample $\tau = \theta_{\mathrm{u}}$ fails the strict signal-direction guard (Definition 4.2) and, the interior (a, b) being empty, $\tilde{Q} = Q$ has no crossing there — so $E_P^{\mathrm{A}} = E_P^{\mathrm{B}} = \varnothing$ in the $(+,+)$ collapse. We establish this statement by exhibiting witnesses for each of the four bracket-sign cases and the length-1 collapse rather than by a single uniform argument; a uniform proof is left to future work.

Remark E.2 (Scope of the verification)**.** The verification above is a four-case verbal walk-through with an informal handling of the $b = a$ boundary collapse (see Remark E.3 below). The four bracket-sign cases and the length-1 collapse are verified individually, but the argument is not yet a uniform formal proof. A four-row case table plus a separate boundary lemma is a candidate route to such a proof, which we defer to a follow-up note.

Remark E.3 (Edge case $b = a$)**.** At $b = a$ the open interval (a, b) collapses to the empty set, so the inflated signal $\tilde{Q}$ is undefined on the interior of the plateau. The proof treats this as the single-step finite-case identity and stipulates that under the symmetric bracket signs $(+,+)$ the Model-B multiset at a is the empty multiset. This case is excluded from the main argument and discharged by stipulation; a formal closure would require a boundary lemma identifying the length-1 collapse with the empty inflation.

Proof sketch. Direct case analysis on the bracket-sign pair (s_-, s_+). Under Edge (C4), Model B fires an event at a plateau edge iff the $\operatorname{sgn}(Q - \theta_{\mathrm{u}})$ flips across that edge; the event type is determined by the direction of the flip (entry-side $-\!\!\rightarrow\!\!=$ gives validate, $+\!\!\rightarrow\!\!=$ gives invalidate). On the inflated signal $\tilde{Q}$, Model A's static-threshold monitor sees the same sign-flip pattern: displacing the interior to the side *opposite* the brackets turns every wall-contact edge of P into a crossing of θ_{u}. For $(-,-)$ the interior sits above the wall, so the entry edge a is an ascending crossing (validate) and the exit edge $b+1$ a descending one (invalidate); for $(+,+)$ the interior sits below, type-swapping both edges to {invalidate at a, validate at $b+1$}; for the straddling half-cases $(-,+)$ / $(+,-)$ the interior stays on the wall and exactly one edge crosses, giving the single Model-A event. The four cases together exhaust the possible bracket-sign pairs.

For the single-step identity $b = a$: the same case analysis applies position-wise to the length-1 plateau, with all events timestamped at a (Model B) or at a and $a + 1$ (Model A) according to the strict-extremum firing rule. The identity is not a metric limit; it is the finite-case specialisation of the inflation construction at $b = a$, which is well-defined because the inflated $\tilde{Q}$ is defined on the open interval (a, b) and that interval is empty when $b = a$. □

Sign-convention symmetry: the validate $\leftrightarrow$ invalidate swap. The five conventions enumerated above and the sub-cases illustrated in Figures 34 to 37 catalogue *detection rules* — when and where an event should be emitted. An entirely orthogonal degree of freedom is the *labelling rule*: which symbol — validate or invalidate — is attached to each detection. Throughout this paper we have followed the natural alignment

$$\theta_{\mathrm{u}} \longleftrightarrow \textsf{validate}, \qquad \theta_{\mathrm{d}} \longleftrightarrow \textsf{invalidate},$$
$$\text{ascending crossing} \longleftrightarrow \textsf{validate}, \qquad \text{descending crossing} \longleftrightarrow \textsf{invalidate},$$

which matches the dominant intuition of "above means validated, below means invalidated". Nothing in the formal definitions of either modality compels this alignment, however: the tuning algorithm may apply a *sign-convention swap* that exchanges validate and invalidate globally, or — finer-grained — per wall, per convention, or per scenario. Every case we

have covered so far admits its swapped twin *verbatim*, with the label of every event reversed and the count inequality of Section 6 preserved (swapping the labels permutes $|E_{\mathrm{B}}|_{\mathsf{validate}}$ with $|E_{\mathrm{B}}|_{\mathsf{invalidate}}$ but leaves the totals intact). The swap is operationally meaningful: a low-water alarm fires $\mathsf{validate}$ at the lower wall (the danger condition), inverting the natural alignment; a danger-zone monitor that should validate when the system *leaves* a safe band similarly benefits from the inverted convention. We have therefore catalogued every case under one labelling; the alternative labelling is obtained by reflecting all event types, and the resulting figures and tables are identical up to the swap. In aggregate, the tuning algorithm's expressive freedom is the Cartesian product of *(timing convention, detection rule)* from the five options $\times$ *(per-wall, per-modality choice)* of which convention to use $\times$ *(per-modality labelling sign)* of the swap above. Any point in this product is realisable; the enumeration in this section makes explicit only the projections that have independent expressive content.

Definition E.4 (Convention algebra)**.** The *convention algebra* of the augmented monitor is the finite product space

$$\mathcal{C} \;=\; \mathcal{T}^{|W|} \times \mathcal{S},$$

where $\mathcal{T} = \{\textsc{Strict}, \textsc{Weak}, \textsc{Sample}, \textsc{Edge}, \textsc{DirRes}_{\mathsf{entry}}, \textsc{DirRes}_{\mathsf{exit}}\}$ is the set of six timing conventions of (C1–C5) above (we split DirRes into its two refinements), W is the set of walls of the modality ($|W| = 1$ for Model A, $|W| = 2$ for Model B), and $\mathcal{S} = \mathbb{Z}/2\mathbb{Z}$ is the sign-convention swap acting on event labels. The cardinalities are

$$|\mathcal{C}_{\mathrm{A}}| = 6 \times 2 = 12, \qquad |\mathcal{C}_{\mathrm{B}}| = 6^2 \times 2 = 72.$$

A point $c \in \mathcal{C}$ is a complete specification of the touch-event detection and labelling rule of the augmented monitor. $\mathcal{C}$ inherits a natural action of $\mathcal{S}$ that reflects event labels globally, and a natural projection $\mathcal{C} \twoheadrightarrow \mathcal{T}^{|W|}$ that forgets the labelling sign.

The convention algebra $\mathcal{C}$ is itself a tuning parameter of the augmented monitor in the same sense as π of Definition 10.2: a complete monitor specification is the tuple $(\theta_{\mathrm{u}}, \theta_{\mathrm{d}}, \pi, c)$ with $c \in \mathcal{C}$. Throughout the rest of this paper we operate at the canonical point $c_0 = (\textsc{Strict}, \textsc{Strict}, \mathsf{id}) \in \mathcal{C}_{\mathrm{B}}$ (strict-extremum on both walls, identity labelling), the choice that preserves the count inequality of Proposition 6.1 and matches the dominant touch-detector literature.

Remark E.5 (Sample(f_s) as a one-parameter family)**.** The Sample entry of $\mathcal{T}$ in Definition E.4 is treated as a single point for cardinality-counting purposes: the cardinalities $|\mathcal{C}_{\mathrm{A}}| = 12$ and $|\mathcal{C}_{\mathrm{B}}| = 72$ count Sample once. Operationally, however, Sample is a *one-parameter family* indexed by the sampling fraction $f_s \in (0, 1]$ (cf. the touch-conventions table of Section 5): each fraction yields a distinct firing rule on a plateau of length $L > 1$, producing $\lfloor f_s L \rfloor$ events instead of one. The canonical reading we adopt throughout is $f_s = 1$ (every plateau sample fires, collapsing Sample to Weak) when calculating count upper bounds, and $f_s = 1/L$ (one event per plateau, collapsing Sample to a chosen instant of Edge) when minimising event count. Counts reported as $|\mathcal{C}_{\mathrm{A}}| = 12$ and $|\mathcal{C}_{\mathrm{B}}| = 72$ are therefore *modulo the Sample parameter*; allowing f_s to vary freely produces a continuum of conventions sandwiched between Edge (f_s near zero) and Weak ($f_s = 1$).

Remark E.6 (Cardinality enumeration)**.** The factorisation $|\mathcal{C}_{\mathrm{B}}| = 6^2 \times 2 = 72$ of Definition E.4 reads as follows. The six timing conventions $\{\textsc{Strict}, \textsc{Weak}, \textsc{Sample}, \textsc{Edge}, \textsc{DirRes}_{\mathsf{entry}}, \textsc{DirRes}_{\mathsf{exit}}\}$ are chosen *independently* on the upper and lower walls, giving $6^2 = 36$ detection-rule pairs; the sign-swap $\mathcal{S} = \mathbb{Z}/2\mathbb{Z}$ then doubles this to 72 by either preserving or exchanging the

validate $\leftrightarrow$ invalidate labels. Model A has a single wall and a single timing choice, so its detection-rule space has only $6 \times 2 = 12$ points. The same independence assumption underlies the SAMPLE subspace collapse noted in Remark E.5: the upper-wall sampling fraction $f_s^{(u)}$ and the lower-wall fraction $f_s^{(d)}$ are also independent parameters when SAMPLE is chosen on each wall.

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